

# Towards Secured Smart Grid 2.0: Exploring Security Threats, Protection Models, and Challenges

Lan-Huong Nguyen<sup>†</sup>, Van-Linh Nguyen<sup>†</sup>, *Senior Member, IEEE*, Ren-Hung Hwang, *Senior Member, IEEE*, Jian-Jhih Kuo, *Member, IEEE*, Yu-Wen Chen, *Member, IEEE*, Chien-Chung Huang, Ping-I Pan

**Abstract**—Many nations are promoting the green transition in the energy sector to attain neutral carbon emissions by 2050. Smart Grid 2.0 (SG2) is expected to explore data-driven analytics and enhance communication technologies to improve the efficiency and sustainability of distributed renewable energy systems. These features are beyond smart metering and electric surplus distribution in conventional smart grids. Given the high dependence on communication networks to connect distributed microgrids in SG2, potential cascading failures of connectivity can cause disruption to data synchronization to the remote control systems. This paper reviews security threats and defense tactics for three stakeholders: power grid operators, communication network providers, and consumers. Through the survey, we found that SG2's stakeholders are particularly vulnerable to substation attacks/vandalism, malware/ransomware threats, blockchain vulnerabilities and supply chain breakdowns. Furthermore, incorporating artificial intelligence (AI) into autonomous energy management in distributed energy resources of SG2 creates new challenges. Accordingly, adversarial samples and false data injection on electricity reading and measurement sensors at power plants can fool AI-powered control functions and cause messy error-checking operations in energy storage, wrong energy estimation in electric vehicle charging, and even fraudulent transactions in peer-to-peer energy trading models. Scalable blockchain-based models, physical unclonable function, interoperable security protocols, and trustworthy AI models designed for managing distributed microgrids in SG2 are typical promising protection models for future research.

**Index Terms**—Smart Grid 2.0, Intelligent Energy Networks, Security Attacks, AI for Smart Grid, AI for Security.

## I. INTRODUCTION

Integrating distributed energy systems is a major topic in the green and renewable energy era with sustainable goals of usage efficiency, autonomous intelligence, and resilience capability against sudden failures. These new energy integration capabilities expects to be the core of Smart Grid 2.0 (SG2) [1], [2]. SG2 aims to enhance energy distribution and usage efficiency with the help of communication technologies [3]–[5]. These interdependent power grid and communication networks can help connect distributed electricity, gas,

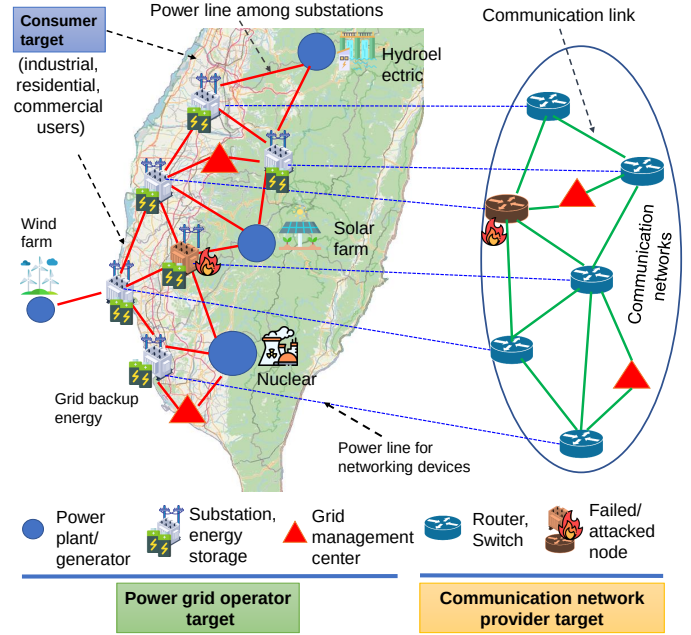


Fig. 1. The illustration shows the crucial role of communication technologies in synchronizing measurement data from substations, enabling remote control capabilities for efficient power distribution. However, the dependency of power grids on communication technology creates fresh threats of security attacks to energy security, e.g., ransomware to disable control systems and denial of services against transmission lines to stop data exchange.

and cooling systems, offering unprecedented opportunities for remote control capability and flexibility [6]. However, with reliance on digital communication technologies, as shown in Fig. 1, SG2 faces significant threats, targeting connectivity among power grid providers, two-way communication network systems, and consumer entities (industrial, residential, and commercial users). For example, ransomware attacks have recently been recorded to cause prominent blackouts in many countries [7].

Recent drone attacks and supply chain risks also threaten critical facilities (economic loss) and even endanger public safety (e.g., power outages in the cooling systems of nuclear plants). Also, providing sophisticated communication networks to millions of charging stations and diverse renewable energy sources in SG2, while not overburdening the distribution network or destabilizing the grid, is also a challenge. Understanding the threats and challenges is the critical step toward developing robust defense approaches for guaranteeing energy security, and further national safety. This article aims

L. H. Nguyen and R. H. Hwang are with College of Artificial Intelligence, National Yang Ming Chiao Tung University, Tainan, Taiwan.

V. L. Nguyen and J. J. Kuo are with the Department of Computer Science and Information Engineering, National Chung Cheng University (CCU), and also with the Advanced Institute of Manufacturing with High-Tech Innovations (AIM-HI), CCU, Taiwan.

Y. W. Chen is with New York City College of Technology, New York, United States.

C. C. Huang and P. I. Pan are with Green Energy and Environment Research Laboratories, Industrial Technology Research Institute, Tainan, Taiwan

<sup>†</sup> L. H. Nguyen and V. L. Nguyen contributed equally to this work

Corresponding authors: R. H. Hwang

to explore the various security threats in SG2, particularly its communication infrastructure and SG2 enabling technologies, such as peer-to-peer energy trading and AI-powered grid network functions. In addition, the study discusses emerging strategies for safeguarding SG2 from cascading failures and for developing effective distribution grid restoration plans in disaster scenarios and severe security attacks in the future.

### A. State-of-the-art literature review

Exploring security threats for smart grids has been a hot topic for years but few studies address SG2's security matters in a comprehensive manner. Fig. 2 presents SG2's essential components and security concerns, together with state-of-the-art relevant studies. Accordingly, most articles focus on security threats in the conventional smart grid that features electricity sources and distribution. For example, the authors in [8] provide a comprehensive survey of typical security attacks and vulnerabilities of authentication and security protocols in the conventional energy model. AI and blockchain for conventional smart grids and related security is briefed in [9]–[11]. However, the paper covers few aspects of security protection or Artificial Intelligence (AI) role in specific functions in each layer or from related stakeholders (power generator, communication provider, and consumers). Similarly, the surveys in [12]–[14] cover a narrow scope of security in specific smart grid networks, e.g., the connection between home and grid supply, metering data collection and transmission [15].

Recently, the survey papers focus on the security threats in communications among power generation and distribution components of the newer smart grid model with battery storage systems [16]–[19]. As technology and infrastructure continue to advance, renewable energy (e.g., wind, solar, geothermal, hydropower) plays an increasingly vital role in the global energy transition toward a cleaner and more sustainable future. Energy storage systems are required to maintain the stability of such distributed sources. Several surveys on the safety of smart inverters [20], battery storage/swap [21], or control systems [22] against remote attacks or physical tampering is also presented. On the other hand, the authors in [23]–[25] provide a holistic view of control and communication strategies in multi-energy generation grids or robust models against cascading failure in interdependent power-communication networks. However, the studies did not address the security threats or specific attacks for each entity (power provider, communication network provider, consumer). Unlike prior studies, this work aims to investigate weaknesses in the interdependence of microgrids that heavily rely on distributed energy sources and communication technologies. Additionally, there will be a focus on identifying new risks associated with AI-powered energy control and novel energy trading/storage models. This research will be particularly important as many new small energy sources (e.g., from solar roofs) are integrated into management networks. Overall, the first goal of our work is to provide a comprehensive view of cybersecurity in these new elements, referring to SG2's energy security principles, which have received little attention in the existing literature.

Besides, many governments considered grid security as a national security matter and proposed measures to improve



SG2's key security matters for three entities (power provider/ network provider/consumer)

Fig. 2. This work addresses energy security principles for SG2 from a view of interdependent power grid communication networks, notably with the introduction of new technologies for three entities (power provider, communication network provider, consumer), such as energy storage, 5G/6G, AI-powered functions, and peer-to-peer energy trading models.

information security protection [26]–[29] and resilience strategies in disaster and crisis scenarios [23], [30], [31]. Grid security refers here to the consistent and reliable availability of all fuels and electricity sources in a timely, sustainable, and cost-effective manner. For example, several standards for information security in smart grids, such as the framework developed by the National Institute of Standards and Technology (NIST), have been developed. The US Department of Energy is carrying out the Cybersecurity for Energy Delivery Systems (CEDs) program [32], aimed to enhance the security and resilience of the country's energy infrastructure. The North American Electric Reliability Corporation (NERC) has established network security standards for the power industry in North America. The European Union Agency for Network and Information Security (ENISA) has issued guidelines for safeguarding EU's smart grids [29]. The International Electrotechnical Commission (IEC) has established standards for network security in power systems (IEC 62443, 62351 standards) [27]. Therefore, this survey's second goal is to determine which features have not yet been defined in the security standards and what standards the nationals apply for their energy management systems based on facility availability, deployment cost, and environment compatibility. Further, understanding energy restoration plans for potentially cascading failures of communication technologies is critical to consult a proper model for deploying SG2.

### B. Review methodology

Given the difficulties of installing from scratch owing to high costs, SG2 will likely inherit many control components, existing facilities, and communication infrastructure from the current smart grid. Inspired by this fact, we present possible security concerns in SG2, as viewed through the mirrors of two aspects' lessons learned. The first phase involves examining energy security principles, identifying the primary risks to the components of a smart grid system, and assessing the security vulnerabilities in legacy technologies with examples of well-known energy crises and blackout events. Additionally,

this involves energy restoration strategies in the event of probable cascade failures resulting from security attacks on communication lines. The rest is to figure out security flaws in emerging technologies and new decentralized energy models that are expected to be the main vehicles towards SG2, such as AI-powered energy control functions, battery storage technologies, and the integration of advanced communication technologies for charging stations and distributed renewable energy sources. Security threats are often the motivating factor behind the need to change countermeasure approaches. These attacks typically reveal system faults or protocol issues that were not expected during the design process. Analyzing and learning from attacks like this gives significant insights into the essential security changes for SG2, particularly in addressing known weaknesses exploited in prior generations. Finally, this work also addresses security matters from a top-down approach where security threats against the interdependent relationship of stakeholders (power grid operator, communication network provider, consumer) will be assessed and suggested with corresponding defense strategies. The unsolved problems become possible targets for SG2 improvements, which serve as the foundation for proposing future solutions.

### C. Contributions

Given the slow transition from legacy to new technologies, it is difficult to predict when SG2 will be in full operation. However, by drawing a line of relative differences between the current smart grid platform and the expected SG2 architecture, this research can help the developers and researchers determine the security weaknesses and find the right starting point of SG2's technologies to improve. The primary contributions in this work are summarized as follows.

- 1) The first attempt to thoroughly investigate the principles of SG2 security for national safety, taking into account the whole perspective of security measures for communication links among power grid operators, communication network providers, and consumers. The study examines the relationship between power grids and communication networks in terms of cascading failures. It identifies potential solutions and necessary improvements for SG2, specifically in energy restoration planning and communication isolation.
- 2) The first attempt to offer a comprehensive perspective on the security risks of SG2 enabling technologies that need to be adapted to meet the evolving requirements of SG2, e.g., blockchain-based energy management/trading, AI-aided grid operations, the networks of electrified transportation systems (EV charging stations) and distributed renewable energy sources. Given that SG2 follows economic trajectory of the technology evolution, a systematic review of the transition process and potential changes in supply chain management and new communication methods can guide power grid operators and network providers in effectively upgrading their security infrastructure and countermeasure techniques in the future.
- 3) This study summarized lessons learned from the limitations of current protection implementations in SG1

and the vulnerabilities of SG2 emerging technologies that can aid researchers and developers in determining the problem formulation for further studies. To the best of our knowledge, this survey represents the initial endeavor to comprehensively assess security threat aspects for SG2, spanning from vulnerabilities in distributed renewable energy sources to EV charging network architecture, and then AI-powered grid management.

### D. Structure of the paper

The rest of this paper is organized as follows. Section II briefs the fundamental information about SG2 architecture, energy security principles, and overall strategies to protect the power grid-communication networks' infrastructure. The security attacks and defense approaches for power providers, communication network providers, and consumer stakeholders are then detailed in Section III, IV, V, respectively. Section VI outlines security risks and some countermeasure techniques in emerging technologies and their role in securing SG2 is detailed in Section VII. Section VIII discusses lessons learned and future research. Section IX concludes this paper. Fig. 3 summarizes the main points of our survey. The acronyms used in this work are listed as follows.

#### ACRONYMS

|       |                                                            |
|-------|------------------------------------------------------------|
| AGI   | Artificial General Intelligence                            |
| AI    | Artificial Intelligence                                    |
| AKA   | Authentication and Key Agreement                           |
| DERs  | Distributed Energy Resources                               |
| DMZ   | Demilitarized Zone                                         |
| DPI   | Industrial Deep Packet Inspection                          |
| ENISA | European Union Agency for Network and Information Security |
| FAN   | Field Area Network                                         |
| HAN   | Home Area Network                                          |
| HSM   | hardware security module                                   |
| ICS   | Industrial Control Systems                                 |
| IDS   | Intrusion Detection System                                 |
| IEC   | International Electrotechnical Commission                  |
| IPS   | Intrusion Prevention System                                |
| LAN   | Local Area Network                                         |
| MTD   | Moving Target Defense                                      |
| NAN   | Neighborhood Area Network                                  |
| NERC  | North American Electric Reliability Corporation            |
| NIST  | National Institute of Standards and Technology             |
| PKI   | Public-key Infrastructure                                  |
| SCADA | Supervisory Control And Data Acquisition                   |
| SG1   | Smart Grid 1.0                                             |
| SG2   | Smart Grid 2.0                                             |
| WAN   | Wide Area Network                                          |

## II. SG2: UPGRADES FROM SG1 AND SECURITY CONCERNS FROM GRID SECURITY STRATEGIES

This section introduces fundamental differences between smart grids (i.e., Smart Grid 1.0 (SG1)) and SG2, key security concerns, and defense strategies. The discussions of SG2 insecurity and national safety are also presented.

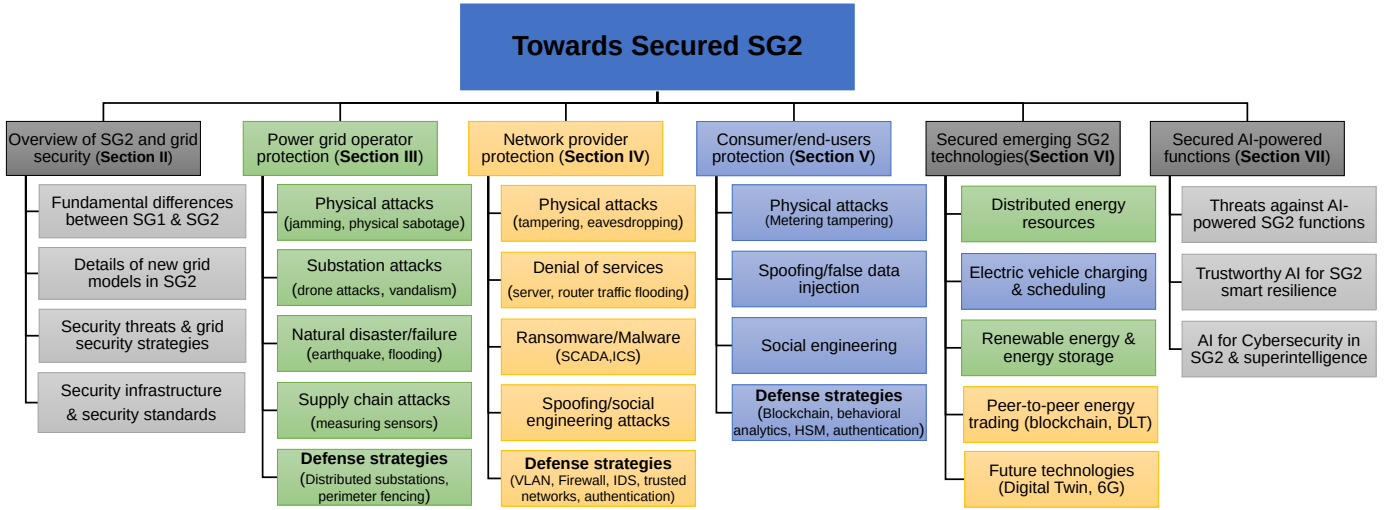


Fig. 3. The following is a summary of the major findings from our survey on security and protection strategies for Smart Grid 2.0. The decorative colors for technologies match those for the three entities (power provider, communication network provider, and customer) illustrated in the previous figures.

#### A. Overview of fundamental differences between SG1 and SG2

SG2 is expected to be a comprehensive framework designed to modernize and enhance the usage efficiency, reliability, and sustainability of energy distribution and management in SG1. SG2 leverages communication technologies and data analytics to transform SG1 into interconnected and intelligent systems that are capable of optimizing energy production, distribution, and consumer demands at peak periods. Fig. 4 illustrates the fundamental differences between SG1 and SG2. Instead of managing energy distribution from large power plants only as in SG1, SG2's architecture will involve both large and small producers in a distributed energy model, where microgrids can automatically be operated to serve city oases and partially managed by AI. In this way, key features in SG2 are distributed intelligence and self-healing grids, highlighting autonomous energy distribution management. Overall, SG2 enhances automated control and centralized management model in SG1 with the new capabilities, such as distributed microgrid model, peer-to-peer energy trading, and self-healing/self-monitoring grids.

SG2 integrates the goal of reducing carbon emissions toward clean/green energy. For example, to achieve the goal of carbon neutrality by 2050, many countries expect the contribution of solar and renewable energy sources to be up to 40% by 2035 or 45% by 2050 [33]–[35], and a Net-Zero emission target by 2070 [36]. As a result, in the next decades, renewable energy sources (hydropower, solar power, wind power, bioenergy, and geothermal energy) will dominate the market. However, given the challenges of far-distance distribution from power plants (e.g., solar farms in the deserts or wind farms at the beach), maintaining the continuity of renewable energy sources, particularly at peak hours, needs the help of advanced technologies, such as energy storage and super grids [37].

#### B. Details of new energy generation, distribution, and consumption model in SG2

Generally, SG2 includes three primary components: power grid providers, communication network providers, and con-

sumers, as illustrated in Fig. 5. Power grid providers/operators have a complicated infrastructure of power plants, power lines, transformers, and substations. Power grid providers/operators play a role in energy generation and distribution. Network providers provide communication technologies for energy measurement and remote control, i.e., connectivity to synchronize data from sensors at substations or consumers' smart metering devices to management applications, e.g., Supervisory Control And Data Acquisition (SCADA). The communication infrastructure includes the Wide Area Network (WAN), Neighborhood Area Network (NAN), and Home Area Network (HAN) with networking technologies like LoraWAN/Zigbee [21]. Generally, the WAN is the primary network infrastructure for establishing a connecting backbone. NAN and Field Area Network (FAN) are used for connecting customer smart meters, substations, and WAN. The HAN facilitates connectivity for devices in the house/building/factory [20]. The diversity of connectivity technologies in HAN and NAN is to maintain flexibility for supply chains in different countries. In this work, we use the general network architecture for research reference.

Regarding specific upgrades from SG1, there are two major upgrades in SG2: (1) upgrading the way of generating, distributing, trading, and consuming energy in power grid providers/operators (2) upgrading the networking infrastructure and consumer devices. For the first major upgrade, the energy landscape evolves from a one-way distribution model to a dynamic, bidirectional flow of energy and information [21], [22]. Decentralized energy resources, such as solar panels and energy storage systems, assume a more pivotal role, enabling consumers to be energy users and contributors to the grid. Consumers evolve from passive users to active participants who can not only monitor their energy consumption in real-time but also contribute excess energy back to the grid (P2P trading), transforming them into prosumers (as illustrated in the right side of Fig. 5). Real-time data from the smart sensors at substations enables utilities to optimize energy distribution and balance demand and supply proactively [6].



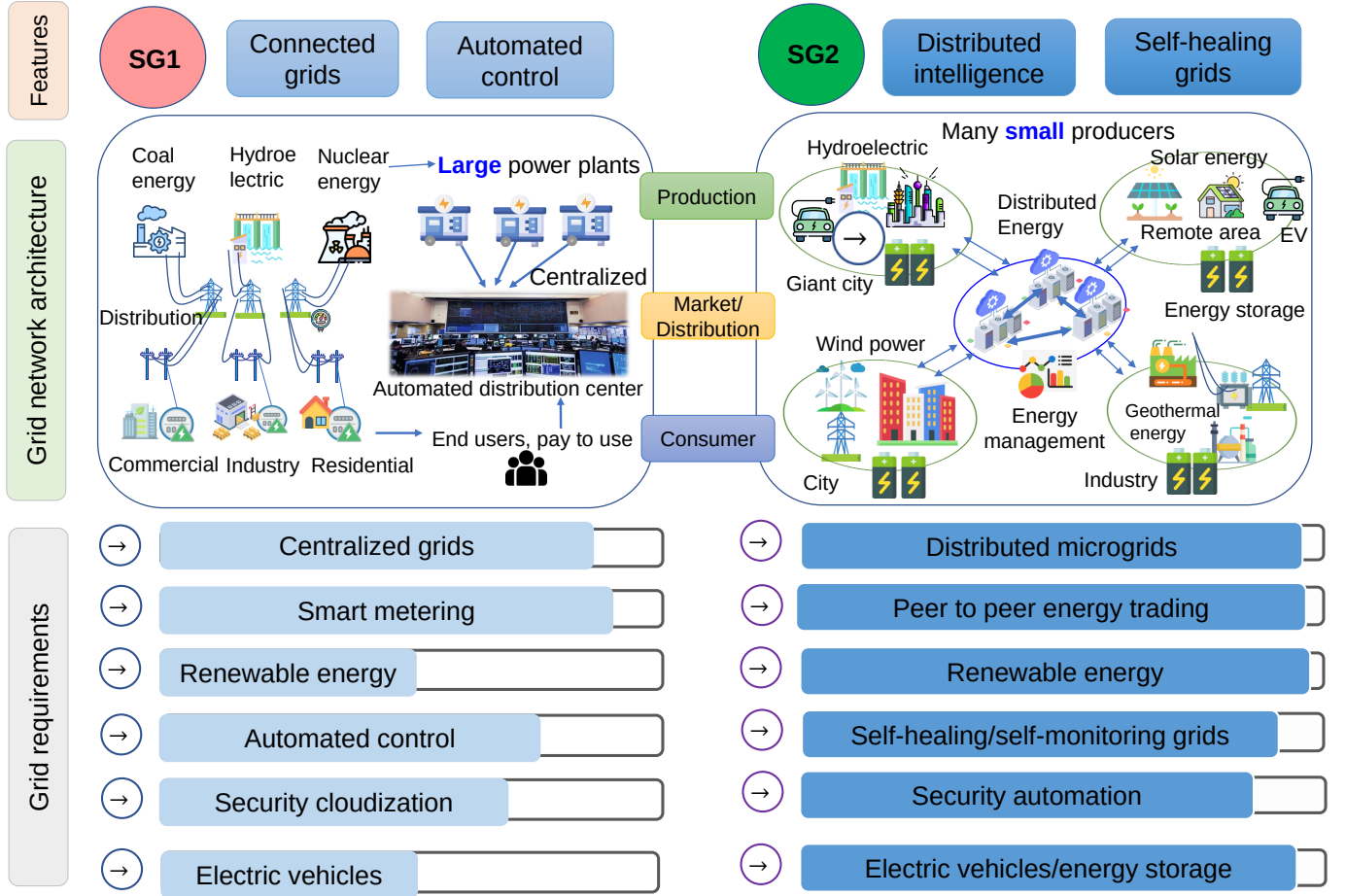


Fig. 4. The illustration of comparison between smart grid 1.0 and smart grid 2.0 along with key features, architecture and main requirement differences.

Moreover, to enhance grid resilience, the introduction of microgrids [38], [39], electric vehicle integration, and predictive maintenance ensures a more resilient, reliable, and efficient energy infrastructure. Microgrids emerge as localized energy systems that can operate autonomously or interdependently with the main grid. By enabling localized generation and consumption, microgrids bolster grid resilience, allowing communities to maintain power during disruptions and emergencies. Through data-driven predictive maintenance and AI-powered control algorithms [40]–[42], grid operators can preemptively identify and address potential faults, minimizing downtime and optimizing system performance. The integration of electric vehicles (EV) also becomes a crucial component in SG2, with the ability to intelligently manage EV charging and discharging [43], not only minimizing grid congestion but also utilizing EV batteries as a form of distributed energy storage [44]. Further, that is the appearance of novel customer-oriented energy management as a service (EMaaS) [45] or a fair demand response with electric vehicles [46]. This evolution necessitates communication and management modernization, a significant upgrade from the conventional smart grid model.

For the second upgrade type, most upgrades of network infrastructure come from new versions of existing WAN/FAN/LAN platforms (e.g., Zigbee, LTE) to be integrated into the smart metering and control management systems

[21]. Recently, the European Union (EU), China, and private sectors in the United States and other countries proposed a novel communication model to exploit the technical features of the fifth-generation (5G) communications for SG2 [47], [48]. By exploiting the broadband capability to collect massive amounts of unstructured data (video, images), 5G can assist energy distributors in improving their energy transmission flow efficiency and delivering personalized services to specific consumers. Implementing cloud offloading, such as optimizing energy production and delivery, is crucial for system security and stability [47]. These activities may be time- and space-intensive. Cloud and edge computing from 5G may reduce computational burden in the center and allow the system function distributed and resiliently. In the long term, the sixth-generation (6G) networks can be the successor of 5G to continue enhancing the efficiency of the operations and enabling the development of new renewable energy sources [49]. 6G with non-terrestrial network segments, e.g., satellite communications, can help utilities monitor and manage electricity distribution over larger areas, which is especially beneficial for microgrid communities in rural and remote areas.

### C. Security concerns in transition networks towards SG2

Electricity has always been crucial for maintaining essential societal functions, including electrified transportation,

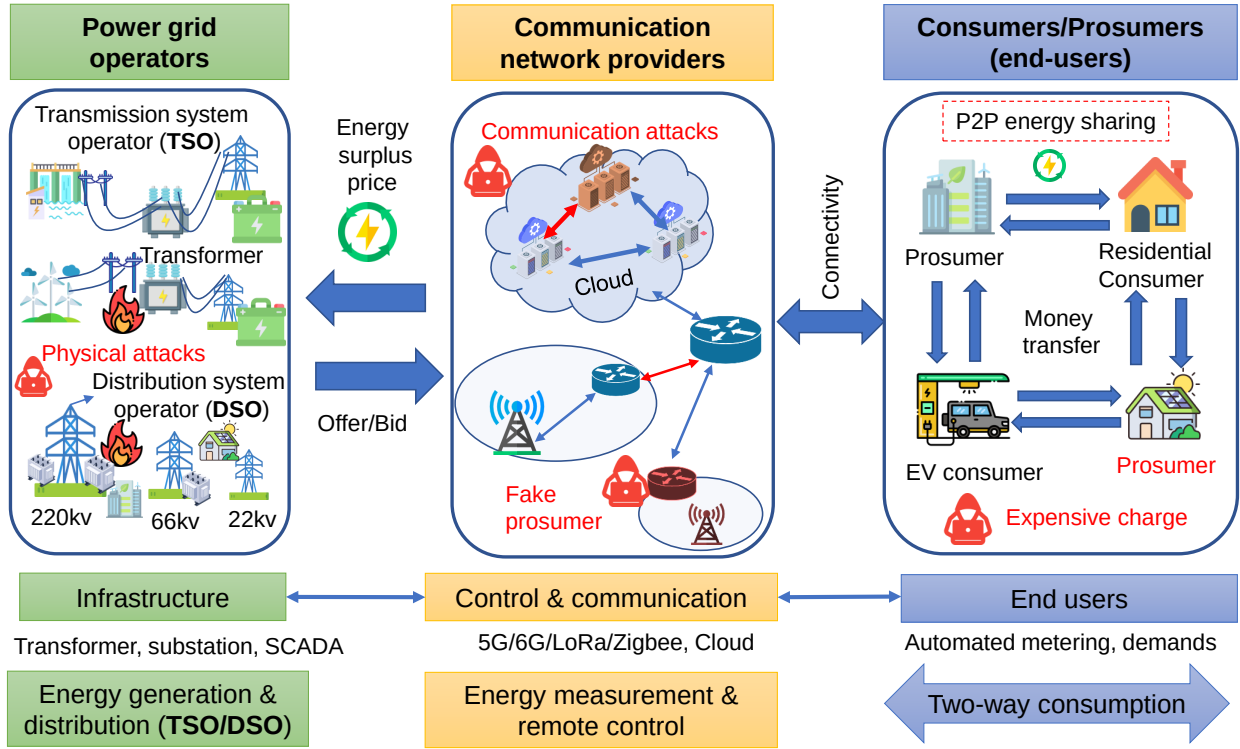


Fig. 5. The illustration of three stakeholders in SG2: power grid providers, communication network providers, and consumers. Power providers have a complicated infrastructure of power plants, transformers, and substations. Communication network providers include several networking technologies (e.g., Zigbee, LoRa), which provide connectivity to connect sensors from energy generation, transmission, and distribution, to consumers and remote centers.

machinery lines, public light, and economic development. The transition trend from fossil fuels to renewable energy in SG2 also motivates many users to use electric vehicles, which increases pressure on power grid infrastructure. Since society and millions of people depend on massive amounts of electric resources, ensuring a timely, sustainable, and cost-effective electricity supply is a priority. Blackouts over several hours, let alone days, can disrupt crucial services in hospitals and nursing homes [50], halt manufacturing lines/transportation, and cost millions of dollars in damages [51]. For example, in December 2015, some versions of the BlackEnergy and KillDisk malware disrupted the functioning of many substations in Ukraine, resulting in the loss of power for around 225,000 citizens for a few hours [52]. Generally, electricity security is a national security matter. Besides civil usage, electricity is also vital for military factories and combat control operations. Table I summarizes key security threats and potential protection approaches. The security threats and typical examples are classified by three target stakeholders, as shown in Fig. 5, i.e., power grid provider/operator, communication network provider, and consumer. The severity and likelihood aspects are assessed using records of well-known attacks or catastrophic economic losses. The following sections detail the security threats and defense for three stakeholders.

For power grid operators (the first target in Table I), one of the major security concerns is sudden hardware failures of power generators, actuators, monitoring sensors, and microcontrollers [23]. Although the severity level is high, the failures of these devices are rare because of the stringent standards

for lasting quality of supply in smart grid infrastructure [28]. The other biggest security concern for power providers is physical security, where there are diverse threats, such as physical sabotage, vandalism, drone attacks (against important substations or high-voltage power lines, e.g., 500kV, 220kV), compromised supply chain, and even natural disasters. Further, since renewable energy sources are used to generate electricity, the stability of these supplies are critical, too. The details of attacks and protection methods are discussed in Section III.

For network providers (the second target in Table I), major security concerns are physical and cyber attacks. Several typical examples are communication link tampering, ransomware/malware injection to internal computers or SCADA, and denial of services against routers and control centers. For example, Enel Group's internal IT network in Europe was temporarily blocked by Snake Ransomware attacks in June 2020, resulting in the disruption of customer service activities [59]. The increased use of digital and networking technologies to optimize energy usage and distribution efficiency and provide one-click remote control capability. However, this interconnected model theoretically creates new space for attackers to access the power grids. In SG2, the communication technologies help to synchronize measurement sensor data at power providers' facilities (e.g., cooling systems, microcontrollers) into overload monitoring applications. The heavy dependence of control systems on data exchange or remote control convenience is vulnerable to cyberattacks. The attacks target two primary purposes: (1) interrupt the data stream exchange or routing devices; (2) falsify the data [75].

TABLE I  
THE MAIN THREATS, TYPICAL EXAMPLES, THEIR SEVERITY LEVEL, LIKELIHOOD AND POTENTIAL PROTECTION APPROACHES IN SG2.

| Target                           | Security threats  | Example                                                                                                                                                                                                                                                | Severity | Likelihood | Protection method                                                                                                                                                                                                                                                                                                       | Reference                                                                             |
|----------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Power grid operator (TSO/DSO)    | Equipment failure | <ul style="list-style-type: none"> <li>► Power generator</li> <li>► Cooling system</li> <li>► Actuators/sensors</li> <li>► RTU/PMU/HTS</li> <li>► PLC microcontrollers</li> <li>► Transformer</li> <li>► Switchgear</li> <li>► Substations</li> </ul>  | High     | Low        | <ul style="list-style-type: none"> <li>✓ Regular maintenance</li> <li>✓ Emergency backup</li> <li>✓ Equipment stockpile</li> </ul>                                                                                                                                                                                      | [23], [50], [53], [54]<br>[21], [24], [55], [56]                                      |
|                                  | Physical security | <ul style="list-style-type: none"> <li>► Time synchronization</li> <li>► Substation attacks</li> <li>► Thief and vandalism</li> <li>► Terrorist/drone attacks</li> <li>► Natural disasters</li> <li>► Supply chain</li> <li>► Aging devices</li> </ul> | High     | High       | <ul style="list-style-type: none"> <li>✓ Perimeter fencing (camera/barrier)</li> <li>✓ Distributed substations</li> <li>✓ Fault detection</li> <li>✓ Mobile transformers</li> <li>✓ Emergency stocking plan</li> <li>✓ Diverse supplies</li> <li>✓ Regular drills</li> <li>✓ Legal actions (thief/vandalism)</li> </ul> | [52], [57]–[59]<br>[21], [24], [55], [56]<br>[28], [60], [61]                         |
| Network provider (ISP)           | Equipment failure | <ul style="list-style-type: none"> <li>► Router/modems</li> <li>► Network cables</li> <li>► Base station/towers</li> <li>► SCADA center</li> </ul>                                                                                                     | High     | Low        | <ul style="list-style-type: none"> <li>✓ Backup routing network</li> <li>✓ Equipment stockpile</li> <li>✓ Distributed towers</li> <li>✓ Diverse supplies</li> </ul>                                                                                                                                                     | [59], [62], [63]                                                                      |
|                                  | Physical security | <ul style="list-style-type: none"> <li>► Physical tampering</li> <li>► Link sabotage</li> <li>► Supply chain</li> </ul>                                                                                                                                | High     | Medium     | <ul style="list-style-type: none"> <li>✓ Blockchain, HSM</li> <li>✓ Legal actions (Copper thief)</li> </ul>                                                                                                                                                                                                             | [52], [64], [65]                                                                      |
|                                  | Cybersecurity     | <ul style="list-style-type: none"> <li>► Denial of services</li> <li>► Ransomware/malware</li> <li>► False data injection</li> <li>► Social engineering</li> <li>► AI-aided attacks</li> <li>► Indirect attacks</li> </ul>                             | High     | High       | <ul style="list-style-type: none"> <li>✓ Network slicing</li> <li>✓ Trusted/VLAN networks</li> <li>✓ Encryption/authentication</li> <li>✓ Endpoint/IDS/IPS</li> <li>✓ DMZ/Firewalls/MTD</li> <li>✓ Blockchain/ledgers</li> <li>✓ Behavioral analytics</li> <li>✓ Secure service edge</li> </ul>                         | [7], [66]–[68]<br>[8], [69], [70]<br>[21], [71]–[73]<br>[60], [74]–[76]<br>[77], [78] |
| Consumer                         | Equipment failure | Smart metering                                                                                                                                                                                                                                         | Low      | Low        | Equipment stockpile                                                                                                                                                                                                                                                                                                     | [28], [56], [79]                                                                      |
|                                  | Physical security | <ul style="list-style-type: none"> <li>► Metering tampering</li> <li>► Energy thief</li> <li>► Data exposure</li> <li>► Malware/DoS (industrial zone)</li> </ul>                                                                                       | Medium   | High       | <ul style="list-style-type: none"> <li>✓ Blockchain, HSM</li> <li>✓ Abnormal detection</li> </ul>                                                                                                                                                                                                                       | [80]–[82]<br>[83]                                                                     |
|                                  | Cybersecurity     | <ul style="list-style-type: none"> <li>► Social engineering</li> <li>► False data injection</li> </ul>                                                                                                                                                 | Medium   | Medium     | <ul style="list-style-type: none"> <li>✓ Blockchain/ledger</li> <li>✓ Behavioral analytics</li> <li>✓ Encryption/authentication</li> </ul>                                                                                                                                                                              | [41], [84]–[86]<br>[40], [87], [88]                                                   |
| Joint stakeholders (ISP/DSO/TSO) | Equipment failure | Communication links                                                                                                                                                                                                                                    | High     | High       | Compatible standards                                                                                                                                                                                                                                                                                                    | [56], [60]                                                                            |
|                                  | Physical security | <ul style="list-style-type: none"> <li>► Cable thief</li> <li>► Monitoring fragmentation</li> <li>► Insider attacks</li> <li>► Hijacked hardware</li> <li>► Malicious supply chain</li> </ul>                                                          | Medium   | High       | <ul style="list-style-type: none"> <li>✓ Trusted supply chain</li> <li>✓ Power switch plan</li> <li>✓ Intrusion detection</li> <li>✓ Joint warning</li> </ul>                                                                                                                                                           | [70], [75], [89]<br>[83]                                                              |
|                                  | Cybersecurity     | <ul style="list-style-type: none"> <li>► DDoS attacks</li> <li>► False data injection</li> <li>► Impersonate stakeholder</li> <li>► Ransomware</li> </ul>                                                                                              | Medium   | Medium     | <ul style="list-style-type: none"> <li>✓ Firewall/IDS/IPS</li> <li>✓ Joint incident response</li> <li>✓ Unified/SSO authentication</li> <li>✓ Mutual trust protocols</li> </ul>                                                                                                                                         | [15], [90], [91]<br>[92]–[94]                                                         |

Disconnecting the data stream or collecting inaccurate data might result in incorrect decisions on the activation or deactivation of individual electricity lines. Further, an attacker might potentially manipulate the power distribution by redirecting the power supply or disabling certain power lines, resulting in widespread power outages affecting several residences. The interdependence of communication networks and the power grid then requires many protection layers (e.g., a combination of firewall, IDS, VLAN, and advanced authentication) and optimized power distribution networks [95]. If several efforts to receive data fail, the power grid may activate its backup network communication systems. These communications (e.g., VPN, leased lines) are theoretically isolated from the Internet

and used for emergencies or authorized access privileges. The attacks and defense for the communication network provider are detailed in Section IV.

For consumers/end-users (the third target in Table I), major security concerns are tampering attacks on smart metering devices and false data in peer-to-peer energy trading models. The meters, which monitor and report electricity usage, can be vulnerable to physical and cyber attacks. Tampering meters can involve physical alterations to the device to manipulate consumption data, leading to inaccurate billing and potential financial losses for consumers [80]. Additionally, smart two-way communication meters can be compromised through firmware updates that inject false data, leading to spoofed

readings that can either inflate or deflate actual energy usage [75]. This false data injection can mislead utility companies and disrupt billing accuracy, posing a significant threat to the integrity of energy consumption data [69], [96]. In the context of peer-to-peer (P2P) energy trading, where consumers trade excess energy directly with others, spoofing and false data injection can have even more severe implications. Malicious actors can falsify transaction data to manipulate energy prices or falsely report energy trades that never occurred [55]. This can lead to financial fraud, undermine trust in the P2P trading system, and destabilize the energy market. Addressing these security concerns is essential for maintaining the reliability and trustworthiness of smart grids, particularly as they become more decentralized. The attacks and defense for the consumers are detailed in Section V.

#### D. Specific security threats in novel SG2 networks

In SG2, the rise of novel decentralized energy markets, private medium-small renewable grid operators, and peer-to-peer energy trading opens up a joint stakeholder model (illustrated in Figure 5), where Internet Service Providers (ISPs), Distribution System Operators (DSOs), and Transmission System Operators (TSOs) cooperate to operate the grid and distribute energy to end users. However, new attack surfaces, such as price manipulation and fake energy transfers in peer-to-peer energy exchanges, are predicted to grow quickly. For example, the secondary control functions of microgrids and small operators via communications increase the complexity of network management [60]. When SG2 incorporates large-scale EV and various charging vendor infrastructures, and private grid operators (particularly from renewable power plants, households, and small providers), fragmentation of grid status monitoring systems or inconsistency of joint security infrastructure among diverse stakeholders open the door for **insiders** and **supply chain attacks**. Inconsistent security protocols among these entities make it easier for attackers to perform Man-in-the-Middle (MitM) attacks and fake data injection (FDI) [70], [75], [89], leading to incorrect grid balancing or energy billing errors. Attackers exploit the fact that ISPs, DSOs, and TSOs may have separate monitoring and management systems that don't fully communicate with each other. This fragmentation creates blind spots, where an attack on one part of the grid or communication network may go unnoticed by the others, leading to undetected breaches and delayed mitigation actions. A **lack of coordination in incident response** can delay action during **Distributed Denial-of-Service (DDoS) attacks** [66] targeting ISPs, which can disrupt real-time data exchange between distributed energy resources (DERs) and grid operators, leading to grid imbalances [97]. Attackers can exploit this delay to intensify distributed attacks, escalate damage, or prolong service disruptions, knowing that each party might expect the other to handle the response.

Lack of cooperation between ISPs, DSOs, and TSOs can allow attacks to propagate from the communication network (e.g., **malware on ISP-controlled systems**) to the grid's operational technology (OT) environment, such as SCADA systems, leading to power disruptions [51], [86]. The disjointed approach to cybersecurity between ISPs and grid

entities also opens up risks for **ransomware attacks**, data manipulation, and insider threats. If there is no unified strategy for incident response, ransomware can quickly spread, locking down critical components of both the communication and power grid systems. Further, without unified authentication schemes and mutual trust protocol standards [82], attackers can gain unauthorized access to critical grid systems by exploiting the weakest link in the network (e.g., private grid operators or small energy distributors with limited security defense capability). In the absence of cooperation, an **insider** from one party (e.g., an ISP or ISP tenants) could bypass inadequate security measures of another party (e.g., DSO) to carry out attacks. SG2's significant expansion in IoT devices to provide real-time data for smart management, such as smart meters, home automation systems, and linked appliances, may introduce new vulnerabilities of lightweight security protocols and authorization [20]. Also, due to the growing complexity of supply chain management, malware or **hijacked hardware components** can be inserted into grids through different stakeholders with a lack of strict plugin-in validation.

Mitigating the security threats in cooperative grid operations requires a comprehensive solution. The straightforward solution is to deploy **intrusion detection** and **endpoint security systems** across both communication networks and operational technology environments [15], [90]. Further, automated responses to attacks (e.g., **isolating compromised network segments through network slice** of the grid, **rerouting power lines** [91]) can shorten the response time and help maintain grid stability. Self-healing solutions like **automatic local switching plans** [92] to switch fault operations and isolate failures within the shortest time interval can prevent wider disruptions. However, given the complexity of multiple ISPs/TSOs/DSOs, particularly the integration of renewable energy sources (RESs) in power distribution systems (PDSs), there should have **joint incident response** plans through API or open interfaces, with clearly defined roles and responsibilities. The restoration plan can be based on non-convex non-linear stochastic optimization formulation with joint probabilistic constraints [93]. The related parties should share **playbooks for distributed coordinated attacks**, periodic drills, and communication strategies during an incident.

Implementing platforms for **real-time data sharing and secured protocols** between DSOs, TSOs, and ISPs can improve situational awareness. The secured protocols (e.g., TLS 1.3-based and quantum-resistant-based [98]) are essential to provide synchronized views of grid operations, cyber activities, and network health, enabling faster detection and better-informed decision-making. However, these solutions may come up costly and require a trust agreement among stakeholders. At this point, **blockchain-based or Zero-Trust verification platforms** can be the solutions [94] to secure communications and maintain the relative trust. For example, by providing a tamper-proof ledger, blockchain can help ensure the integrity of grid-related communications and improve coordination during incident response. Further, by **decentralizing energy generation and integrating more DERs** (like solar panels, batteries, and microgrids), TSOs/DSOs can improve their ability to react to vandalism, drone attacks,



or natural disasters. In this model, DERs or microgrids can maintain power at a local level while the larger grid is being stabilized. Finally, **unified authentication** frameworks (e.g., single sign-on) using physical uncloneable functions for cooperative microgrids [99] can be a good idea to mitigate the unbalanced security capability of each party.

#### *E. Discrepancy of security standards used in the grid sector and security threats of remote control models in SG2*

Securing the power grid is a global concern, and various countries have established security standards and implementations to safeguard their energy infrastructures. Unlike civil applications, security standards for SG2 are based on Industrial Control Systems (ICS) and SCADA standards (e.g., IEC 62443-3-2 and NIST SP 800-82), which are well-designed for large-scale and critical infrastructure. Theoretically, the SCADA for the integrated security operation center or in energy management systems is isolated by industrial Demilitarized Zone (DMZ) and protected with many protection layers of firewalls, anomaly detection, and Intrusion Prevention System (IPS) platforms from remote access. Meanwhile, the WAN and Local Area Network (LAN) in substations are bounded by security perimeter solutions, e.g., IPS, Industrial Deep Packet Inspection (DPI). For multiple networking technologies (e.g., 5G, Zigbee, LoRaWAN), the protection is implemented from the devices and gateways to the core networks (WAN).

Each nation also uses the ICS security architecture in smart grids differently. For example, the US North American Electric Reliability Corporation (NERC) implements critical infrastructure protection requirements via its CIP framework [28]. This framework covers physical and cybersecurity security for power generating, transmission, and distribution systems. To maintain grid dependability and resilience, NERC CIP guidelines include strong access controls, incident response plans, and frequent security evaluations. The European Network for Cyber Security (ENCS) collaborates with energy companies to establish cybersecurity requirements tailored to the energy sector [29]. Japan prioritizes energy security but follows local standards. China's National Energy Administration (NEA) [113] has outlined their own measures for energy network security, addressing aspects like data protection, risk assessment, and response plans. The Central Electricity Regulatory Commission (CERC) in India has introduced their own guidelines for cybersecurity in the power sector [112].

Generally, the countries have developed their comprehensive security standards and implementations to protect their power grids. Most standards and guidelines are aligned with international best practices (e.g., ISO/IEC 27001). However, given the national security, many implementations are adapting to their unique energy landscapes to follow local laws and create efficient energy ecosystems (appropriate to the country's available resource capacity and supply chains). Due to many reasons (sloppy implementations, employee mistakes, poor security design), the power grids in many countries may not be entirely equipped with the best protection mechanisms as expected in the security standards [78]. This is why the attackers may successfully launch an attack against the power grids, even with many well-known attacks listed in this paper.

Further, the growing centralization and remote control capabilities of SG2, especially for large consumers, offer substantial grid security vulnerabilities. The foreign-made devices have raised concerns about potential vulnerabilities that could allow unauthorized access and control. Imagine that installation technicians can access websites that control large numbers of inverters, creating a scenario where the grid could be destabilized with a single command. This underscores a broader problem of security and control in grid-connected devices, regardless of their origin. To prevent the one-click control attacks or dependence of a single vendor, there are several defense approaches: (1) network slicing, VLAN, and firewalls isolate communication networks; (2) strict authentication and remote access permission with secure protocol/information security standards and strong oversight of a balance responsible party (BRP); (3) decentralizing control and communication network systems, (4) diverse supply chains through trusted international partners. Finally, we believe, to maintain the good of all these technical issues, human resource training and regular stress tests are also important too. However, all these protective mechanisms may be too expensive for certain stakeholders (power grid providers, communication network providers, consumers). To mitigate the damage, the approaches (1) and (2) can be affordable to implement. The following sections detail the security attacks and threats in general that can impact the platforms in many countries. Based on these threat examples and lessons learned, we also highlight the importance of collaboration among stakeholders and local laws to enhance SG2's security.

### III. SECURITY THREATS AND PROTECTION MODELS FOR POWER GRID PROVIDERS/OPERATORS

As stated earlier and summarized in Table I, major security concerns for power grid operators come from hardware failures and physical security. Since the grid infrastructure plays a critical role to nation energy security, equipment used in power plants is often subject to strict maintenance and quality management cycles. However, given the complexity of the smart grid operations and aging devices, the grid operators are vulnerable to many security matters and also the target of many adversaries [18], [70]. This section summarizes typical security threats against the power providers. Security protection mechanisms are then presented. Finally, we summarize the remaining challenges of protection models.

#### *A. Typical security concerns against power grid providers*

Security threats to electricity suppliers come from both inside and outside. The **inside security concerns** refer to problems in the internal grid control networks or broken devices. For example, disgruntled employees or those with malicious intent can sabotage operations or leak sensitive information. Human error is another significant internal threat, with mistakes by employees potentially leading to operational disruptions or security breaches [8]. **Outside factors** might include unforeseen events (natural disasters), physical attacks (drones, terrorism, vandalism), and cyber attacks (malware/ransomware) by adversaries. Further, failures in software, hardware, and older equipment cause less resistance to attacks.

TABLE II  
SUMMARY OF SEVERAL SECURITY STANDARDS USED IN THE POWER GRID SYSTEMS IN SOME COUNTRIES.

| #  | Security standard                      | Features                                                                                                         | Countries   | Application cover                                         | Latest version | Type       | Issuer*       | Study |
|----|----------------------------------------|------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------------------------------|----------------|------------|---------------|-------|
| 1  | NISTIR 7628                            | ►Describe security controls, risk management, and privacy                                                        | ✓ US        | All components of Smart Grid                              | 2014           | Guidelines | NIST          | [28]  |
| 2  | NERC CIP                               | ►Describe critical infrastructure protection, incident response, and recovery planning                           | ✓ US        | Secure industrial critical Systems                        | 2021           | Guidelines | NERC          | [28]  |
| 3  | IEEE C47.230                           | ►Standard security requirements for substation automation, protection, and control systems)                      | ✓ Worldwide | Substation & ICS                                          | 2018           | Standard   | IEEE          | [100] |
| 4  | FIPS 140-3                             | ►Describe standards for cryptographic modules                                                                    | ✓ US        | Hardware modules                                          | 2022           | Standard   | NIST          | [28]  |
| 5  | IEC 61850                              | ►Describe communication protocols & data models for substation automation systems (including for energy trading) | ✓ US        | Grid communications                                       | 2016           | Standard   | IEC           | [101] |
| 6  | ISA/IEC 62443                          | ►Describe network segmentation, access control, and security monitoring                                          | ✓ Worldwide | Industrial automation & control systems (+energy storage) | 2021           | Standard   | ISA/IEC       | [102] |
| 7  | IEC 62351                              | ►Describe securing communication protocols used in power systems automation                                      | ✓ Worldwide | Network protocols of power systems                        | 2023           | Standard   | IEC           | [27]  |
| 8  | ENISA                                  | ►Describe emerging threats, risk assessment methodologies, best practices for securing smart grid deployments    | ✓ EU        | All components                                            | 2019           | Guidelines | EU            | [29]  |
| 9  | ISO/IEC 27001                          | ►Describe information security management risks (including for energy trading)                                   | ✓ Worldwide | Information Security                                      | 2022           | Standard   | ISO/IEC       | [103] |
| 10 | ISO/IEC 15408                          | ►Assess the security of smart grid components and systems                                                        | ✓ Worldwide | Product security                                          | 2022           | Standard   | ISO/IEC       | [104] |
| 11 | ISO 15118                              | ►Describe security standards for Vehicle-Grid Communications                                                     | ✓ Worldwide | Product security                                          | 2022           | Standard   | ISO           | [105] |
| 12 | IEEE 1686                              | ►Describe security standards for Intelligent electronic devices                                                  | ✓ Worldwide | Substation security                                       | 2023           | Standard   | IEEE          | [106] |
| 13 | IEEE 2030                              | ►Guide for the interoperability of energy storage systems with the grid infrastructure                           | ✓ Worldwide | Energy Storage                                            | 2019           | Standard   | IEEE          | [107] |
| 14 | CEN-CENELEC-ETSI                       | ►Describe interoperability, protocols, data exchange formats                                                     | ✓ EU        | Smart Meters                                              | 2018           | Framework  | EU            | [73]  |
| 15 | Privacy Act                            | ►Address the privacy of user data                                                                                | ✓ Local Law | Regulation                                                | -              | Policy     | Local country |       |
| 16 | Smart Grid Security Guidelines         | ►General guidelines                                                                                              | ✓ China     | Custom Smart Grid                                         | 2012           | Guidelines | SGCC, CEC     | [108] |
| 17 | Smart Grid Security Guidelines         | ►General guidelines                                                                                              | ✓ Japan     | Custom Smart Grid                                         | 2019           | Guidelines | METI          | [109] |
| 18 | JISEC Framework                        | ►The Japan information security evaluation and certification, certifies the security of IT products & systems    | ✓ Japan     | Product security                                          | 2023           | Framework  | CCRA          | [110] |
| 19 | Industrial Control Security Guidelines | ►Security measures, vulnerability management, access controls, and incident response                             | ✓ Worldwide | ICS Security                                              | 2015           | Guidelines | NIST          | [111] |
| 20 | Cyber security in Power Sector         | ►Guidelines for grid security                                                                                    | ✓ India     | Custom Smart Grid                                         | 2022           | Guidelines | CERC          | [112] |

\* NIST: National Institute of Standards and Technology; NERC: North American Electric Reliability Corporation; ISO: International Organization for Standardization; ISA: International Standards on Auditing; IEC: International Electrotechnical Commission; CERC: Central Electricity Regulatory Commission; EU: European Union; METI: Ministry of Economy, Trade and Industry; CCRA: Common Criteria Recognition Arrangement; SGCC, CEC: State Grid Corporation of China, China Electricity Council.

Several security threats with specific damages are summarized as follows.

① **Time synchronization attacks:** This attack threatens the precise operation of phasor measurement units (PMUs) in grid operators by disrupting their time synchronization processes [114], [115]. Since PMUs rely on accurate time-stamping from Global Navigation Satellite Systems (GNSS) like GPS, any interference with these signals can compromise the reliability of synchrophasor measurements. The attack can cause significant disruptions in phase angle monitoring (PAM) by corrupting total vector error (TVE), leading to inaccurate phase angle calculations and erroneous power transfer dynamics. These attack types can also trigger false alarms, unwarranted trip commands, and degrade voltage stability

controllers, compromising the overall reliability and protection of the smart grid.

② **Equipment failure:** The failures can come from aging microcontrollers and electricity circuits in operation systems of power plants [23], [54]. According to Statista, between 2000 and 2023, the large states in the US (e.g., Texas, California) has an average of 251 blackouts in the 23-year period due to aging broken devices or equipment failures from thunderstorms, wind, hurricanes, or just a brownout during a heat wave. The failures cause substantial direct economic costs, e.g., insurance claims up to 3 billion USD [53]. Fig. 6 illustrates an electric distribution model for different consumer types. Accordingly, failures of extra-high voltage or primary/secondary substations can cause devastating

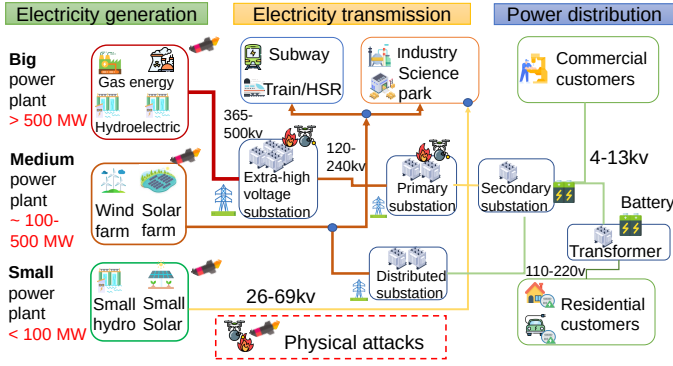


Fig. 6. The illustration of an electric distribution model to different consumer types, from public transportation, industry clients, medium-small commercial consumers, and residential users. Failures of extra-high voltage or primary/secondary substations can cause devastating consequences, i.e., large blackouts. They are also the targets of many physical attacks.

consequences, i.e., large blackouts. Besides, they also generate other substantial damages, including metro/subway passenger delays and emergency vehicle stops owing to traffic signal failures in urbanized cities. The amount, duration, location, and time of day of a blackout affect its societal effect. Furthermore, equipment failures due to aging infrastructure or technical malfunctions can compromise the reliability of the electric supply. For example, the large blackout in the northeastern United States in 2003 was partly attributed to aging infrastructure and system failures [116].

③ **Physical/substation attacks:** In this attack, attackers might tamper with equipment, disable security measures, or even manipulate operational settings, leading to widespread disruptions. For instance, in 2013, attackers cut fiber optic cables and used sniper rifles to damage transformers at a PG&E substation in Metcalf [58], California, causing significant damage up to 15 million USD worth of repairs and raising concerns about an unprecedented and sophisticated terrorism attack on an electric grid substation with military-style weapons [117]. Another emerging threat is manipulating energy storage systems, such as batteries, essential for grid stability and load balancing. Malicious actors could exploit vulnerabilities in these systems to manipulate energy flow, leading to grid instability or even overloading components [118]. For example, the 2015 cyber attack on Ukraine's power grid involved hackers gaining access to control systems and changing operational settings, leading to power outages for hundreds of thousands of people [119]. The other threat is the attack from cruise missiles or military drones that target essential power transformers (extra-high voltage or primary/secondary substations as shown in Fig. 6) to cause fire and power outages over large areas, as illustrated in Fig. 7. Another variant is to ram vehicles into substations or transmission lines, which can cause significant damage and outages. For example, an attack on two power substations left more than 40,000 people without power in Moore County, US, on December 3, 2022 [120].

④ **Supply chain attacks:** The attacker intentionally can introduce compromised or malicious hardware (remote mon-

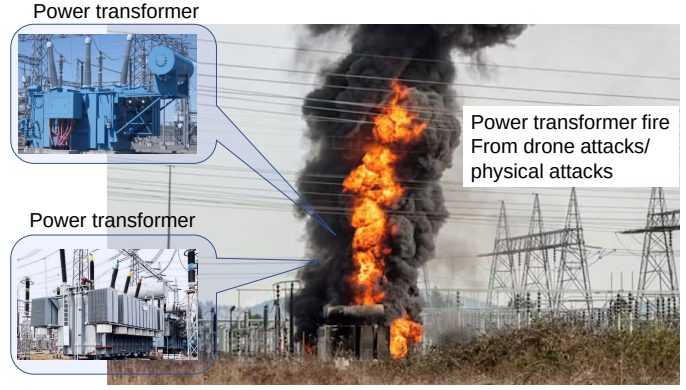


Fig. 7. The illustration of a substation transformer attack by drones/vehicles or vandalism that causes fire and electric outage.

itoring sensors [121]) into the infrastructure, which can later open a backdoor for physical intrusion or remote control [57]. This type of attack can be particularly insidious, as it may go undetected until the compromised hardware is activated. Theft of critical infrastructure equipment, such as heating monitoring sensors in power plant's cooling systems or copper cables [63], is a significant concern. We can imagine how devastating it would be for emergency infrastructures if power systems went down owing to a loss of control at the management center from stolen devices. Further, through recent trade wars, pandemic/conflict crisis, and the decarbonization goals toward 100% clean electricity, a severe shortage of key commodities, such as key minerals/materials (rare earth elements) and gas supplies (in gas-electric power plants), can create an inflationary cost environment for companies, resulting in increased costs of transformers, wire and cables, batteries, and solar panels. Also, the clean energy transition will require a large expansion of transmission and distribution infrastructure, including new and end-of-life line replacements, power transformers, high-voltage direct current systems, digital relays, and smart inverters [57]. Many of basic supplies rely on a few nations, leaving them susceptible to interruption.

In general, physical and supply chain attacks can have severe implications for grid stability, functionality, and even public safety. However, these attack types often require the presence of the attackers near the attack areas, limiting the attacker's capability if the grid infrastructure is physically well-protected (e.g., by building walls). Finally, insider threats cannot be overlooked. Human errors or individuals with authorized access to the physical infrastructure could intentionally misuse their privileges to compromise system integrity. As SG2 becomes more reliant on digital controls and interconnected devices, the risk of cyber-physical attacks grows [21]. Major security concerns in SG2 will be severe shortages of key minerals and materials in the supply chain to produce solar panels and electric vehicles or the risks of long-range cruise missiles/drones targeting critical transformers and substations, given the popularity of UAV/drones everywhere. The volatile movements in the geo-political landscape towards a new multipolar order, coupled with changes in global economic dynamics, especially concerns about trade wars, are prompting

many nations to establish a roster of reliable partners for sources of essential components in SG2 as detailed below.

### B. Specific threats for new power grid models in SG2

Unlike in SG1, the energy market in SG2 may open for thousands of grid operators, including private sector and small-scale renewable energy producers. In this distributed energy resources (DERs), solar panels and wind turbines in small producers with limited security capability are often remotely controlled and may lack robust security mechanisms [30], [33], making them vulnerable to **brute-force cyber attacks** with default credentials as in current IoT devices. Further, the growing use of IoT devices in SG2 by smaller grid operators increases the risks, where the IoT devices have inadequate ability (e.g., no enterprise firewall or intrusion prevention system) to protect themselves from being attacked. If those devices are hacked, then the extensive devices can be used to attack the core network (e.g., DDoS) or disturb the electricity flow and information exchange. Note that, smaller operators may not invest adequately in infrastructure or maintenance or even fail to fully comply with technical grid codes and operational standards (due to lack of resources, knowledge, or intentional disregard), leading to substandard equipment or outdated technologies in grid-connected systems. With many small operators involved in energy trading, attackers could target exploiting vulnerabilities of the integrity of financial transactions (e.g., through **51% attacks** or **consensus mechanism manipulation** [122], [123]) and automated energy bids. This could lead to significant financial losses for specific operators if the collusion of a portion of the network or the incorrect pricing (due to false data attacks [75] during synchronization or energy trading) occurs for a certain time. Small DSO operators, including small private sector participants, might have **disgruntled employees or contractors** who could abuse their access in energy bidding systems and grid operation to compromise the grid to initiate local cascading grid failures. In extreme cases, grid operators in a competitive market, especially those relying on intermittent renewable energy sources, may not always align their energy generation with grid demand [21] but price bidding (**selling only when prices are favorable**). Some might **intentionally withhold or oversupply energy** to manipulate market prices or gain an advantage in energy trading markets, particularly in crisis situations and peak demand periods. The time synchronization attacks [114] can be a threat in SG2. This could stress or destabilize the grid, leading to outages. In the contrary, several operators, especially those using solar and wind, may produce excess energy during periods of low demand without proper coordination with grid operators. This **overproduction** can lead to voltage rises, grid congestion, and potential damage to grid infrastructure, especially if energy storage systems are not in place to absorb the excess.

### C. Security protection models for power grid operators

There are various countermeasures to mitigate the magnitude of security threats against power grid providers. This

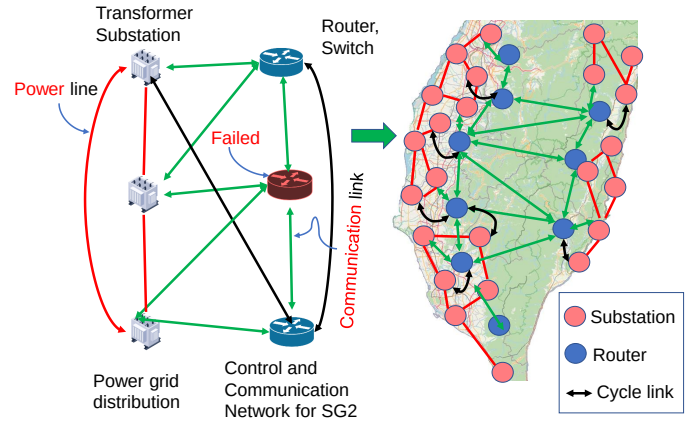


Fig. 8. The illustration of a power grid model in Taiwan and a design for connection lines of grid control systems to prevent cascading failures. Links or optimal coupling between the nearest neighbors are created to guarantee that no failure substation/router will cause the failure of the whole communication control network or a blackout of the whole grid.

subsection and the protection method column in Table I summarizes several typical models.

To mitigate equipment failures stemming from aging micro-controllers and electricity circuits, power plants can employ several key strategies, such as regular maintenance and timely replacement of outdated components [124]. Additionally, implementing redundant systems and automated failover mechanisms ensures that operations continue seamlessly in the event of equipment failure. For example, in 2013, the Tennessee Valley Authority upgraded its grid infrastructure in a way, that if one line or component fails, the system automatically reroutes power through alternate pathways, preventing outages and maintaining continuous operation [56]. The implementation also includes special circuits and fiber networks to prevent electromagnetic pulse attacks and geomagnetic disturbance [125]. Fig. 8 illustrates a study model on mitigating the minimum number of node failures to prevent cascading failures. Accordingly, the authors use graph structure to model equipment failures or offline routers and suggest a star-topology-based heuristic algorithm to find a near-optimal routing solution. Recent studies on the flow dynamics in both the grid and the failure rollover reinforced that increasing the power/communication link coupling is beneficial to the resilience against cascading failures [23], [24]. The utilization of advanced data analytics and AI models, along with datasets containing information on equipment failure rate and natural events (e.g., earthquake/heavy rain/thunderstorm risks), enables real-time monitoring and predictive maintenance [21]. This enhances the capability to detect early signs of failure and facilitate quick repairs. For internal grid operations, developing misbehavior detectors to check abnormal operational data (e.g., unscheduled power on/off key devices, suddenly increased voltage levels) can also mitigate insider attacks [55].

For physical attacks, primary solutions are detection intrusion, equipment tampering detection, video monitoring, camera analytics, and using new cut/climb/ram-resistant fences. Table I in the previous section summarizes several prospective protection solutions against physical communication links in



the grid operators. The most common idea is to equip the metering devices and substation components with perimeter fencing (camera/barrier), hardware security module (HSM), and card-based entry systems for on-site security personnel. HSM can provision cryptographic keys for critical functions such as encryption, decryption, and authentication [126] in relevant metering devices. The authentication methods help limit access to authorized individuals only and prevent physical access/tampering since any physical manipulation of the devices (e.g., firmware/operational settings modification) will be detected by HSM's tamper-evident and intrusion-resistant safeguarding capability. For transformer-targeted attacks, additional spares inventory is critical to speed emergency replacements [56]. Mobile transformer fleets and mobile switch houses with racks and portable truck-mounted GIC switchgear can offer the greatest flexibility for restoration in the case of a series of missile attacks and substation failures.

The most challenging task is to avoid substation attacks, given the diversity of attack sources (e.g., intentional sabotage by trucks and flammable substances). Multiple physical measures are required to protect this critical infrastructure. For example, besides perimeter fencing around the substation with physical barriers (concrete walls) to restrict unauthorized access, implementing access control measures such as gates, locks, card readers, and biometric systems is also an efficient method to ensure that only authorized personnel can enter the substation [54], [127]. The sensor alarms can be equipped to trigger shout alarms or notifications to security personnel once the substation is intruded. Installing robust fencing, anti-climbing devices, barriers around substations, or high-definition cameras with night vision capabilities at strategic locations around substations can deter theft and provide evidence for investigations [128]. Besides the strict regulations and increasing penalties for copper theft, educating the public about the impact of the theft activities on critical infrastructure and encouraging them to report suspicious activities can enhance community vigilance [63].

For complicated attacks (e.g., by remote bombs and cruise missiles), building backup substations/distributed microgrids or anti-drone technologies and missile defense systems can be a way to mitigate the damage of physical attacks and ensure the substation can continue operating in case of an attack or equipment failure [60]. Accordingly, instead of constructing 5-10 major power plants with great production capacity and a substantial portion of the energy balance, there should be dozens to hundreds of small power plants spread throughout the nation that can power a city if one fails. However, to implement this decentralized electric management model, the AI-based automated operation capability and distributed security management are critical. Section VII clarifies several security concerns in AI and distributed security management. The other way is to build risk-aggregated substation testing cases to assess potential security matters for future defense [61]. If a local station is flooded, the mobile transformer or backup one from the emergency stocking plan can be used.

To mitigate supply chain attacks, materials/fuel strategic reserves (gasoline, diesel, aviation fuel and propane, copper, rare earth elements) are necessary in certain conditions where

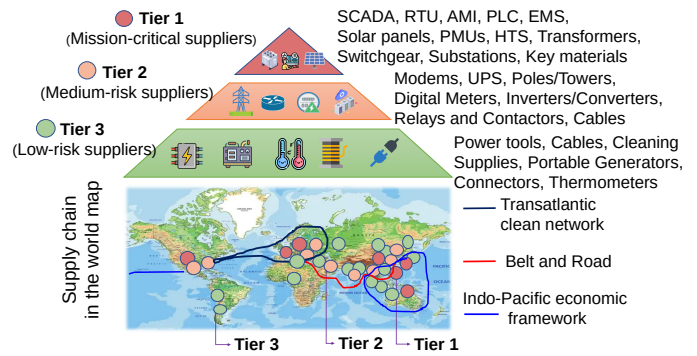


Fig. 9. The hierarchy model of suppliers by their important role in the supply chain's resilience and sustainability. Tier 1's suppliers are critical to the supply chain while tier 2 and tier 3 companies are medium-to-low risk suppliers. There are three typical supply chain model examples through trade initiatives: Transatlantic Clean Network [131] (now a part under U.S.-E.U. Trade and Technology Council supervision), Belt and Road Initiative [132], Indo-Pacific Economic Framework (IPEF) [133].

post-storm, pandemic, and conflicts may restrict the supply access [57]. Establishing a varied supply chain consisting of a range of smart grid components and services, organized in a hierarchical list of trustworthy partners, as shown in Fig. 9, may also provide a clear strategy for determining which suppliers should be given priority in order to enhance collaboration and reduce reliance on a single country. The cooperation can upgrade to build intercontinental super grids linking renewable sources, e.g., across North Africa, the Middle East, and Europe [37] or connecting China, South Korea, Taiwan, Mongolia, Russia, Japan, and India in Asian [129] as illustrated in Fig. 10. However, national security concerns, country rules, the safety of power transmission routes (e.g., underground), geopolitical and environmental sustainability [130], and expensive capital costs of ultra-high voltage (UHV) power lines (up to 756 or 1100kv) and transformers (up to 800 metric tons) limit this global super grid venture to market demand. Several countries like the US and China are considering building a better distribution system to connect renewable sources in remote areas, e.g., a West-to-East grid in the US, hydropower plants in Mongolia, and solar/wind power plants in the Gobi Desert to the coastal regions in China.

To mitigate the specific threats against the DERs model with thousands of small producers or microgrids, a multi-layer protection model should be applied. First, government agencies may require regulatory standards and periodic inspections to **ensure small operators comply with grid codes, hardware integrity, security audits, and operational standards** [28]. Further, a **real-time monitoring and demand-response system** to ensure that energy generation from private operators aligns with grid demand can be critical. The system can provide price signals and grid conditions to operators to avoid overproduction or underproduction. Any **violation or abnormal demand-response activity** or energy withholding/oversupply during peak demand can face severe penalties. **Local power switch optimization** for grid balancing in energy production from small operators [92] can be helpful, particularly during periods of high or low demand.



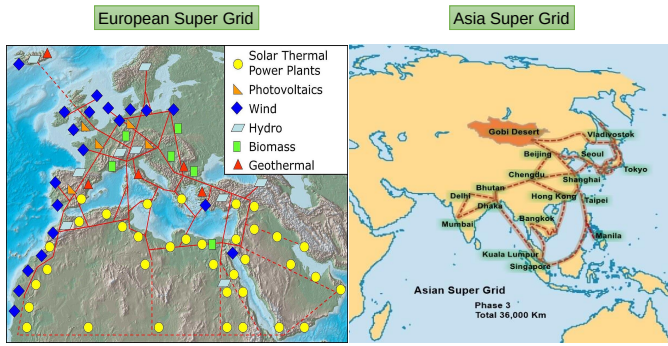


Fig. 10. Two conceptual plans for maintaining sustainable energy supplies: a super grid linking renewable sources across North Africa, the Middle East, and Europe [37]; a super grid to establish a power transmission network connecting China, South Korea, Mongolia, Russia, Japan, and India [129].

Finally, according to the NIST guidelines [28], saving electricity and building emergency backup power lines can be a high degree of priority if most electricity infrastructure is often damaged by unplanned extreme events or wars. The resiliency in this case requires well-planned information to the local communities and integration of the grid infrastructure with restoration plans in emergency situations. Further, electricity restoration for critical infrastructures (hospitals, nuclear plants) can be then prioritized. Regular national drills and employee training for countering security threats and power outage scenarios are extremely helpful in guaranteeing grid security.

#### D. Remaining challenges

There are many remaining challenges to preventing security threats against grid operators in SG2. One of the biggest challenges is to protect substations, transformers, and critical transmission lines against intentional physical attacks, such as vandalism, sabotage, or theft of critical components. Since those attacks can come from insiders, civilians, or enemies, predicting “when and where the attacks potentially occur” remains a complicated technical matter. Equipping anti-drone and sabotage activities for many substations may cause significant capital costs and operational expenditures. Second, many existing power grid infrastructure components are legacy systems with limited or outdated security features (probably up to five or ten years old). Retrofitting or securing legacy systems can be costly and complex. Third, the state-sponsored attack groups may launch cyber-physical attacks (e.g., Stuxnet [134]) that exploit vulnerabilities in programmable logic controllers of SG2 to damage SCADA and PLC systems of power plants. Fourth, many utilities in small operators face resource constraints, including budget limitations, that may hinder their ability to equip HSM for all metering devices or backup components (e.g., transformers). Fifth, supply chain vulnerabilities can introduce security risks, as compromised or counterfeit components may be integrated into the grid infrastructure, potentially enabling attacks or equipment failures. Finally, detecting and responding to physical security incidents in real time can be challenging. A delayed response can lead to significant damage or electric disruptions for many households.

#### IV. SECURITY THREATS AND PROTECTION MODELS FOR COMMUNICATION NETWORK PROVIDERS

Communication technologies, such as wired and fiber networks, help enable real-time data exchange between various components of the grid, such as power plants, substations, microgrids, and consumer devices. This facilitates enhanced monitoring and control, allowing for better management of energy flow and quicker response to issues like outages. In SG2, communication networks support the integration of decentralized renewable energy sources and demand response programs, thereby increasing grid reliability and resilience. However, the increasing dependence of SG2 on communication technologies poses severe security threats to the reliability and integrity of energy management and distribution systems. Generally, security attacks on communication network providers target vulnerabilities of transmission and security protocols (e.g., wireless, TLS, TCP/IP protocol). This section summarizes typical security threats against the network providers in SG1 vs SG2 and protection mechanisms.

##### A. Typical security attacks against network providers

Security threats to network providers mostly come from many vulnerabilities of Internet technologies. Given the diverse range of security attacks on the Internet, we summarize some of the most severe threats that have been recorded via specific damage events and discuss several hints on SG2.

① **Denial-of-service (DoS):** This attack can overwhelm the communication network infrastructure with a massive volume of traffic, rendering it unavailable for legitimate users and hindering critical communications. Various well-known DoS attacks at the network layer can be low-rate HTTP requests or DNS/NTP amplification. DoS attacks can disrupt the grid’s operation by causing power outages or blackouts or disrupting communication between smart grid devices. For example, in March 2019, a DDoS attack hit electrical system operations in Kern County, California, and Converse County, Wyoming, USA and temporarily disrupting the online services [66]. DDoS attacks can lead to a chain breakdown of indirect problems such as inaccurate readings or preventing critical information exchange and real-time grid management [8], [21].

② **Malware/Ransomware attacks:** Malware is one of the most popular source threats against smart grid infrastructure. Their target is to disrupt or disable grid operations and steal sensitive data. For example, in June 2018, Ingerop company in France was hacked by malware and lost 65 GB of data with 11,000 files from planned nuclear-waste dump projects, details of the Fessenheim nuclear power plant, and personal details of more than a thousand Ingerop employees [67]. Malware can be installed on smart grid devices in various ways, such as phishing emails, infected websites, or USB drives. Ransomware is the other variant of malware. Ransomware encrypts the victim’s data and demands a ransom payment in order to decrypt it. The Colonial Pipeline ransomware attack in May 2021 was the largest US energy infrastructure attack [7]. The hack gang entered Colonial Pipeline Co.’s network using a stolen VPN password of an employee from a previous data theft. The hackers took 100 GB of data in two hours and

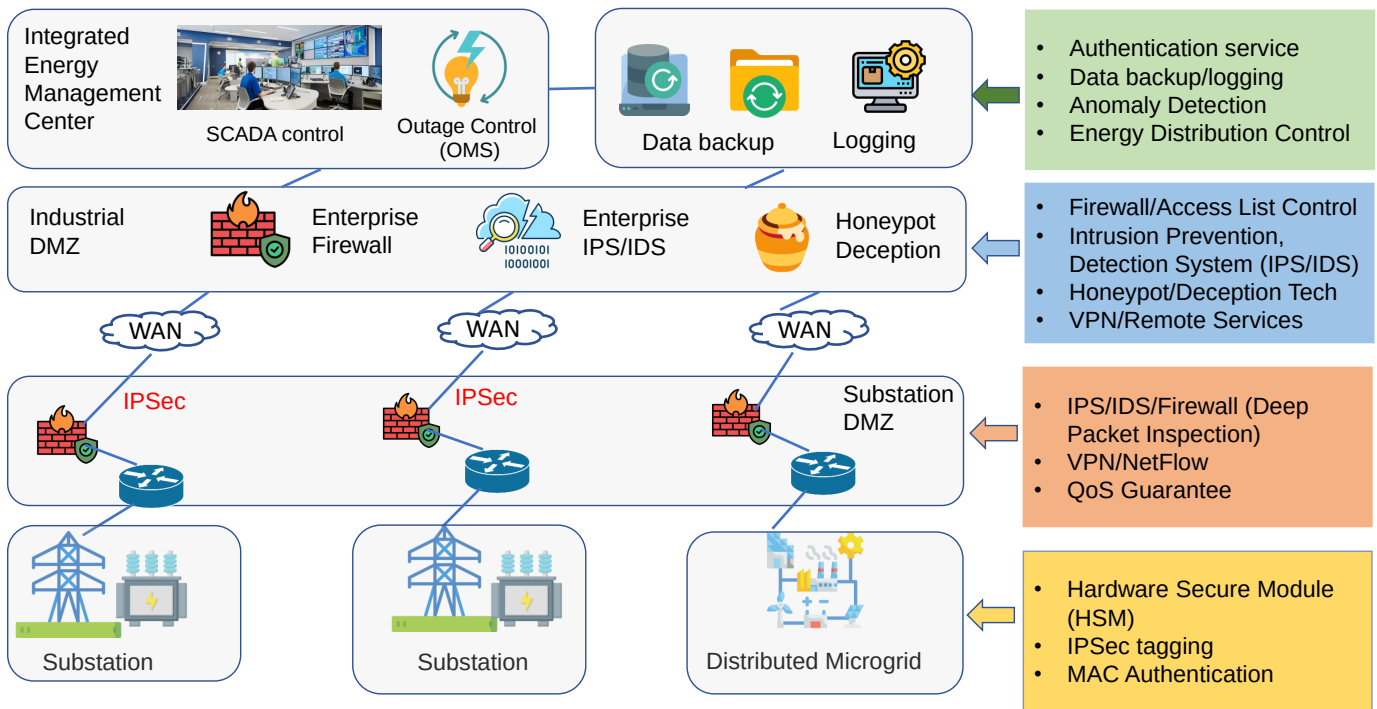


Fig. 11. An example of security implementation for smart grid where networks are connected and protected by industrial DMZ, enterprise firewalls, and IPS/IDS platforms.

threatened to release it. Colonial paid a 75-bitcoin ransom (4.4 million USD [7]) and restored operations five days later. In a separate incident, Ryuk ransomware encrypted Volue's front-end platforms, impacting 2,000 customers in 44 countries [7]. Ransomware attacked City Power, a major electricity supplier in Johannesburg, South Africa, leaving more than a quarter of a million people without power [68]. Ransomware is also used in many recent attacks, e.g., Enel Group [59].

③ **False/spoofing data injection attacks:** This attack introduces fraudulent information into the network, such as false packets/untrusted MAC sensor nodes, further undermining data integrity and leading to erroneous energy distribution or energy theft [69], [70]. False/spoofing data may include inaccurate measurements, improper timestamps, or incorrect sender IDs [135]. Further, the attacker can eavesdrop on sensitive data transmissions and manipulate control signals, potentially leading to incorrect grid operations, energy theft, or even physical damage [136]. The attacker can advance the attacks by impersonating the metering devices' IP addresses/MAC identities to pretend to be legitimate nodes and compromise the authenticity of the communication, enabling unauthorized access and data breaches [18].

④ **Communication link sabotage attacks:** Attackers destroy the communication cables to disrupt the connectivity between smart meters and the grid management centers. On the other hand, copper theft in communication cables is a significant concern. The copper theft cost Lumen, a global communications service provider, up to 500,000 USD to fix business landlines [62]. According to S&P Global Market Intelligence, copper demand will triple by 2035, leading to supply shortages in 2025 [62]. Copper phone lines and wire

thefts have skyrocketed as criminals may sell their hauls for cash. The organized theft rings may target mobile towers and landlines.

⑤ **Indirect attacks:** Besides the above direct attacks (which damage the infrastructure), the hacker can launch the indirect attacks, e.g., manipulate the network routing paths of data, causing data to be directed to unauthorized destinations, potentially exposing sensitive information or causing disruption. Generally, the attack goal is often used to inject malware and ransomware into the grid's internal networks. This attack type has many variants, e.g., routing information spoofing, alteration or replay, blackhole and selective forwarding attacks, sinkhole attacks, and Sybil attacks [8], [18]. Another variant is social engineering/phishing which exploits human psychology to trick users into taking actions that are harmful to the grid. For example, an attacker might pose as a customer service representative from an electric distributor and ask users to provide their personal information to update billing information; otherwise, the electricity will be cut off. The hacker can bait users to use discount services and download fake software [84], [85]. Finally, the remote terminal units in decentralized energy resources (DER) controllers or SICAM substation automation systems can be the targets of stack buffer overflow or firmware vulnerability exploitation [18].

Unlike the attacks against the power grid operators, the attacker can remotely launch cyberattacks against the network providers. Besides, individuals in trusted private networks for smart grids could misuse their network access privileges to compromise network security in the other network segments. In SG2, the communication technologies are expected to expand for connecting multi-energy generation grids [23]. Major

security concerns against this network model in SG2 will be still the malware/ransomware and blackmail attacks. However, as cyberwarfare becomes integrated into national security strategies and cyberattacks have become a lucrative money-making industry, these attacks are expected to be considerably intensified and, in some instances, supported by governments. Further, networking technologies for specified SG2 generation, such as renewable energy sources and distributed energy resources, are the new targets.

### B. Specific threats for new communication models in SG2

In SG2, thousand of grid operators and millions of consumers may connect and rely on the connectivity provided by network operators. The complexity of maintaining the connectivity for such a sophisticated grid exponentially increases and building a leased line or dedicated network infrastructure for individual grids can be a challenge. Further, the real-time data exchange and high communication traffic could potentially lead to network congestion, latency, or insufficient bandwidth, affecting the timely management of grid stability. The other threat is poorly synchronized communication among thousands of operators that could lead to mismatches in grid data, causing imbalances in energy distribution. In such complicated and diverse communications, five attacked identified in the previous subsection (e.g., DoS, communication link sabotage [62], false data injection [69], [70], ransomware [71]) are expected to be more intensive. Potential other attacks can be routing attacks (blackhole) and Sybil attacks [8], [18] (fake multiple grid operators from a single identity to create false coordination). The attacks could result in imbalanced grid operations or unreliable energy trading. Finally, the use of lightweight communication protocols for IoT-enabled SG2 like MQTT [137], [138] may introduce security gaps (without using TLS protocols) if they are not properly secured from data tampering and transaction interception [137].

### C. Protection technologies for network providers in SG2

Securing the communication networks requires a comprehensive approach of multiple protection models at different network layers, as illustrated in Fig. 11. Ideally, for each essential component in SG2, the protection models involve a DMZ (e.g., substation/industrial DMZ). The DMZ is integrated with many advanced protection techniques, e.g., IPS/IDS, authentication, honeypot, at ingress/egress routers/gateways. Table III summarizes several key protection models (the fourth column) for five security attacks above. An approach, e.g., IDS/IPS, can be used for prevent various threats.

① **Network slicing:** In this paradigm, SG2 communications are technically separated from civil networks [72]. According to this approach, any possible security interference/breach that happens in one slice will not affect the others. Ideally, SG2 communications, particularly SCADA and substation networks, are isolated from the Internet through private communication links or VPN connections. Fig. 12 illustrates a 5G network slicing concept for SG2 where different grid services are expected to be isolated based on their service priority and response time requirement. Accordingly, end-to-end network

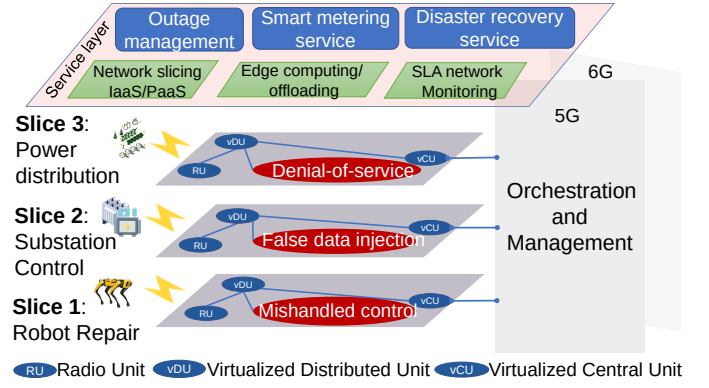


Fig. 12. Illustration of a 5G network slicing concept for SG2 [71] where each service, depending on their security requirements, can be dedicated in a slice to protect the communication secrecy. For example, communications between substations and the remote center or between power distribution control sensors and the center.

slicing for SG2 [71] is a promising technique. It is vital in network slicing to define isolation characteristics for SG2 individually, specify KPI criteria, and enforce them. A high-volume DoS attack, on the other hand, may make this exceedingly difficult to contain. Defining isolation properties for each slice, as well as defining and enforcing KPI standards in smart grid environments, is crucial. A network slice manager, such as Network Slice Management Function (NSMF) [72], is expected to be in charge of abstract virtual network functions.

② **Virtual local area networks (VLAN):** Similar to network slicing, however, instead of applying for wide area communication networks, VLANs help isolate different types of traffic in a single physical communication network infrastructure. For instance, in a SG2, VLAN 10, VLAN 20, VLAN 30, and VLAN 40 are assigned for the communication networks of power generation plants, substation control systems, administrative offices, and backup/recovery systems, respectively. The isolated communication networks logically prevent unauthorized access against unassigned communication network areas, limiting the impact of potential cyberattacks [74]. For example, control teams in power distribution and staff for equipment/communication link maintenance can be placed on different VLANs, ensuring that only authorized personnel can access critical operational systems. However, while VLANs can offer some security benefits by isolating sensitive data and limiting the spread of broadcast domains, the complexity of configuring large-scale VLANs can bring up new threats. For example, the assigned network segments may access each other if there is a misconfiguration [74]. In this way, relying solely on VLANs for security can be misleading. For robust network security, VLANs should be used in conjunction with other security measures such as authentication, IDS/IPS, and strong access control policies.

③ **Secure control and communication protocols:** For network access control, stronger authentication between energy distributors and the consumers is required to prevent spoofing attacks and man-in-the-middle attacks. For example, 5G Authentication and Key Agreement (AKA) and future 6G AKA [78] will be core services to unify the authentication for

mobile users and IoT metering devices, although the metering subscriber identity (i.e., National Meter Identifier) must likely synchronize with mobile identifiers (eSIM). In this way, users can use one unified identity with a single point of access to manage all electricity service fees, car charging, battery swap, or P2P electricity trading. Users benefit from this single sign-on (SSO) capability, allowing them to access multiple systems with one set of credentials, thereby improving convenience and reducing password fatigue. Unified authentication generally facilitates interoperability among different energy providers, enabling the integration of multi-energy generation grids and optimizing cost in peak time. This interoperability is crucial as the smart grid expands, enabling efficient load sharing of new energy sources. This unified model also helps simplify the administration of user credentials and the authentication protocol upgrading process for all devices.

However, a private Public-key Infrastructure (PKI) infrastructure for SG2 authentication [73] can still be necessary to utilize secure communications for special services (defense industry). Further, cryptographic algorithms in secure communication protocols such as Internet Protocol Security (IPsec) and Transport Layer Security (TLS) must be adopted to prevent unauthorized interceptors from accessing communications. Toward the age of quantum computing, adopting quantum-safe cryptographic schemes (e.g., extending the key length of AES-128 to AES-256/AES-512) [139], quantum-resistant algorithms (e.g., NTRU/AES) or Quantum Key Distribution schemes [140], [141] can be a benefit for communications between SCADA and substations. End-to-end encryption and decreasing the cost of protection (e.g., energy consumption, deployment cost) are critical challenges for next-generation communication security, which will serve as the foundation for SG2 communications. Because metering data transfer between substations might be enormous, adopting end-to-end encryption may be prohibitively expensive. Currently, 3GPP and standards organizations advocate the optional use of security measures based on their capacity to fulfill necessary services. If SG2 needs mandatory end-to-end encryption, it is unclear how to implement this requirement over the multiple communication subnetworks, given their co-existence of legacy and modern technologies.

**④ Advanced endpoint security and intrusion detection systems:** As shown in Fig. 11, endpoints play a guard role in protecting communication gateways and the borders of the core network infrastructure of the smart grid (i.e., substation DMZs). To guard against external threats, SG2 core networks have historically required DMZ and control nodes. To prohibit unwanted traffic from core network parts, a security gateway may examine bi-directional traffic against the operation rules. The types of security gateways include IDS, service-oriented architecture API protection, antivirus programs, VPN, and so on. In a traditional smart grid, as illustrated in Fig. 11, security gateways located at substation DMZ and Industrial DMS are responsible for inspecting all traffic between Internet and Industrial Control System (ICS) [76], [77]. Such gateways will need to enhance their capacity in SG2 significantly. Many experts anticipate that the enhanced capabilities for the next-generation endpoint security will include (1) in-line deep

packet inspection (DPI), (2) encrypted traffic inspection, (3) next-generation firewall, (4) next-generation intrusion detection/prevention, and (5) unified identification [78]. Current AI-powered engines will need significant improvements in detection capabilities, such as higher ability for online traffic training and less impact from imbalanced datasets, as well as resilience for defending varied communication protocols. As suggested by artificial general intelligence, one viable option is to increase the generative learning capabilities of deep learning models [142], [143].

**⑤ Deep packet inspection and next-generation firewall:** Along with Endpoint Security Gateways, Deep Packet Inspection (DPI) and Firewall are vital components to protect substation DMZs, as illustrated in Fig. 11. DPI technologies provide advanced packet analysis capability (e.g., analysis of packet headers and protocol fields of source data, sending frequency, network protocols, malicious payload patterns) for distributed microgrids and substations [21]. This capability is particularly useful, given the diversity of the management models in SG2 microgrids. On the other hand, stateful firewalls offer additional features like the smart metering application or buffer overflow attack detection, fragmented worm or intrusion detection, and access list control to prevent vulnerability penetration or keep unauthorized users out, particularly in SG2 microgrids [60], [75]. The primary challenge is that many next-generation security designs are still only concepts. For example, many AI-driven DPI and firewall technologies also struggle with privacy preservation in training/learning, scalable online learning, false alarms, and vulnerability to many adversarial attacks.

**⑥ Physical attack defense solutions:** Unlike substation protection, affordable and effective protection (e.g., camera, fencing) against copper theft and communication cable sabotage attacks is a challenge for communication network providers since communication cables are often placed in many residence areas. One of the most effective methods is to educate the public about the impact of copper theft on critical infrastructure and encouraging them to report suspicious activities can enhance community vigilance [63]. The other is to build backup lines to prevent cascading failures, as introduced in Section III.B and Fig. 8. In the worst case, several companies consider pursuing legal action against thefts and scrap metal businesses that knowingly purchase stolen copper. For example, Bell Canada has taken legal action against individuals stealing copper wire from their communication networks. The thefts typically involve cutting down telecommunication cables from poles and selling the copper for cash, with each incident taking 10 to 12 hours to repair and significantly impacting service reliability [65].

**⑦ Emerging defense solutions:** Software-defined Networking is deploying widely over the Internet. By separating the packet forwarding (data plane) and routing (control plane) operations, SDWAN attempts to increase network control performance and intelligence. In SG2, SDWAN is expected to be utilized for power utility distribution automation [145], [146]. DoS/DDoS attacks and insider adversaries are the most severe risks to SDN/SD-WAN. The most common mitigation strategies are to exploit AI models to classify abnormal traffic



TABLE III  
COMPARISON OF SECURITY THREATS AND PROTECTION MODELS FOR THE COMMUNICATION NETWORK PROVIDER IN SG1 VS SG2

| Security threats                           | Severity | Likelihood | Protection model                     |                                                | Efficiency | Deployment cost | Open issues                                                                          | Reference    |
|--------------------------------------------|----------|------------|--------------------------------------|------------------------------------------------|------------|-----------------|--------------------------------------------------------------------------------------|--------------|
|                                            |          |            | SG1                                  | SG2                                            |            |                 |                                                                                      |              |
| Denial of service (HTTP, NTP, DNS)         | High     | High       | IDS/IPS/MTD                          | Intelligent IDS/IPS/MTD (Superintelligence)    | High       | Low             | - High complexity<br>- Deal with encrypted data<br>- Comprehensive dataset for SG2   | [144]        |
|                                            |          |            |                                      | Distributed microgrid                          | High       | Medium          | - Complicated management                                                             | [78]         |
| Malware/ransomware                         | High     | High       | IDS/IPS                              | Intelligent IDS/IPS (new AI models)            | High       | Low             | - Adversarial attacks<br>- Deal with fragment malware<br>- Lack of rigorous datasets | [67]         |
|                                            |          |            | Layer 3 VLAN                         | Layer 3/7 VLAN                                 | Medium     | Low             | - Misconfiguration<br>- Complicated management<br>- Limited VLAN IDs                 | [74]         |
|                                            |          |            | Network isolation                    | Deep slicing                                   | Medium     | High            | - High cost<br>- Scalability                                                         | [71], [72]   |
|                                            |          |            | Firewall/DPI                         | Intelligent Firewall/DPI                       | High       | Low             | - Adversarial attacks                                                                | [142], [143] |
| False data injection Impersonation         | High     | Medium     | Abnormal detection                   | Intelligent IDS/IPS (new AI models)            | High       | Low             | - Comprehensive dataset for SG2                                                      | [21]         |
|                                            |          |            | Authentication                       | Unified authentication                         | High       | Low             | - Single point of failure<br>- Provider acceptance                                   | [78]         |
|                                            |          |            | Encryption (TLS 1.3)                 | Blockchain (Quantum-safe TLS) Trusted networks | High       | Medium          | - High computation, scalability<br>- Zero-Trust implementation                       | [23], [70]   |
| Communication link sabotage (copper theft) | High     | Medium     | Educate public on theft consequence  | Educate public on theft consequence            | Medium     | High            | - Lack of awareness on high severity, sustained efforts with many partnerships.      | [65]         |
|                                            |          |            | Legal actions                        | Legal actions                                  | High       | High            | - Detection and damage proof<br>- Available law provision                            | [62], [65]   |
| Social engineering                         | High     | High       | Educate public Identity verification | Educate public/ Identity verification          | Medium     | Medium          | - Fast changing phishing tactic<br>- Sustained efforts to teach                      | [84], [85]   |
| Indirect attacks (jamming/Sybil)           | Medium   | Low        | IDS/IPS                              | Intelligent IDS/IPS (physical layer)           | Medium     | Low             | - High complexity<br>- Unclear efficiency                                            | [21]         |

**Severity/Likelihood:** **High** (occurred and caused damage/finance loss); **Medium** (occurred but the damage is small; **Low** (threats tested in research lab))

**Deployment cost:** **High** (require many extra devices and efforts to implement); **Medium** (require several extra devices or efforts); **Low** (Easy to deploy)

in Intrusion Detection System (IDS) [18] and Moving Target Defense (MTD) [144]. If an anomaly is discovered, the detection system may instruct the SDN controller to rewrite the data plane (programmable switches) to reduce the magnitude of the attack [147]. A promising solution is to use MTD-based solutions by masking or altering critical network features (e.g., true IP addresses) on a regular basis to avoid DDoS attacks and reconnaissance scans. Secure access service-edge (SASE) architecture [148] can be used to provide protection for SDWAN in SG2. Another promising technology is substation virtualization [149], where critical grid management applications are virtualized into the cloud or edge servers. Network functions and control applications can be optimized and deployed quickly at the stability and scalability advantage of the cloud.

In a complex network to connect thousands of grid operators, to prevent cyberattacks, all strategies above should be included properly, from enforcing strong encryption and mutual authentication to secure data and identities; deploying IDS/IPS/DPI/DMZ [60], [75] to detect and mitigate threats like DDoS; implementing network segmentation (VLAN, VPN) and slicing [71] to isolate and protect grid communication, adopting blockchain node for each grid operator, and ensuring regular security updates for IoT devices [27]. Additionally, other classical methods, such as security audits, adaptive synchronization protocols [76], [77], and multiple communication links [23], can still enhance overall grid resilience.

#### D. Remaining challenges

Table III summarizes the potential changes of security threats and defense solutions against the communication net-

work providers in the smart grid and SG2. One of the biggest remaining challenges for security matters for communication technologies is standardizing and implementing commercial post-quantum cryptographic schemes. This information security model will impact not only the security communication protocols in SG2 but also the Internet, given its essential role in every communication. Furthermore, emerging technologies such as deep slicing and blockchain technology have the potential to be game changers, but their complexity and high energy consumption may make widespread adoption difficult. Furthermore, if the technologies' overhead computation and security vulnerabilities are not addressed, they will most likely be disregarded. However, seamless upgrades for many networking technologies at the same time pose significant dangers. One such option is to include AI in the update automation [150] (e.g., federated learning-based automated firmware/protocol version upgrade systems for thousands of routers, switches, and firewalls in the networks of distributed grids and substations). However, given the numerous known adversarial attack concerns of AI-based models (e.g., evasion attacks, data poisoning, model poisoning attacks, and label-flipping attacks in federated learning [150], [151]), it is questionable if AI can considerably enhance the situation or, on the contrary, it makes the poisoned updates spreading faster. In this case, the integration of blockchain for protecting the integrity of AI models and trustable AI may help.

#### V. SECURITY THREATS AND PROTECTION MODELS FOR CONSUMER/END-USERS

Consumers in SG2 means individuals, households, public infrastructure, or commercial groups consume electricity generated by the grid. Generally, since the consumer types are

diverse, security defense capabilities vary. In the traditional power grid, given that there is no major financial reward if the attack is successful, hackers may be less motivated to target people or families consumers for money than for retribution or other wicked purposes. Commercial electric consumers, such as public transportation and industrial complexes, are the favorite hacking targets. However, in SG2, with the rapid change to distributed generation, heating of residential homes with heat pumps, need for charging EVs and UAVs, home batteries, households are rapidly becoming the ideal targets. For example, sudden charges or discharge of energy from millions of connected households and EVs from rerouting energy flow or malicious energy demands of compromised vehicles can be a threat to the stability of grid operations. Generally, the small consumers are not professionally secured with enterprise firewalls and intrusion prevention systems to prevent attacks. In this case, remote home batteries and connected EV cars can become targets for malware, potentially being turned into zombie systems to carry out malicious tasks under the control of a remote hacker. This section summarizes several typical threats and protection models for consumers.

#### A. Typical security attacks against the consumers

Security threats to consumers mostly come from vulnerabilities of smart metering devices and internal information technology (IT) networks (of commercial consumers). Based on the literature, we summarize several typical attacks and threats as follows.

① **Metering device interference/Energy theft:** Attackers destroy the communication cables to disrupt the connectivity between smart metering sensors and the grid management centers. These sabotage attacks can lead to sensor/link failures, prevent the SG2 from responding to operational demands, and potentially cause economic loss or electric disruption. The other variants can be time synchronization attacks or smart meter tampering attacks to bypass counting and electricity theft [79] and cut out power bills [81]. As illustrated in Fig. 13, the attacker can illegally use metal objects (magnets) to push against the terminal block and bypass a substantial percentage of power bills [80]. Generally, a smart electricity meter calculates energy consumption by measuring voltage between the input and neutral lines, and current across a shunt between the input and output lines. The total energy consumed is derived from the sum of the voltage-current products over time. This tampering method includes reversing the line connections, which can result in negative power readings, and using metal objects to bypass the current sensor, thus reducing the recorded power. Another method involves applying magnets to affect transformers and current sensors, leading to lower power readings. The attacker can also use a fake cover to mislead electric readers. The physical attacks against a meter can be changing code or retrieving a pre-shared key (PSK) (e.g., in LoRaWAN-based) to clone the meter [78].

② **Social engineering:** Social engineering attacks are designed to exploit human psychology to trick users into taking actions that are harmful to the grid. For example, an attacker might pose as a customer service representative

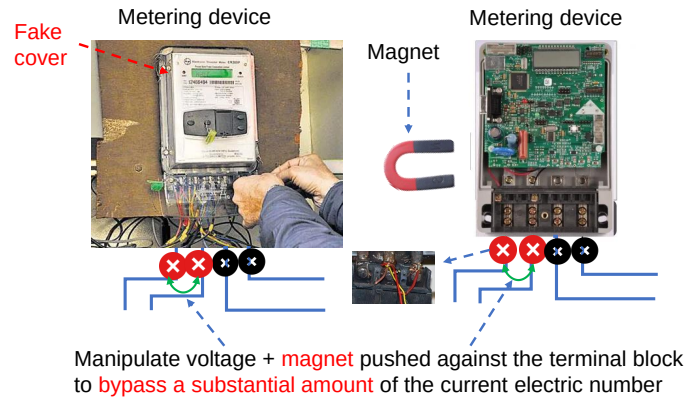


Fig. 13. The illustration of a fake cover and smart meter tampering attack where magnets and voltage interference can bypass energy numbers by the metering device [80].

from an electric distributor and ask users to provide their personal information to update billing information; otherwise, the electricity will be cut off. Also, an attacker impersonates a maintenance worker or technician from the utility company and contacts users, claiming they need remote access to the user's smart meter for maintenance purposes. The hacker can also bait users into using discount services and download fake software [84], [85]. All personal details can be used to launch secondary attacks, e.g., using the stolen credentials to log into the consumer's smart meter account, manipulate energy usage data, or gain access to broader network segments within SG2.

③ **Malware attacks:** Malware attacks mostly target commercial consumers. For example, Industroyer is a sophisticated malware designed specifically for industrial control systems (ICS) in industry complexes [86]. Industroyer has multiple components, including a backdoor, data wiper, and payloads that control electric circuit breakers, causing power outages by repeatedly opening and closing breakers. The malware communicates with its command and control servers over the Tor network, making it difficult to detect and mitigate. Some variants from Industroyer exploited CVE-2015-5374 vulnerabilities to launch a denial-of-service (DoS) and cause Siemens SIPROTEC relays unusable [86]. Malware can be installed in the consumers' IT networks in various ways, such as phishing emails of online electric bills or from compromised smart metering devices. Note that if consumers' smart meter devices do not often support two-way communications or direct data write capabilities, spreading malware via compromised smart metering devices is challenging.

Security threats against consumers in SG2 are a significant concern, particularly due to the varying levels of protection among different consumer groups. Individual and household consumers often utilize less well-protected methods (e.g., firewalls, IDS, barriers), making them more vulnerable to attacks. The attack damage can be reduced if the consumer devices do not support bidirectional communications as in SG1. Devices that only support one-way communication frequently send data to a remote server but do not receive commands or updates from it, limiting the potential for malware to propagate or for attackers to manipulate the devices directly. For commercial

consumers, such as industrial complexes and science parks, the stakes are higher. These entities typically rely on more sophisticated metering devices that support bidirectional communications. While this capability allows for more efficient and responsive local energy management, it also opens up more vectors for potential attacks, e.g., malware/ransomware, DoS. The attacks can cause the consumer's IT systems substantial operational disruptions and safety hazards [152].

### B. Specific threats for the new consumer models in SG2

Unlike in SG1, consumers in SG2 can also play the role of the grid providers, i.e., both consume and produce energy. With the rise of DERs, such as solar panels, wind turbines, home energy storage systems [19], and electric vehicles (e.g., battery-based), consumers can generate their own electricity and feed surplus energy back into the grid. Big prosumers (e.g., a factory with solar/wind turbine farms) can adjust their energy usage in response to grid demands, helping balance load and maintain grid stability during peak periods or energy shortages (e.g., for households in extreme weather or crisis situations). A well-coordinated plan with many prosumers (e.g., big prosumers, millions of EVs, home portable electric generators) can reduce grid congestion, provide local energy solutions for households, and improve grid resilience (against physical attacks, grid outage) [93]. However, **compromised consumers and prosumers** could also target demand response systems, **artificially inflating or reducing consumer demand signals** to disrupt grid stability or manipulate energy prices for financial gain [153]. As consumers in SG2 gain more control over energy production (e.g., via solar panels) and consumption, uncoordinated energy flow or production can cause grid imbalances if not properly managed or synchronized with the grid. Besides, **energy theft and fraud** through altering consumption data, inaccurate billing, or fraudulent energy trading practices, can become common, given the difficulty of managing thousands of contracted prosumers. Further information on the specific threats in energy trading models are detailed in Section VI-B.

### C. Specific security protection models for the consumers

In essence, protecting the consumer requires several comprehensive techniques. For individual consumers, physical security measures such as tamper-evident seals and secure meter enclosures can help prevent unauthorized access. Regular inspections and monitoring can detect any signs of tampering early. For commercial consumers, **multiple protection layers** may require authentication, data encryption, application security protocols, firewalls, service identity access management, operation/kernel systems reinforcement, abnormal behavior detection, and so on. For example, industrial consumers may need an IT team and should be educated about recognizing phishing attempts and the importance of not sharing sensitive information. Implementing **multi-factor authentication (MFA)** can add an additional layer of security. The data transmission and secure communications in smart electricity-related tasks should follow IEC 62056-21, an international standard for reading utility meters [82].

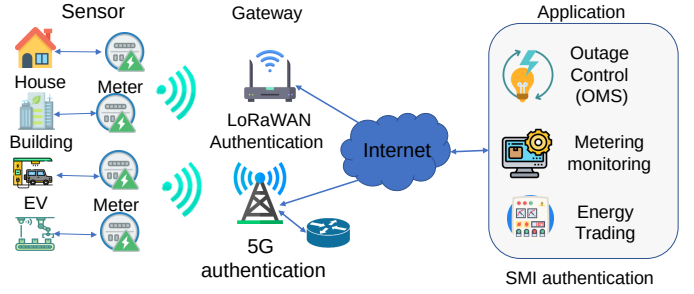


Fig. 14. The illustration of multiple authentication mechanisms for carrying data from smart metering in smart grids. The authentication infrastructure will be tailored to the communication technologies equipped for meter devices.

The connections between commercial customers' smart meters and the grid supplier can be authenticated through the PKI or custom **authentication mechanisms** of network carriers, e.g., LoRaWAN gateway, as illustrated in Fig. 14. The credentials used in authentication can vary, such as pre-shared keys/tokens/certificates. For 5G-based smart grids, the authentication is based on 5G PKI architecture. However, each smart meter or end device is required to be equipped with a SIM card or eSIM (which may cause a high cost). Authentication, among other services in smart grids, e.g., camera surveillance and sensor monitoring, is also followed by custom models [99]. To prevent energy theft from consumers, several emerging technologies have been promising to be used in smart grids, e.g., blockchain [83], [154], [155] or zero-sum peer-to-peer transaction settlement verification [156]. This can create an immutable and transparent record of energy transactions and prevents fraudulent alterations to consumption records and ensures accountability for energy usage. For secure communications, **encrypting all data transmissions** can protect against interception. Implementing secure communication protocols such as transport layer security (TLS) can further enhance security. In SG2, AES can be upgraded with a new key length, e.g., 256 bits. WiFi or Ethernet-based HAN may still rely on TLS 1.3 [78].

**Firewalls, intrusion detection systems (IDS), and abnormal behavior detection** can help detect and mitigate DoS attacks and malware/ransomware targeting commercial consumers' communication networks. Implementing machine learning algorithms to analyze smart meter data for anomalies and alerting authorities are the hot topics currently. Further, with the trend toward virtualization of networking technologies, the security-as-a-service (SECaaS) paradigm will be crucial. In this paradigm, service prosumers or energy distributors may contract with Internet security providers to do all security services (malware detection, cyber attack prevention). The other is the enhancement of AI models to allow proactive learning and response against numerous risks [40], [41], [87]. To prevent imbalances in the network of large-scale prosumers, distributed cooperative grid synchronization mechanisms [157], [158] for smart energy management will be essential to coordinate and maintain seamless handover between energy production and consumption. Smart meters can be equipped with tamper detection sensors [126] that alert utility

companies in real-time if any attempt is made to physically or digitally interfere with the meter. By continuously monitoring energy consumption patterns with advanced data analytics and machine learning, utilities can detect anomalies that indicate potential theft. Sudden drops or spikes in usage that deviate from benign patterns can trigger further investigation.

#### D. Remaining challenges

For consumers in SG2, security attacks as energy thief or malware/ransomware/DoS attacks become more common, given the complexity of managing bidirectional communications in smart meters. Among these, metering sensor interference or artificial energy price inflation attacks are specific ones for the integration of consumers and prosumers in SG2. The unique difference is, in the context of SG2, successful attacks could allow the attacker to manipulate energy consumption patterns or intentional interference on consumer/prosumer devices, impacting power bills. Securing these consumer systems requires both physical deployment with boxes, robust authentication mechanisms or cooperative grid synchronization mechanisms, and IDS at connected multi-source networks. Furthermore, unified authentication and end-to-end data encryption techniques can be critical but have not yet been implemented in SG2. The hierarchy protection by service priority or the relevance of data transfer should be properly constructed. In the next part, we cover the security risks and protection of many upcoming technologies that are projected to be prevalent in SG2, such as distributed energy resources and peer-to-peer energy trading.

### VI. SECURITY ATTACKS AND PROTECTION MODELS IN SG2 ENABLING TECHNOLOGIES

SG2 will involve a series of enhancements from the conventional smart grid model, e.g., the efficiency, reliability, and sustainability of energy distribution and management. This section summarizes the security problems and corresponding solutions of enabling technologies that are expected to be popular in SG2: distributed energy resources (DERs), peer-to-peer energy trading, and energy storage technologies.

#### A. Security threats and defense for DERs

DERs denote small-scale or micro units of energy generation connected to the electricity grid at the distribution level. DERs can be operated automatically or through a remote cloud center [159]. This application concept revolutionizes the smart grid by permitting decentralized, community-generated energy from energy resources near electricity usage locations. DERs minimize power system augmentation, consumer costs, and emissions. To maintain energy stability without long-distance transmission power lines, DERs depend on rooftop solar panels, small wind turbines, small hydroelectric generating, and battery storage as more people avoid living near toxic power plants. However, to maintain the distributed microgrids, DERs rely much on the communication network systems. Therefore, one major threat in DERs is **cybersecurity attacks**, which can exploit **vulnerabilities in TCP/IP-based protocols**, as

summarized in Section IV. Further, due to low cost, many local grids of DERs may not be equipped with advanced security protection, e.g., substation DMZ, enterprise firewalls [97]. According to [136], **data injection and physical layer attacks** in DERs may be less challenging to launch than in well-protected large-scale smart grids. Further, many DER components (e.g., metering devices and authentication gateways) may come from various suppliers, and some may sacrifice security features for low cost and simplicity. If these compromised components make their way into a larger interconnected DER, they could be used as entry points for attackers to infiltrate the larger energy network or initiate disruptions. Finally, **the lack of standardization in security practices across various DER technologies** increases the risk of these assets being exploited by malicious actors [160].

Securing the physical infrastructure of DERs is vital for SG2. With the strict requirement on cost, deploying physical access controls, advanced surveillance, and tamper-evident measures can be difficult. As summarized in Table V (the first application domain), preventing physical compromises by wall fences and integrating HSM modules on key components of DER is then the most economical protection solution. Further, the elements connected to the Internet should be equipped with **basic firewalls** (e.g., access list control, limit the control from dedicated IPs, or **enable peer-to-peer authentication** [161]). Incorporating **lightweight anomaly detection and intrusion prevention systems** in every DER gateway [136] can proactively identify suspicious activities or unauthorized access attempts. IDS/IPS technologies are still the main forces to alert grid operators to potential security breaches swiftly, enabling rapid isolation and damage mitigation. Additionally, **maintaining purchased devices from reliable sources, regular stress tests on security**, and updates to the software and firmware of DER devices help identify vulnerabilities and ensure that security measures stay current against evolving threats. Finally, building solutions for cascading failures of several DERs as introduced in Section III (protecting power providers) and the study [162] are essential.

#### B. Security threats and defense in new energy trading models

A significant difference between SG2 and its predecessor is the appearance of a new pro-consumer energy trading model, where the customer can be both an energy surplus supplier (i.e., sell energy surplus to energy distributors) and a consumer [6]. As stated in [161], [163] and [164], the peer-to-peer (P2P) energy management system, also known as bilateral contract networks, can facilitate the coordination of prosumers for higher efficiency and flexibility. Generally, the main energy trading techniques can be divided into four categories: game theory [165], auction theory [166]–[168], and constrained optimization [169]. For example, the Stackelberg game theory method was proposed to coordinate prosumers [161]. To control the demand response issues in residential homes, an hour-ahead and intraday P2P technique were used [170]. A multiclass P2P energy management platform was designed to coordinate prosumers with diverse preferences for energy source/destination [164], [171]. A consortium blockchain solution can be used for localized P2P power trading, with energy



price determined by an auction process, while privacy and transaction security were enhanced [166], [167], [172]. One of the primary benefits of blockchain is enhanced transparency and trust. Blockchain's decentralized and public ledger ensures that all transactions and contract executions are visible and verifiable by all network participants, which fosters trust among parties who do not need to know or trust each other. An example of blockchain technology for peer-to-peer (P2P) energy trading is the Brooklyn Microgrid project [173]. This initiative was designed to allow residents in Brooklyn, New York, to generate, buy, and sell energy directly to one another using a blockchain-based platform.

However, the major security threats in P2P electricity trading are **Byzantine attacks** and **vulnerabilities of blockchain techniques**. Table IV summarizes several typical security attacks and protection models for blockchain-based SG2 models. For example, 51% attacks or Byzantine attacks introduce misleading or noisy input into several nodes of distributed energy management systems, resulting in nonconvergence of transaction models. For the blockchain-based energy trading models, **vulnerabilities of the blockchain architecture** itself are the most concerning. Smart contracts, for example, are often self-executing contracts that are maintained on a blockchain. They are used to automate the execution of peer-to-peer energy trading contracts. However, smart contracts can be complex to implement and configure correctly. This can lead to replay attacks or software flaws (e.g., mishandled exceptions, unhandled errors [174]) that can be exploited by attackers to steal energy or overestimate the transmitted energy surplus (siphon funds) [175]. For instance, the Parity wallet bug in 2017 caused by a coding error led to the freezing of 280 million worth of digital coin Ether, illustrating how even small mistakes can have severe financial repercussions [176]. The consequence will be devastating if the unvetted mistake occurs for the blockchain network of millions of consumers.

Blockchain vulnerabilities, further, include incorrect arithmetic operations, unchecked external calls, poor randomness generation, and lack of formal verification [177]. Particularly, the immutability of smart contracts will have a negative impact if errors or vulnerabilities are discovered after deployment. For example, when a vulnerability is discovered after deployment, rectifying it can be challenging to maintain due to the long chain of dependent transactions. One prominent example is the Decentralized Autonomous Organization (DAO) hack in 2016 [178], where a vulnerability in the DAO's smart contract code was exploited, resulting in the theft of approximately 60 million worth of Ether. Due to the immutable nature of the Ethereum blockchain, the developers could not simply alter the contract to fix the vulnerability or reverse the fraudulent transactions. This incident led to a contentious hard fork in the Ethereum blockchain, splitting it into two separate chains: Ethereum (ETH) and Ethereum Classic (ETC). This example illustrates that while immutability ensures data integrity, it can also make addressing unforeseen issues and errors exceedingly difficult, sometimes requiring drastic measures such as a hard fork. Also, P2P energy trading systems are vulnerable to a variety of **conventional cybersecurity attacks**, such as **denial-of-service attacks** [122], and **phishing attacks** [179].

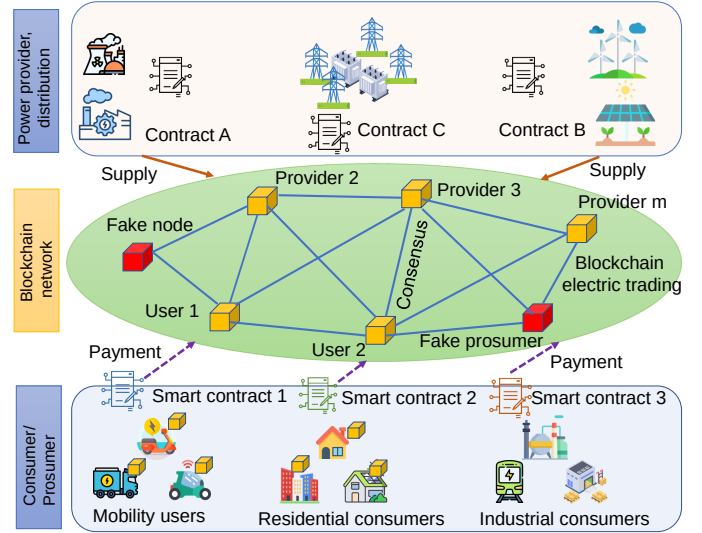


Fig. 15. Illustration of security threats in peer-to-peer energy trading via a blockchain network where the attackers can create fake prosumers to steal energy or overestimate the transmitted energy surplus (siphon funds).

**Privacy concerns** also arise in new energy trading models. While blockchain and other decentralized technologies can enhance transparency and traceability, they can also **expose sensitive consumer data to the public domain**. Unauthorized access to participant information, energy consumption patterns, and financial details could lead to privacy breaches and misuse of personal data [180]. For example, the attacker can follow the energy consumption patterns to identify the consumers' habits and target them with phishing attacks [172]. If charging stations are put in medical clinic parks, businesses, and hospitals, sensitive information about EV drivers, such as their driving habits and visited places, can be revealed [181].

**Collusion** is a major security concern in game-theory-based and auction-theory-based P2P energy trading models. For example, a group of malicious prosumers (consumers who also produce energy) collude to manipulate the energy market. They agree to coordinate their bidding strategies in order to drive up the price of energy to their advantage on a particular day, resulting in higher prices for consumers and lower profits for other energy producers [155]. Sybil attacks also pose high risks to P2P energy trading. In the Sybil attack, an attacker creates multiple fake identities to gain undue influence in a system. In the context of P2P energy trading, the Sybil attacker could use their fake identities to manipulate the energy market (drive up the price due to fake high demands), launch collusion attacks, or steal funds from other participants [122].

For protection solutions, as summarized in Table V (the second application domain), to prevent the negative consequences of immutability matters and vulnerabilities on blockchain-based energy networks, smart contract audit and bug-free verification are essential. For example, the researchers in [182] propose a formal verification method for smart contracts. Accordingly, it is a rigorous mathematical process that can be used to prove that a software system satisfies its specifications. This can help to identify and fix any vulnerabilities in smart

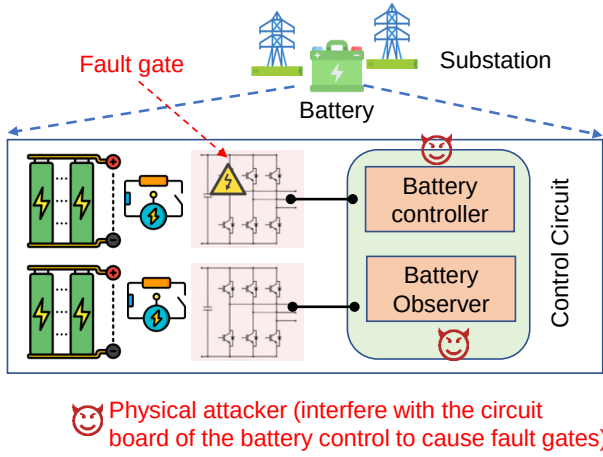


Fig. 16. Illustration of energy storage technologies and potential security attacks where the attackers can launch cyber-physical attacks against sensors or external devices of cyber system controls. Their goal is to create abnormal on/off gates that cause faults in key components of smart grids (power converter/transformers).

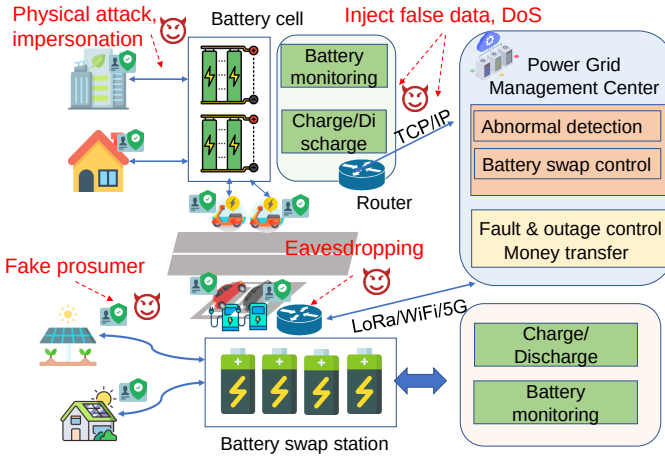


Fig. 17. Illustration of security attacks against battery storage technologies where TCP/IP-based communications between battery cell pack management and integrated energy management center can be intercepted and injected false data for incorrect control commands, e.g., disable battery charge/discharge functions, or to be fake prosumers. The attacks against EV charging systems are detailed in Fig. 18 and Section VI-D.

contracts before they can be exploited by attackers. Further, robust governance frameworks and smart contract controls that allow for flexibility and error correction, e.g., multi-signature protocols for approving changes [178], establishing legal and technical mechanisms for dispute resolution. Additionally, privacy-preserving technologies, such as zero-knowledge proofs, can help protect sensitive participant information while maintaining the benefits of decentralized energy trading [181], [183]. Besides, using reputation-based networks with mutual verification [182] or block alliance consensus mechanisms [184] to identify and prevent malicious actors can be a promising approach. Maintaining energy pricing regulations and the eligibility of participants via additional authentication can mitigate the magnitude of the attack.

In summary, protecting new energy trading models within

SG2 requires a holistic approach that combines technological measures, cryptographic techniques, and regulatory compliance. Addressing these matters still requires time and these topics have not yet been well-explored in the vision of SG2.

### C. Security threats and defense in energy storage technologies

As we mentioned earlier, battery energy storage systems (BESS) are core components of DERs and future SG2 by enabling grid stability through the integration of renewable sources and discharging at proper times. However, these technologies also introduce specific security threats. One significant security concern is the potential for **battery-related incidents or fault gates of energy coordination**, including thermal runaway, leakage, or explosions from the physical attacks [19], [26] as illustrated in Fig. 16. Fig. 17 illustrates four detailed cases of **physical attacks (e.g., fake prosumer, impersonation)** against signal exchanges between sensors and estimation observers to mislead the voltage/power-line controllers (giving the wrong on-off gate decision). This can lead to incorrect energy flow adjustments or even shutting down a critical power line for various consumers and rivals. Note that BESS control functions rely heavily on telemetry capabilities such as load frequency control, damping oscillations, and power imbalance measurement. Any disconnect from faulty communication networks might result in a loss of control of BESS functions. Besides, manufacturing defects, poor maintenance, or external factors such as temperature variations can trigger the incidents [21]. In recent years, lithium-ion-based BESS cell failures that resulted in thermal runaway and explosion accidents have also raised serious concerns, e.g., the BESS explosion in Surprise Arizona, US, 2019 [187].

Moreover, the increasing reliance on **lithium-ion batteries**, which dominate the energy storage landscape, raises concerns about **supply chain vulnerabilities** [188]. As the demand for these batteries grows, so does the potential for compromised components entering the market. Malicious actors could insert **counterfeit or tampered components into battery systems**, leading to reduced performance, safety risks, or even intentional battery failure. Attackers may take advantage of vulnerabilities introduced into devices prior to shipping or during **firmware upgrades**. Fig. 17 illustrates physical attacks on IoT gateways to disrupt data exchange between battery cell packs and remote controllers or consumers. Accordingly, embedded devices and communication modules are highly sensitive to physical attacks, e.g., side-channel attacks. BESS near substations are complicated systems with hundreds of gateway controllers and circuit breakers. Because of this complexity, the operator may buy components from many vendors. Even if most of those suppliers are dependable, an operator may have to use lesser-known vendors due to a sudden shortage of particular components. Even dependable providers might be exposed to supply chain hijacking. For example, in the SolarWinds security issue, malicious code is introduced into the developer's update channel, which is subsequently distributed to all impacted devices [188]. Another attack version is that a malicious attack may send a bogus signal to consumers telling them to charge/discharge their devices when they should not,

TABLE IV  
SUMMARY OF SECURITY THREATS, PROTECTION MODELS, AND OPEN ISSUES IN POTENTIAL BLOCKCHAIN-BASED SG2 MODELS

| Security threat              | Blockchain/Damage                                          | Severity | Likelihood | Potential examples in SG                                                                                                                                                                          | Protection model                                           | Efficiency | Open issues                                                                                                    | Reference |
|------------------------------|------------------------------------------------------------|----------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------|-----------|
| 51% attacks                  | - Bitcoin fork (2018)<br>- Loss: \$18 million worth of BTG | High     | High       | A coalition of DER operators or prosumers gains control over more than 51% of the total energy generation capacity could manipulate the energy supply, creating artificial shortages or surpluses | Proof of Stake                                             | Medium     | A concentration of power among a few wealthy stakeholders or high-energy consumption                           | [185]     |
|                              |                                                            |          |            |                                                                                                                                                                                                   | Periodic checkpoints, abnormal detection                   | High       | Require substantial changes to the consensus mechanism & model for behavior monitoring                         | [185]     |
|                              |                                                            |          |            |                                                                                                                                                                                                   | Strong governance                                          | Medium     | Voter apathy can lead concentration of power among a few active participants                                   | [122]     |
| DAO attacks                  | - Ethereum (2016)<br>- Loss: \$60 million worth of Ether   | High     | High       | Attacker can withdraw funds before the contract updates the balance, resulting in financial loss and disruption of grid                                                                           | Smart contract audit                                       | Medium     | Depending on the efficiency of audit model and verification engines.                                           | [177]     |
|                              |                                                            |          |            |                                                                                                                                                                                                   | Multi-signature wallets<br>Hard fork                       | Medium     | Complexity in setup and use that requires coordination among multiple parties                                  | [178]     |
| FDI attacks                  | - Bitcoin (2016)                                           | High     | Medium     | Attacker can manipulate energy consumption figures, incorrect energy production data from DERs, or falsified transactions                                                                         | Fault Tolerance<br>Proof of Stake<br>End-to-end encryption | High       | Low scalability, complexity in configuration, voter apathy, complexity in key exchange, computational overhead | [186]     |
| Internal fraud               | - Bitcoin (2014)<br>- Loss: \$450 million at the time      | High     | High       | A utility company employee might alter blockchain entries to record false energy consumption data, resulting in lower bills for certain customers                                                 | Smart contract audit<br>Role-based access                  | Medium     | Depending on the efficiency of the audit model, verification engines, and access control.                      | [185]     |
| Parity wallet (Software bug) | - Ethereum (2017)<br>- Loss: \$30 million in Ether         | High     | High       | Attacker can could drain funds allocated for energy transactions, leading to financial losses and operational disruptions                                                                         | Smart contract audit                                       | High       | - High complicated blockchain comes with potential bugs                                                        | [176]     |

resulting in network frequency degradation. DoS attacks on the central controller in the event of centralized management may result in the loss of energy control by BESS and the instability of power operations [196].

Regarding protection models for BESS, the first and foremost is to install IDS/IPS that can analyze the abnormal charge/discharge behavior from BESS [21]. Supply chain verification, end-to-end encrypted communications, P2P authentication, and source traceability are also essential. Besides, physical protection like fencing and buildings may help, as summarized in Table V (the third application domain). There is a lack of studies in this field, given the difficulty of building a testing or simulation platform. A summary of holistic protection approaches to address the causes of failure in BESS, e.g., full BESS system installation testing, hazard and gas chromatography, and composition testing, can be found at [187].

#### D. Security threats and defense for electric vehicle charging and scheduling in SG2

Electric vehicles (EVs) are a crucial part of the green transition to zero-carbon alternatives. EVs help reduce greenhouse gas emissions and reliance on fossil fuels, contributing to cleaner air and a more sustainable environment. The adoption of EVs is supported by advancements in battery swap technology and renewable energy integration. For example, electric scooters now account for 12% of all scooter sales in Taiwan, a figure that continues to grow with the convenience of battery swapping technology and the high density of charging stations [197]. In China, EVs comprised 60% of new car sales in 2023, while in Europe and the US, the figures were around 25% and 10%, respectively [198]. By 2024, electric automobiles may have a market share of up to 45% in China, 25% in Europe, and over 11% in the US [198].

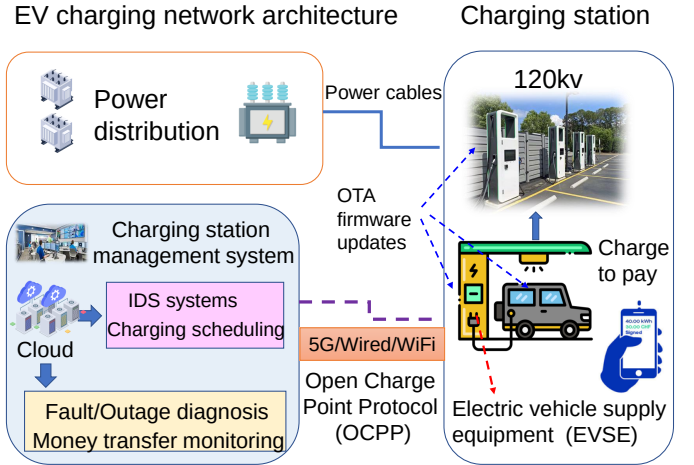


Fig. 18. Illustration of an EV charging network architecture and its core components, e.g., electric vehicle supply equipment (EVSE), charging station, charging station management system (CSMS), and network connection between EV and CSMS using Open Charge Point Protocol (OCPP).

A prominent security threat in EV charging systems is the potential for data breaches. EV charging stations collect and transmit a significant amount of user data, including payment information and personal details. If the EVs or charging stations use old generation OCPP or their firmware is not updated frequently, attackers can intercept communication links between the EV, the electric vehicle supply equipment (EVSE), the charging station, and the charging station management system (CSMS) (illustrated in Fig. 18), leading to identity theft or financial fraud [21]. For instance, researchers in Sandia National Laboratories, US, highlight that several public EV charging networks lacked encryption or some plug-and-charge functions may abort charging sessions by disrupting

TABLE V  
COMPARISON OF SECURITY THREATS AND DEFENSE IN EMERGING TECHNOLOGIES FOR SG1 VS SG2

| Application domain          | Security threats (attack target)                               | Severity | Likelihood                    | Protection models                                      |                                                                      | Efficiency | Open issues                                  | References                       |
|-----------------------------|----------------------------------------------------------------|----------|-------------------------------|--------------------------------------------------------|----------------------------------------------------------------------|------------|----------------------------------------------|----------------------------------|
|                             |                                                                |          |                               | SG1                                                    | SG2                                                                  |            |                                              |                                  |
| Distributed energy resource | False data injection (data collection)                         | High     | Medium                        | Encryption IDS/IPS                                     | Quantum-safe Encryption Intelligent IDS/IPS                          | High       | Different security capability of the sources | [171]                            |
|                             | Compromised source (federated learning)                        | Medium   | Low (insider)                 | HSM, IDS Authentication                                | HSM, IDS Authentication Blockchain                                   | Medium     | Unreliable supply chain                      | [161]                            |
| P2P energy trading          | DoS attacks (Network infrastructure)                           | Medium   | Low (distributed model)       | IDS/IPS Load balance                                   | Distributed ledger                                                   | Medium     | Real-time response, scalability              | [122]                            |
|                             | Impersonation/Phishing (Fake prosumer)                         | High     | High                          | Identity authentication                                | Smart contract audit Behavior analysis                               | High       | Scalability of verification                  | [22]                             |
|                             | 51% attacks (blockchain-based power grids)                     | Medium   | Low (require large collusion) | Proof of stake Consensus mechanism                     | Periodic checkpoint Strong governance                                | High       | Transparent nature of blockchain             | [154]                            |
|                             | Collusion attacks (Auction theory-based energy trading models) | High     | Low (Insiders, fraud)         | Anomaly detection                                      | Anomaly detection Blockchain                                         | High       | Coordinated behaviors of insider attackers   | [155]                            |
|                             | Byzantine attacks (Electric controllers)                       | Medium   | Medium                        | IDS/IPS                                                | Intelligent IDS/IPS                                                  | Medium     | Complex to detect in a large scale grid      | [175]<br>[161]                   |
|                             |                                                                |          |                               |                                                        |                                                                      |            |                                              |                                  |
| Energy storage systems      | Physical attacks Fuel leakage (Battery fault, explosion)       | High     | High                          | Fencing/Barrier                                        | Battery swap Fencing/barrier Legal action                            | Medium     | Economical solutions                         | [26]<br>[19]                     |
|                             | Spoofing attacks (Battery operations)                          | Medium   | Low (Insider)                 | Encryption                                             | Anomaly detection Blockchain                                         | Medium     | Insider attacks                              | [21]                             |
|                             | Firmware update tampering                                      | High     | Low (compromised)             | HSM, secure boot                                       | End-to-end encryption HSM, blockchain                                | High       | Affordable solutions                         | [21]                             |
|                             | Supply chain (Lithium-ion battery)                             | High     | High                          | Trusted vendor list                                    | Blockchain-based management                                          | Medium     | Limit of options, supply quality             | [188]<br>[57]                    |
|                             | Adversarial attacks (AI-powered function)                      | High     | Medium                        | No mechanism (yet be applied)                          | Adversarial training, Robust AI models                               | High       | Diverse attacks                              | [136]                            |
|                             |                                                                |          |                               |                                                        |                                                                      |            |                                              |                                  |
| EV charging and scheduling  | Data breaches (Payment, personal data)                         | Medium   | Low (encryption)              | OCPP 1.0 TLS 1.3                                       | OCPP 2.0 End-to-end encryption Quantum-safe encryption               | High       | High cost                                    | [21]                             |
|                             | Identity theft (Charge card)                                   | High     | Medium                        | Two-factor authentication                              | Biometric authentication                                             | High       | User-friendly authentication                 | [21]                             |
|                             | Physical sabotage (CSMS links)                                 | High     | High                          | No mechanism                                           | Anomaly detection                                                    | Medium     | Long links exist in residential areas        | [189]                            |
|                             | Collusion attacks (Imbalanced grid)                            | High     | Low (large collusion)         | No mechanism                                           | Blockchain Anomaly detection Trusted networks                        | High       | Collusion of insider attacks                 | [118]                            |
|                             | Adversarial attacks (pricing management)                       | High     | Low                           | No mechanism                                           | Adversarial training Robust AI models                                | Medium     | Diverse attacks                              | [190]                            |
|                             |                                                                |          |                               |                                                        |                                                                      |            |                                              |                                  |
| Digital twin                | False data injection (Measurement sensor)                      | High     | Medium                        | Anomaly detection                                      | Intelligent anomaly detection, blockchain                            | High       | Real time detection Diverse data             | [191]                            |
|                             | DoS attacks (Data transformation)                              | High     | High                          | CDN/Cloudflare                                         | Distributed management systems                                       | Medium     | Many variant attacks                         | [191]                            |
| 5G/6G for SG2               | Signal jamming Energy depletion Pilot contamination            | -        | -                             | -                                                      | Multiple models as summarized in Table III                           | -          | -                                            |                                  |
|                             | Adversarial attacks (AI-powered functions)                     | High     | Low (black-box attacks)       | Adversarial training Data sanitization Model hardening | Adversarial training Data sanitization Model hardening (enhancement) | Medium     | High cost of training and data preprocessing | [192]<br>[193]<br>[194]<br>[195] |

the PLC communications. This makes it easy for attackers to steal credentials or influence charging sessions via the EV-to-EVSE connection [189]. Lastly, the integration of distributed energy resources (DER) and demand response systems (i.e., P2P energy trading) into SG2 may introduce new attack vectors. Hackers could potentially manipulate the scheduling algorithms for EV charging, causing imbalances in the grid. For example, a coordinated attack could instruct numerous EVs to charge simultaneously, creating a sudden spike in demand that the grid cannot handle, leading to blackouts. In 2020, a study [199] by the National Renewable Energy Laboratory (NREL) demonstrated how malicious actors could exploit demand response signals to destabilize the grid.

To defend against these threats, as summarized in Table V (the fourth application domain), implementing **advanced anomaly detection** is crucial [200]. These systems can monitor network traffic for unusual patterns indicative of a DDoS attack, allowing for immediate countermeasures to be deployed. Additionally, adopting more **secure versions of communication protocols, such as OCPP 2.0**, which includes enhanced security features like encrypted communi-

cation and certificate-based authentication, can mitigate the risks of such attacks. Regularly **patching the software of charging stations** is also essential to protect against known vulnerabilities [200]. To safeguard against data breaches, robust encryption protocols must be employed for all data transmissions between EVs, charging stations, and the central grid. Furthermore, implementing multi-factor authentication (MFA) can add an additional layer of security, ensuring that only authorized users can access sensitive systems and data. Regular security audits and compliance with data protection regulations such as the General Data Protection Regulation (GDPR) can also help ensure that data handling practices meet the highest security standards. Finally, adopting a zero-trust architecture, where every transaction and communication within the grid is verified, can significantly reduce the risk of unauthorized manipulations from firmware tampering. Implementing AI/ML algorithms to predict electric demand more effectively can enhance the resilience of the grid against threats. IDS may use these techniques to identify charging abnormalities and react fast to avoid grid imbalances. The AI/ML solutions for SG2 technologies are detailed in Section V.



### E. Security threats and defense in future technologies for SG2

Digital twin is expected to be one of the most promising future technologies to appear in SG2 [191]. This technology maintains a virtual world of real-world physical products, systems, or processes through updates from real-time data over time. The straightforward advantage of the digital twin is that it provides a robust and affordable environment for simulation, integration, testing, monitoring, and maintenance, improving efficiency and decision-making. As a result, technological advancements in the monitoring and control functions of energy distribution systems may evolve faster than in the conventional smart grid. However, as summarized in [201] and [186], a huge volume of data and real-time analysis requirements are two of the major concerns for any security system. Table V summarizes two typical attacks in digital twins for SG2 (the fifth application domain). For example, the attacker can launch a false data injection that creates inaccurate representations of the grid state, potentially leading to misguided decisions. Further, DoS attacks target disrupting the real-time functionality of AI/ML models in real-to-virtual transformation functions, degrading the grid's responsiveness [186]. Since large-scale data processing and real-time reaction are also challenges for many current defense technologies, these attacks are still open issues to address. Distributed IDS or edge-assisted detection systems can be promising approaches that have yet to be explored, particularly in Distributed Energy Resources (DERs), battery storage technologies, and electric vehicle integration environments.

Upgrading network infrastructure towards 5G and further 6G networks is expected to be the second significant change in SG2, as discussed in Section II-C. However, these networks have their security problems that remain concerns. As pointed out in the study [78] and summarized in Table V (the sixth application domain), signaling DoS attacks/jamming attacks at the physical layer, energy depletion attacks, or exploiting no integrity protection of the user data plane to launch impersonation attacks at the network layer, deep fake/biometric authentication data leakage at the application layer are typical security risks. Furthermore, new risks to 5G/6G-enabled technologies have emerged, such as pilot contamination attacks against massive MIMO-based networks or adversarial attacks against AI-aided network operations (resource allocation, slicing, service offloading, and semantic communications) [78]. Explainable AI, quantum-safe communications, distributed ledgers, and differential privacy can significantly reduce the severity of attacks and personal data breaches. However, the research topics of these technologies specified for smart grid environments have not yet been well explored. Given the importance of AI in many features of new smart grid generation, the following section details the research efforts on security attacks and defense in AI-powered functions, including references to smart grid issues.

## VII. SECURITY ATTACKS AND PROTECTION MODELS IN AI-ENABLED POWER GRID MANAGEMENT

AI-empowered energy management represents an essential space for SG2's autonomous and self-healing features,

enabling grid distribution optimization and usage efficiency. However, these systems are not immune to security threats. The major concerns are **data poisoning, adversarial attacks, and model interference**. In the context of SG2 management, these attacks could lead to inaccurate load forecasts, improper demand response decisions, or faulty optimization strategies, potentially causing grid instability. The following subsections outline several typical threats and defense methods against AI-powered power grid management for SG2.

### A. Security threats against AI-enabled power grid management in SG2

While AI in computer vision and natural language processing rapidly grows, AI for SG2 has just gotten the attention [202], [203]. In essence, smart grid management functions can benefit much from AI learning capability, e.g., reliable electric distribution scheduling, autonomous energy management operations, voltage stability assessment, optimal dynamic pricing trading, and accurate outage forecast. Fig. 19 shows the summary of typical AI-aided functions in SG2 infrastructure. In these applications, AI models can help to **increase the accuracy**: on/off switchgear decisions, electric price charging, electric generating rate limit (to avoid grid overload), and network congestion avoidance.

However, data poisoning, model interference, and adversarial attacks are the biggest security concerns. Fig. 20 summarizes three attacks and their target in each phase of AI learning process. Given AI's reliance on enormous amounts of data for training, data poisoning is a cyberattack aimed at tainting training datasets with outlier data points. Further, model interference attack involves an adversary manipulating the inputs but can also, outputs, or parameters of a machine learning model to degrade its performance or to cause it to produce erroneous results [78]. Both attacks assumes that the attacker must have full permission to access the dataset/learning model or control over the data pipeline, i.e., the write privilege to insert new data points/classes (the public repository or compromised equipment). By contrast, adversarial attacks generate adversarial samples on electricity reading to increase an unanticipated outcome such as misclassification. There are two adversarial attacks: black-box and white-box. White-box adversarial attacks have full access to the target model's architecture and parameters, allowing precise crafting of adversarial examples, while black-box adversarial attacks only interact with the model's inputs and outputs, lacking detailed knowledge of the internal workings. For example, the authors in [192] proposes two adversarial attacks, namely nearest neighbor perturbation (NNP) and nearest neighbor distance (NND), to fool electricity reading and steal energy by iteratively generating adversarial samples. The authors in [204] introduce a signal-agnostic adversarial attack method to generate adversarial examples and degrade the performance of CNN-based power quality classification in smart grids.

Recently, the researchers in [205] proposed a novel method to generate adversarial perturbations successfully against power allocation in communication networks with few random samples. In another study, the authors [206] create adversarial

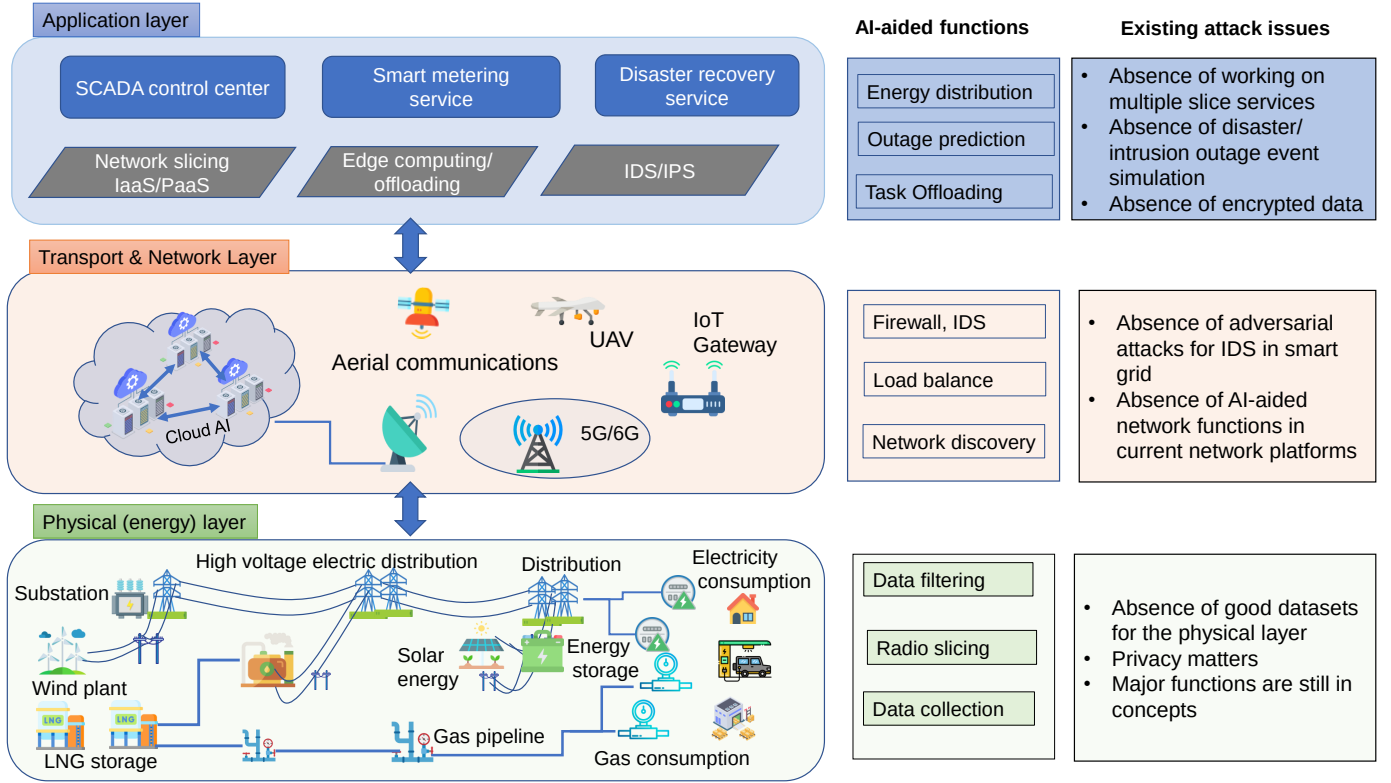


Fig. 19. Summary of typical AI-aided functions in SG2 infrastructure with 5G/6G core networks, and remaining issues for the attacks/defenses. Accordingly, many functions of all layers in SG2, such as radio slicing, outage prediction, and energy distribution, will be equipped with intelligent capability from advanced AI models.

reinforcement learning against dynamic channel access and power control by manipulated state information in deep Q-network (DQN). The work [207] proposed to use the perturbation generator model (PGM) to attack grid-based wireless systems. The PGM is strong enough to generate input-agnostic perturbations, with constraints like power, undetectability, and robustness to ensure effective and stealthy perturbations. Also, there is a significant negative correlation between attack power and channel estimation model performance. As attack power improves, the model's vulnerability decreases. The authors in [203] and [208] found that universal adversarial perturbations can damage the power grid state estimation with the same magnitude as the widely studied false data injection attacks do. In another case, the authors in [194] introduce a novel destabilizing attack to target inverters in microgrids by altering the power control gains and their impact on small-signal stability. The studies in [209] and [204] present two different stealthy adversarial attacks that target deep learning-based smart metering to inject false data or bypass the energy consumption counters. To enhance the stealthy of the attacks, the attacker can propose to infiltrate the demand side management [210] by a combination of false data injection, such as changing the thermostat set points of heating, ventilation, and air conditioning (HVAC) systems and electric water heaters (EWHs), and adversarial perturbations. Unlike adversarial attacks on image classification tasks, adversarial attacks on smart metering may mislead not just human intuition but also power system error-checking procedures [211].

The common point of all the adversarial attacks is that the attacker tries to influence the deep learning model's decision by manipulating environmental factors (noise) or spoofing signal requests (transmission power). By degrading AI-based power grid management and scheduling or radio resource allocation performance, the stability for the SG2 control is negatively affected, particularly in SG2 interdependent control and communications, particularly with digital twin features [191]. The authors in [212] and [192] showed both white-box and black-box adversarial attacks against electricity theft detectors. They created universal adversarial perturbations based on an electricity reading and its nearby readings. Compared to the prior attacks, the attacker significantly needs less transmit power to create misclassification, which reveals a fundamental flaw in DL-based solutions to communication systems. The authors in [190] propose variational auto-encoder-based adversarial attacks with the masqueraded malicious dataset to mislead dynamic pricing systems.

#### B. Trustworthy AI for power grid management functions and smart resilience in SG2

The best protection model against the AI-targeted attacks is to provide a comprehensive approach, from protecting data for AI training to improve a neural network's or AI model's robustness. Fig. 20 summarizes three fundamental defense approaches to maintain trustworthy AI: protect input data against data poisoning (enhance data quality), protect AI design models against adversarial attacks (model protection),

and protect running AI engines against model interference (output restoration). Each protection model is associated with a special stage: data preparation, AI development, and deployment. Since the attack methods are evolving, the defense methods are also expected to develop in parallel or beyond to keep up with the adversary. This subsection overviews defense strategies encompassing the optimization of model robustness, manipulation of model inputs, and assessment of the functional repercussions arising from neighboring weight interactions.

1) *Data quality enhancement*: Enhancing data quality is essential for developing trustworthy AI and preventing data poisoning attacks. High-quality data ensures that AI models are trained on accurate, relevant, and unbiased information, which improves their reliability and performance [78]. The basic idea behind data quality enhancement is to apply pre-processing strategies to the input examples to remove the influence of adversarial perturbation or outlier data points without changing the objects in the original picture of the examples. There are several methods for data quality enhancement: data sanitization (anomaly detection) and verified supply chain. For example, the authors in [213] introduced Structure-To-Signal Network (S2SNet), a protection mechanism based on gradient masking. The primary goal of this method is to convert category-related information into structural information, manage the information in the gradient of input data, and then encode the structural section utilized for classification and erase unrelated sections to remove adversarial perturbations. This method can be applied to the data layer with the feature extraction aspects or a part of the digital twins or AI-aided controllers in smart grids with interactive scheduling aspects. For verified supply chain, blockchain technology can be used to create a secure and transparent record of data provenance, ensuring that all data used for training models comes from verified and trustworthy sources [168].

Data masking/denoising and supply chain verification are also efficient approaches. For instance, in smart grids, personal data of consumers and payment bills can be masked to ensure no exposure of social security numbers and consumer home addresses while still allowing for the analysis of consumption patterns. Errors during the data preparation and collection should be avoided. The researchers [214] introduced a novel denoising approach, the so-called High-level Representation Guided Denoiser (HGD), to address the predicament of propagating residual errors. This approach uses the U-Net architecture to prevent mistakes from escalating in the input data. The HGD method outperforms an alternative approach when subjected to a limited subset of training data, thanks to the exclusive use of fine-tuning, a targeted mechanism to mitigate error amplification in the presence of adversarial examples. Another similar work is [215] where the authors propose to exploit continuous wavelet transform to build resilience models for mitigating adversarial attacks. Some other techniques are available in public outlier detection libraries [216]: Z-score, Isolation Forest, Variational Autoencoding Gaussian Mixture Model (VAEGMM), Prophet Time Series Outlier Detector.

2) *Model protection: optimize the model robustness and explainable AI*: There are many methods to optimize the learning model's robustness: model hardening, adversarial

training/certifiable training. For example, the authors in [217] introduced a new Bit Plane Feature Consistency (BPFC)-based technique. Their primary goal is to generate a general impression from features in the higher-bit planes and then fine-tune their predictions using the lower-bit planes. By enforcing consistency within the lower bit planes to enhance predictive accuracy across diverse quantified images, the efficiency of deep neural networks (DNN) in adversarial contexts attains higher performance relative to conventionally trained models. Nonetheless, a constraint inherent to this approach pertains to the careful selection of specific lower-bit planes to tune their predictions to align with the required robustness of the DNN model. In another work, Kannan et al. [218] offer a mixed-minibatch PGD (M-PGD) adversarial training approach that combines a logit pairing method with PGD adversarial training. M-PGD includes two pairing methods: (1) pairs a clean sample with an adversarial sample, and (2) pairs a clean sample with another clean sample. In SG2, the adversarial training could mean training AI models for electric demand or metering data that includes possible faults or anomalies.

For non-adversarial training approach, the authors in [193] introduce a causal structure learning model that recognizes the causal links in observational data and exploit the causal relationship between samples, outputs, and ground truth labels to limit the effect of adversarial attacks. Another prominent adversarial training technique is defensive distillation [219], which aims to reduce the effectiveness of adversarial samples on DNNs through a teacher-student learning process. This technique involves training a teacher model on benign data to produce soft probability distributions as targets. These soft targets are then used to train a student model, which helps the model generalize better and become more resistant to adversarial perturbations. The other studies in this category are [220], [221] where adversarial training anomaly detectors are robust against false data attacks and data poisoning. Another technique is certifiable training that refers to a set of techniques aimed at providing formal guarantees about the robustness of deep neural networks (DNNs) against adversarial attacks. Unlike traditional adversarial training methods that empirically improve robustness by training on adversarial examples, certifiable training focuses on creating models that can be mathematically proven to withstand certain types of perturbations within a specified range. For instance, the authors in [222] developed a method that uses convex relaxations to create an outer polytope that encloses all possible perturbations of the input. By optimizing this polytope during training, the model learns to maintain correct classifications within this bounded region, even processing with adversarial samples.

Besides adversarial training, ensemble ML models may increase resilience by pooling the findings of varied models to make better predictions. Because an attack that bypasses one model does not always bypass the others, ensembles can increase the learning's robustness against adversarial samples [223]. Finally, Explainable AI (XAI) can indeed play a crucial role in preventing adversarial attacks and enhancing the trustworthiness of AI systems. For SG2, explainability can assist operators in understanding why a certain decision was made, such as adjusting power distribution in response

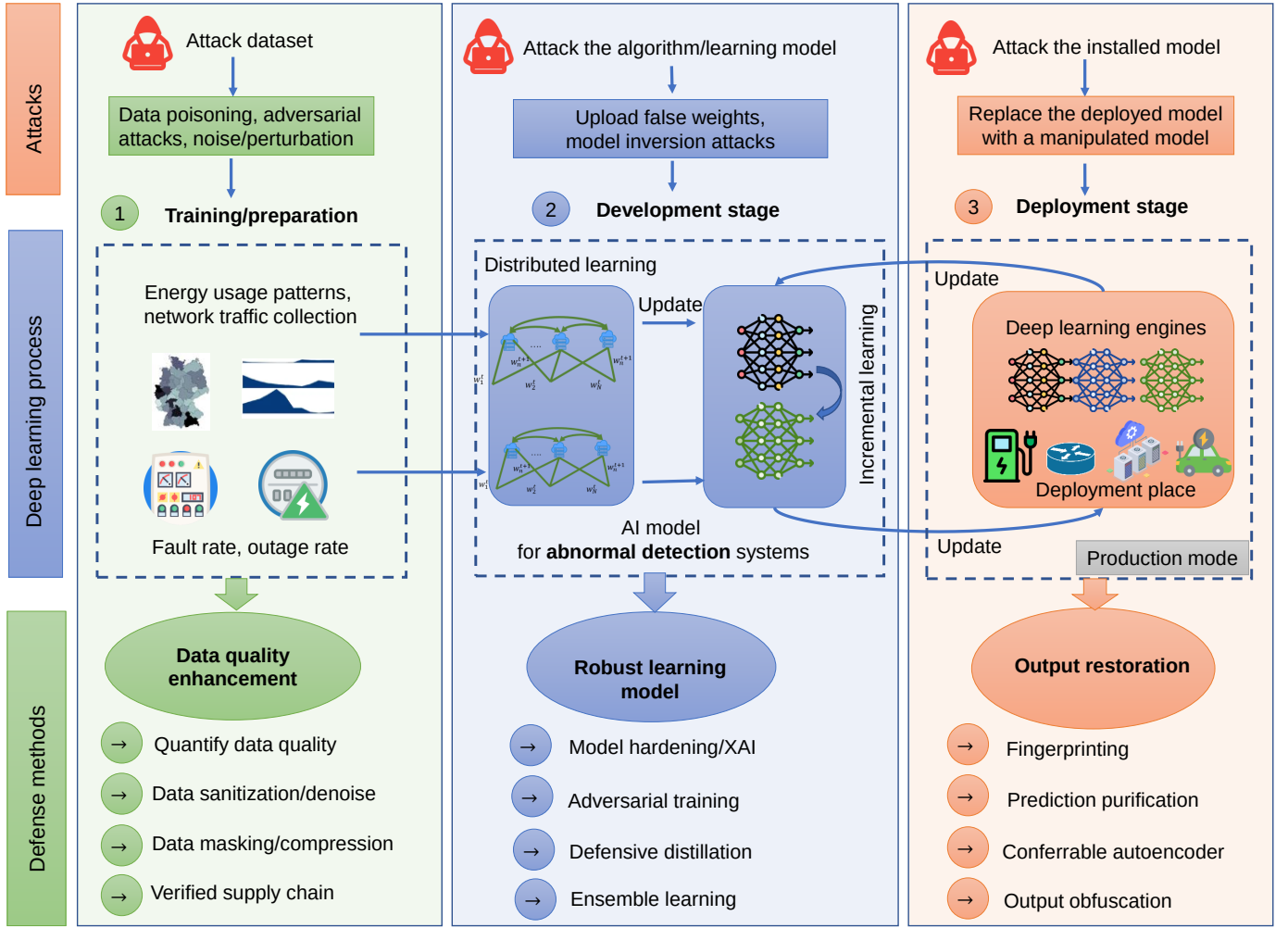


Fig. 20. Typical protection approaches against security attacks against AI-aided functions and energy management in SG2. In essence, there should have a comprehensive protection strategy for all stages, from input data, training, to deployment stage (partially referred to materials in our prior work [78]).

to demand fluctuations or charge/discharging decisions. This transparency can help detect and mitigate adversarial attacks by revealing unexpected or unusual model behaviors, thus enhancing the overall security and trust in SG2. For example, the authors in [224] summarized and emphasized that XAI methods contribute to building more robust and trustworthy AI systems by elucidating model behavior and improving human-AI interaction, particularly in US Defense Advanced Research Projects Agency (DARPA)'s programs. As a critical infrastructure to national security, in SG2, XAI should be prioritized in AI-powered management tasks to balance the trade-off between the reliability/trust of decision-making and the high accuracy of the AI technique.

3) *Output restoration: weight optimization:* Output restoration is a key technique that aims to prevent adversarial attacks by ensuring the reliability of system outputs through fingerprinting, prediction purification, and output obfuscation. For example, fingerprinting involves embedding unique, traceable markers within the output data of grid systems, allowing operators to detect unauthorized changes or anomalies. The authors in [225] proved a case to verify whether a suspected AI policy is an illegitimate copy of another (victim) policy. In SG2, by

embedding fingerprints in power usage reports, any tampering can be quickly identified and addressed. Prediction purification corrects outputs that are suspected of being influenced by adversarial inputs, particularly in model inversion and membership inference attacks [226]. For example, if an unexpected spike in energy demand is detected, purification methods can cross-verify with historical data and auxiliary models to filter out any potential malicious input. Output obfuscation adds controlled randomness to the outputs, such as slightly varying the reported energy usage data, making it difficult for attackers to predict and exploit the system's behavior [227]. These techniques collectively enhance the security and robustness of smart grid operations against adversarial threats.

### C. AI and Superintelligence for enhancing security protection models: intelligent IDS/IPS and abnormal detection

As summarized in previous sections, many conventional solutions, such as firewalls and intrusion detection systems, have been the main forces to prevent security attacks. AI makes such systems more powerful and intelligent. While older security measures (such as signature-based intrusion



detection) have been widely used, they have difficulty dealing with sophisticated threats in SG2 distributed energy resources and microgrids. Several studies [41], [44], [87] highlighted that AI will be a core player in enhancing the efficiency and accuracy of intrusion detection in smart battery storage management of SG2. AI can considerably assist in security defense by enhancing the detection engines' performance. For example, the authors in [228] advocated using deep reinforcement learning (DRL) to enhance secure peer-to-peer energy trading models. AI is a favored technique for detecting eavesdropping and DoS attacks in the network layer of the smart grid. With the large-scale learning, AI is particularly suitable for the following tasks:

- Abnormal behavior/intrusion detection in SG2, such as convolutional neural networks (CNNs)/region search to detect electricity theft [229]–[231], Spiking Neural Networks (SNN)/Temporal Failure Propagation Graph (TFPG) and federated learning to detect anomaly metering traffic and network intrusion activities [90], [160], [232] or in moving target defense (MTD) [233].
- Isolating attack traffic and unauthorized access in SG2 enabling technologies, such as unsupervised machine learning to detect DDoS attacks [234] or ensemble learning to detect intrusion traffic [235] in software-defined SG2 networks. Another method is to exploit neural networks to enhance false data detection against state-of-charge estimation in energy storage systems [44], [70], [89].
- Optimizing fault diagnosis and electrical power outage production in SG2, such as distribution transformer fault prediction using deep neural networks [236] and multi-data-source hybrid deep learning-based predictors to accurately predict energy usage [237].
- Price charging prediction and load balance scheduling, such as deep reinforcement learning-based pricing strategy for profit maximization [238] and smart isolated microgrids [239].

Inspired by recent progress in AI [240], AI for smart grid and renewable energy sources [9], [40], [41], [241], and related security [242], Fig. 21 summarizes our vision of AI promising models for cybersecurity, including for SG2, in the coming years. Given the impending arrival of Artificial General Intelligence (AGI) [243], AI models in the aforementioned applications should be able to self-learn from their surroundings and adjust their capability to deal with unknown inputs. AGI for smart grids is likely a milestone in applying AI to critical infrastructure. Given strengths in learning from extremely large inputs and automatically optimizing choices based on continuous feedback from the environment, distributed learning, multimodal and incremental learning, quantum machine learning, meta-learning/zero-shot learning, and the combination of DRL and GAN [244] will be top choices in this area. Similar to 6G, generative learning will be a crucial step toward the idea of “AI-building AI” or “security design by AI” vision [78]. However, the vision for AI and AGI popularity in SG2 raises new fears that the damage can be more catastrophic if AI-based functions are out of control due to adversarial attacks. For example, AGI can be used to

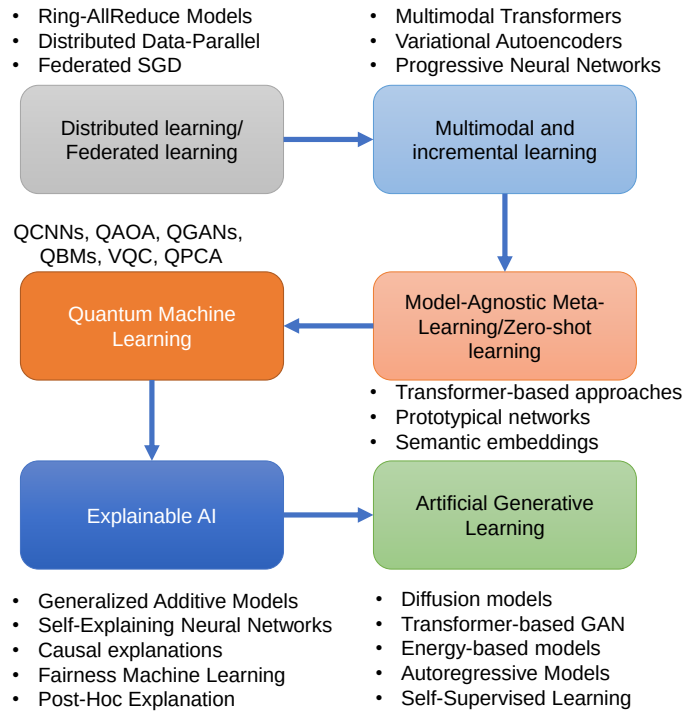


Fig. 21. Summary of current AI models and future vision for intelligent cybersecurity in SG2. As a result, the most dependable AI models will be favored for essential applications, and SG2 could be a primary objective. Several concepts in this vision have yet to be implemented fully as comprehensive solutions or security defense for power grid systems, e.g., quantum machine learning, AGI, explainable AI.

design a sophisticated phishing attack by exploiting a public information and reports of the companies. AGI can expose the grid's operational protocols and identifies critical nodes through careless questions from the employees.

Robust AI models with adversarial training and AGI safety and governance practices should be followed [240], [243]. Furthermore, current AI models require large-scale data and expensive hardware accelerators with Graphics Processing Units (GPU) for training. To balance net zero and environment-friendly goals, training models may require a new approach. As a result, distributed learning and federated learning, which can coordinate the learning process across millions of distributed devices (local agents) to improve the quality of the centralized learning model (global agents), will likely be the top candidates. Finally, the blackbox of how a deep learning model works under specific conditions has been a concern for applying AI in critical fields like smart grids. As explained earlier, explainable AI (XAI) models [242] have been a major target for continuing work in cyber trust to reduce the risk of AI making unjustified judgments, particularly in controversial decisions, such as turning off crucial electric gates for particular communities to save for others or cutting off the electricity to prevent AI out of control. AGI for designing explainable AI can be a promising research direction.

## VIII. LESSONS LEARNED AND DISCUSSIONS ON FUTURE RESEARCH

SG2 will be a significant upgrade of the conventional smart grid framework, which passed through decades of research. The upgrade will highlight the automation, efficiency, and intelligence. Three primary trends to realize these goals are (1) empowering AI into many core components of smart grids, (2) upgrading communication technology infrastructure, and (3) developing new energy sources and trading, e.g., DERs, BESS. However, new advancements create new areas of security risk. This section summarizes key lessons learned through the survey and highlights promising research directions.

### A. Lessons learned from the survey

From the survey, unified authentication, consistent security solutions and standards, and AI-aided security for SG2 emerging technologies such as new energy sources, electrified transportation, and P2P trading are the primary major upgrade targets and the central of ongoing research efforts. In summary, ten lessons learned from the review on security issues and defense methods for SG2 are as follows.

① *The transition from fossil energy to renewable energy era requires an evolution of digitizing management systems and enabling connectivity for all stages from energy generation, transmission, and distribution to consumption in SG2.* Meanwhile, the networkization comes up with cybersecurity concerns, e.g., cascading failures of connectivity, DDoS. Once SG2 elements are connected, data collection from each stage can provide rich information for optimizing the operations of the entire power grid, e.g., by exploiting the computing capability of cloud-based platforms and software-defined control methods. However, the interdependence of control and communication technologies creates a new route for the attackers to remotely access and sabotage the power grids, particularly in cyberwarfare.

② *The diversity of smart metering devices can mitigate the problem of supply chains but potentially cause security fragmentation due to their incompatibility in using security protection standards.* For example, the meter devices can be built based on NB-IoT, LoRa, WiFi, or Zigbee [18]. However, these communication technologies are incompatible with all security designs, let alone their own vulnerabilities. Maintaining consistency and compatibility for communications between consumers and energy distributors is challenging but should be carefully considered in modernizing a smart grid platform. Maintaining a reliable supply chain for vital materials in smart grid infrastructure is also an important issue, given that grid security implies national safety.

③ *The efficient authentication matter is one of the biggest challenges in SG2:* Given the diversity of communication technologies for smart metering and industrial control standards, the authentication mechanisms are expected to vary. However, the heterogeneity in the protection level (e.g., weak authentication in LoRAWAN vs strong authentication in 5G core networks) may pose risks for the entire connected smart grids. The weak security network segment can become a vehicle

to intrude on the core smart grid system. A unified authentication framework, e.g., via authentication, authorization, and accounting (AAA) services in a well-protected control center, can be a promising model. For example, unified authentication can significantly simplify the administration of user credentials in the era of multiple energy providers. The authentication server and protocols can be upgraded quickly for all devices.

④ *Unlike IT systems, holistic protection in SG2 requires both digital and physical aspects of security.* For example, firewalls/IDS/Encryption (digital) are to secure communications, while wall/fencing/shield cages/supply chains (physical) are to protect power lines and components against physical attacks (remote drones, jamming vehicles). Most recorded attacks in US and other countries are software-based (malware, ransomware). However, with the popularity of drones and autonomous vehicles, non-traditional attacks like suicide UAVs, jamming vehicles, or supply chain breakage have been new severe threats that cannot be ignored while building the protection schemes for SG2. In this way, building distributed microgrids and emergency stocks of critical grid components will be critical. Further, national regular drills/rigid testings of collaborative efforts among industry stakeholders in energy crisis scenarios are essential.

⑤ *Evolution for communication systems in smart grids often lags behind that for IT and mobile networks:* Given the importance of the energy systems in our society, technologies like networking and security protocols must usually be verified, particularly at stable and durable capacity, before they can be deployed in SG2 commercially. For example, while 5G technology is rapidly being deployed in mobile networks, its deployment in smart grids is delayed. Smart grids continue to rely on older 4G or even wired technologies [245] due to the extensive verification required to ensure 5G meets the high stability and reliability standards critical for energy systems, potentially slowing the adoption of advanced communication solutions in the energy sector. This slow-but-sure strategy has many advantages for maintaining stability and reliability in a critical field like SG2. However, the lengthy verification and certification processes can delay the deployment of new technologies. This can hinder innovation and the ability to implement cutting-edge solutions quickly, potentially slowing down progress in rolling out new technologies. Finally, networking and attack techniques evolve quickly, and when a defense and communication technology is thoroughly verified and certified, it may already be somewhat outdated. Striking the right balance between thorough testing and timely deployment is crucial. Recently, the security-as-a-service and move-near-cloud models seem to be getting much attention, given their advantage of enabling fast innovation.

⑥ *AI-aided solutions for applications and core functions in SG2 are still in development with much potential for exploration.* Unlike computer vision and natural language processing, machine learning and deep learning research for improving performance in the smart grid's core functions, e.g., energy distribution, outage prediction, energy usage forecasting, and related security matters, are just at the early research stage. As we summarized in Fig. 19, there are many functions at all three layers where AI can help. However, currently, there

are few research on this problem. The primary challenge is the lack of rigorous security datasets for key activities of SG2 and potential attacks, particularly in SG2 emerging technologies, e.g., electric vehicle charging. This is because of the concern for system safety or data leakage. Adversarial attacks and defense approaches against AI-powered energy management are still at the early research stage. Finally, the potential for using AI to enhance SG2 security is great, but it is not a magic wand that can solve all security challenges. Non-AI innovations, such as quantum-safe security protocols, continue to get significant industry support and are a part behind SG2's success.

⑦ *Information interoperability has emerged as a vital factor in securing the heterogeneous networks of SG2:* Energy distributors, energy generators, regulatory bodies, and technology providers often act in their interests, given the concern of secret trade and national security. However, cross-sector collaboration can help to identify emerging threats and vulnerabilities and enable the development of standardized security frameworks that can adapt to evolving challenges. The main challenge is that it is unclear how these sectors can share the information safely, given the privacy concerns and responsibility litigation. To address this, a secure API service can be created, where each sector has controlled access to specific data based on their role and privilege level. For instance, energy distributors can access threat intelligence data provided by cybersecurity firms, while regulatory bodies/policymakers can oversight compliance and incident reports without accessing sensitive operational details. This system is akin to Lawful Interception Services in 5G networks, where law enforcement agencies can access communication data through secure, predefined channels without compromising the privacy and security of the network as a whole.

⑧ *Blockchain can be the key player in securing communications and optimizing data exchange for peer-to-peer energy trading and distributed energy sources in SG2.* The goal is to maintain the trust among distributed components of smart grids along with privacy preservation. Indeed, there have been many blockchain-based studies [91], [94], [123], [166], [167], [172] for this purpose. However, such a vision is unlikely to be realized until blockchain's expensive computing and security flaws (e.g., 51% attacks, DAO attacks [178]) are addressed [179]. Another major challenge is the scalability of the blockchain. Imagine that the blockchain is deployed in managing energy transactions in a large urban area such as New York City. In such a setting, millions of households and businesses would participate in a decentralized energy market, trading electricity produced from renewable sources. A blockchain network, e.g., Ethereum, has a limit of transaction processing, can process about 7 to 15 transactions per second [64] only. This is small compared to the thousands of transactions per second required in a large-scale busy smart grid. Additionally, the heavy energy consumption of blockchain networks, especially those using proof-of-work-based consensus mechanisms, would require significant upgrades on high-performance computing infrastructure, which is costly to invest in. Further, the latency issues of transaction confirmation can be up to 10 minutes [64] which is not suitable for the real-

time processing needs of a smart grid. Furthermore, the storage requirements for storing a full ledger on every node would be massive, making it impossible for nodes to keep up with the fast development of transaction data. Currently there are several prominent solutions, such as sharding and layer-2 scaling protocols [64], their architecture is complicated and has yet to be tested for large-scale deployment. As a result, blockchain in its present form faces severe scaling issues for SG2, further studies on this matter will be an interesting topic.

⑨ *Firewalls, IDS, DMZ, will not lose their roles in SG2, but intelligence capabilities are mostly the most-looking-forward features.* For years, these platforms have established their reputation for safeguarding networks from many threats and network intrusions. They are still significant players in SG2 and beyond. However, in order to retain detection efficiency in a complex environment with various connection technologies, these legacy systems need substantial enhancements in both predictive and adversarial attack resistance capabilities. AI-aided IDS/IPS are then the central efforts for the coming years.

⑩ *Privacy concerns, mainly when energy systems collect and process consumer data, remain a pressing challenge.* For example, the AI system requires rich data collected from consumers (energy consumption/charging habits) and infrastructure (electricity load in each region) to optimize their prediction models. However, these data can be potentially leaked, probably by sloppy management or insider attacks. Striking the right balance between data utilization for optimization and preserving consumer privacy demands robust data anonymization, encryption, and consent mechanisms. Addressing these challenges requires close collaboration with regulatory bodies to develop and enforce policies that protect consumer data without hindering energy innovation.

In conclusion, reaching SG2 with full operation can be a long road ahead. Lessons learned from failures of the current security architecture along with security issues in new energy trading, new bidirectional relationship among distributed microgrids of SG2 highlight the need for comprehensive security frameworks tailored to the unique requirements of SG2, such as security interoperability for multiple energy sources, local power switch optimization for grid balancing, strong authentication mechanisms for decentralized microgrids, trustworthy AI for all AI-powered energy management systems, and enhancing well-proven technologies like IDS with AI capability.

## B. Open challenges and promising research directions

Given the ongoing transition towards SG2 at the early stage, there are several open challenges and promising research directions for future studies. We summarize several typical ones as follows.

① *AI-aided solutions with resilience from blackouts or sudden outages:* As we mentioned earlier, there is a lack of AI research on core functions of smart grids (e.g., slicing, outage prediction, intelligent energy distribution, network balancing, firewall/IDS), given the challenges of collecting datasets and simulating the real environment once the system is operating. The AI-targeted attacks and defense research in this field are even less. Rigorous datasets in this field,

particularly those that reflect the comprehensive activities or network traffic in metering, energy coordination, and disaster response, will be extremely welcomed by the research community. Without well-established datasets, enhancing the performance of AI-based solutions will be difficult. Besides, based on this baseline model, adversarial attacks can be further conducted to verify the resilience performance of AI models and then propose corresponding defense updates. Advanced AI models (as summarized in Fig. 21), such as few-shot learning/federated learning/multimodal incremental learning, can be another interesting topic to enhance AI capability on small data cases or privacy preservation.

② *Unified authentication for multiple microgrids*: The diversity of communication technologies for HAN/FAN (e.g., between the consumers and several energy providers) or distributed microgrids/energy storage technologies creates several different authentication techniques that may result in an inconsistency in accessing the system. A unified authentication can prevent this. For example, consumers may handle power service payments, car charging, battery swap, and P2P electricity trading with one identity and access point. Consumers can access different systems with one set of credentials via single sign-on (SSO), boosting convenience and minimizing password fatigue. However, the solution may need to consider many constraints, e.g., interoperability issues arise due to the varying network protocols and systems used by different grid operators, making seamless integration challenging. Secondly, the risk of a single point of failure increases, as a breach in the unified system could compromise the security of the entire grid. Additionally, the complexity of managing and synchronizing authentication data across multiple providers can lead to inefficiencies and vulnerabilities. The willingness of grid suppliers to share customer authentication information with competitors is a concern in business. Integrating EAP-TLS authentication to enable API calls and creating a uniform authentication system via these API calls might be a nice start.

③ *Developing custom 5G/6G private networks for SG2*: Custom 5G/6G private networks can revolutionize network infrastructure's security in SG2 significantly, given the reputation of solid security protection and fast innovation in cellular network technologies. While 5G/6G promises enhanced connectivity and security, the complexity and scale of these networks demand for distributed energy sources in large areas, making it challenging for communication operators to justify the private investment for different grid operators. Those high costs can be the initial investment on private base stations, network components/routers, and 5G-based smart meters for millions of consumers. Additionally, the ongoing operational expenses, such as network management and periodic upgrades, further strain the financial resources of power grid operators. However, in SG2, the practicality of 5G/6G private networks is enhanced due to the maturity of end-to-end network slicing technologies, which allows for the efficient building of private networks (under the same hardware infrastructure). For now, operators may still rely more realistic solutions by implementing radio slicing and VPN along with endpoint security on public communication networks that effectively balance cost, complexity, and security.

④ *Security defense for autonomous systems and green energy transactions*: The recent transition from gasoline cars to electric vehicles, automated cars, and new charging infrastructure creates a unique chance for software and network-based solutions to optimize the efficiency of data transmission as well as energy distribution. This will remind us of the importance of in-vehicle security protection techniques (abnormal behavior detection, thief charging detection) and cooperative verification (identity verification, Sybil attack detection) in well-connected smart vehicles and power grid infrastructure. However, this topic has not yet been well-explored.

⑤ *New hardware security techniques for preventing physical attacks*: Physical attacks are common in SG2, and the consequence can be a catastrophe if important transformers are victims. Besides physical shields, hardware security techniques, such as trusted execution environment (TEE), physical unclonable function (PUF), post-quantum cryptography, chip-to-chip authentication, and tamper-evident packaging, can be critical solutions that can be enhanced in the future. Further studies and prototypes on these topics are welcome.

⑥ *Enhancing privacy technologies with affordable costs*: Smart grids collect a vast amount of data about energy consumption and generation, and this data can be used to identify individuals and their activities. It is important to protect this data from unauthorized access and use, but doing so can be expensive. Promising solutions can be using secure multi-party computation (allows multiple parties to compute a function on their data without revealing their data to each other), differential privacy (adding noise to data without sacrificing its accuracy), or homomorphic encryption (allows computations to be performed on encrypted data) [78].

⑦ *Digital Twin for speeding up security research innovation for SG2*: Digital Twin (DT), a digital representation of a physical object, person, or process, contextualized in a digital version of its environment, can speed up security research innovation in SG2 by providing a safe and controlled environment for researchers to experiment and test new security solutions. DT can be used to simulate and analyze the behavior of the control systems, too. In the context of SG2, DTs can be used to create virtual representations of the grid's infrastructure, including its power plants, transmission lines, and distribution systems. These DTs can then be used to simulate various security threats and test the effectiveness of different security solutions. However, this research direction is still in its infancy [191].

⑧ *Supply chain security for microgrids*: As we mentioned earlier, the weakness in any part of the supply chain, e.g., shortage of electric transformers, could encourage the attackers to attack specific components (physical attacks or malware for the specific targets) to disable the grid operators. Given the growing fears of many trade wars, trade protectionism, and global conflicts, developing an efficient mechanism to monitor supply chain changes and suspicious activity to avoid relying on a single supplier for any critical component or service can help mitigate the risks and enhance the resilience. Besides technical solutions, building a list of trusted suppliers through vendor risk management protocols and trade initiatives can be a very helpful approach, particularly with microgrid operators.



## IX. CONCLUSION

SG2 is still on the way to shaping its unique characteristics vs the current smart grid model (i.e., SG1). Through lessons learned from the limitations of SG1 security implementations and the remaining technical challenges in SG1 infrastructure, this study highlighted that security protection and many techniques in SG2 lag behind the fast innovation of overall networking and security technologies. Further, the market of core security functions (e.g., authentication/firewalls) is rich but fragmented in standards. Given the high dependence on communication networks to connect distributed microgrids in SG2, potential cascading failures of connectivity poses many severe security concerns. Besides suffering conventional attacks such as ransomware/DoS attacks, there are a growing number of new threats in full operation SG2, such as physical attacks against substations and energy storage or adversarial attacks against intelligent AI-empowered energy management. This work provided an overview of key security threats and prospective solution comparisons in the vision of three stakeholders in SG2: power grid operator, communication network provider, and consumer. Security threats and protection models for emerging technologies that are expected to be deployed widely in SG2 are also presented. We found that many advanced security models (e.g., network slicing, scalable blockchain, post-quantum encryption, trustworthy AI for SG2) are still in the early stage of the research-prototyping loop. Without these new features, SG2 likely relies on the existing platforms, which are mostly outdated and consist of security vulnerabilities. Finally, we believe holistic protection in SG2 requires collaborative efforts to implement the aforementioned protection and power switch optimization models from many stakeholders, policymakers, cybersecurity experts, and standardization bodies. Further, information interoperability has emerged as a vital factor in securing the heterogeneous networks of SG2.

## X. ACKNOWLEDGEMENTS

The authors would like to thank anonymous reviewers for their helpful comments. The authors also thank the Energy Administration of the Ministry of Economic Affairs in Taiwan for its support. This work is also sponsored by the National Science Technology Council under Grant No 111-2222-E-194-007-MY2, 112-2221-E-194-017-MY3, and in part by the Advanced Institute of Manufacturing with High-Tech Innovations (AIM-HI) through the Featured Areas Research Center Program of the Ministry of Education.

## REFERENCES

- [1] Nanyang Technological University, "Grid 2.0," 2023.
- [2] P. Porambage, M. Liyanage, C. Yapa, and C. De Alwis, *Smart Grid 2.0 Security*, pp. 167–184, 2023.
- [3] F. Orecchini and A. Santiangeli, "Beyond smart grids – the need of intelligent energy networks for a higher global efficiency through energy vectors integration," *International Journal of Hydrogen Energy*, vol. 36, no. 13, pp. 8126–8133, 2011.
- [4] Q. Sun, H. Li, Z. Ma, C. Wang, J. Campillo, Q. Zhang, F. Wallin, and J. Guo, "A comprehensive review of smart energy meters in intelligent energy networks," *IEEE Internet of Things Journal*, vol. 3, no. 4, pp. 464–479, 2016.
- [5] Z. Ma, J. Xie, H. Li, Q. Sun, Z. Si, J. Zhang, and J. Guo, "The role of data analysis in the development of intelligent energy networks," *IEEE Network*, vol. 31, no. 5, pp. 88–95, 2017.
- [6] M. Ali, K. Prakash, M. A. Hossain, and H. R. Pota, "Intelligent energy management: Evolving developments, current challenges, and research directions for sustainable future," *Journal of Cleaner Production*, vol. 314, p. 127904, 2021.
- [7] I. Lütkebohle, "Ransomware attacks in the energy industry," <https://www.cdw.com/content/cdw/en/articles/security/ransomware-attacks-energy-industry.html>, 2022. [Online; accessed 19-June-2024].
- [8] Y. Yan, Y. Qian, H. Sharif, and D. Tipper, "A survey on cyber security for smart grid communications," *IEEE Communications Surveys and Tutorials*, vol. 14, no. 4, pp. 998–1010, 2012.
- [9] B. K. Bose, "Artificial intelligence techniques in smart grid and renewable energy systems—some example applications," *Proceedings of the IEEE*, vol. 105, no. 11, pp. 2262–2273, 2017.
- [10] A. Zibaeirad, F. Koleini, S. Bi, T. Hou, and T. Wang, "A comprehensive survey on the security of smart grid: Challenges, mitigations, and future research opportunities," 2024.
- [11] A. Ayub Khan, A. Ali Laghari, M. Rashid, H. Li, A. Rehman Javed, and T. Reddy Gadekallu, "Artificial intelligence and blockchain technology for secure smart grid and power distribution automation: A state-of-the-art review," *Sustainable Energy Technologies and Assessments*, vol. 57, p. 103282, 2023.
- [12] N. Komninos, E. Philippou, and A. Pitsillides, "Survey in smart grid and smart home security: Issues, challenges and countermeasures," *IEEE Communications Surveys and Tutorials*, vol. 16, no. 4, pp. 1933–1954, 2014.
- [13] P. Kumar, Y. Lin, G. Bai, A. Paverd, J. S. Dong, and A. Martin, "Smart grid metering networks: A survey on security, privacy and open research issues," *IEEE Communications Surveys and Tutorials*, vol. 21, no. 3, pp. 2886–2927, 2019.
- [14] M. Z. Gunduz and R. Das, "Cyber-security on smart grid: Threats and potential solutions," *Computer Networks*, vol. 169, p. 107094, 2020.
- [15] J. Sakhnini, H. Karimipour, A. Dehghantaha, R. M. Parizi, and G. Srivastava, "Security aspects of internet of things aided smart grids: A bibliometric survey," *Internet of Things*, vol. 14, p. 100111, 2021.
- [16] N. Ogino, "Robust design of interdependent networks considering intra-network support flow," *IEEE Transactions on Network and Service Management*, vol. 20, no. 4, pp. 4170–4185, 2023.
- [17] M. Amin, F. F. M. El-Sousy, G. A. A. Aziz, K. Gaber, and O. A. Mohammed, "Cps attacks mitigation approaches on power electronic systems with security challenges for smart grid applications: A review," *IEEE Access*, vol. 9, pp. 38571–38601, 2021.
- [18] T. T. Khoei, H. O. Slimane, and N. Kaabouch, "Cyber-security of smart grids: Attacks, detection, countermeasure techniques, and future directions," *Communications and Network*, vol. 14, pp. 119–170, 2022.
- [19] R. D. Trevizan, J. Obert, V. De Angelis, T. A. Nguyen, V. S. Rao, and B. R. Chalamala, "Cyberphysical security of grid battery energy storage systems," *IEEE Access*, vol. 10, pp. 59675–59722, 2022.
- [20] Y. Li and J. Yan, "Cybersecurity of smart inverters in the smart grid: A survey," *IEEE Transactions on Power Electronics*, vol. 38, no. 2, pp. 2364–2383, 2023.
- [21] M. K. Hasan, A. A. Habib, Z. Shukur, F. Ibrahim, S. Islam, and M. A. Razzaque, "Review on cyber-physical and cyber-security system in smart grid: Standards, protocols, constraints, and recommendations," *Journal of Network and Computer Applications*, vol. 209, p. 103540, 2023.
- [22] S. Vahidi, M. Ghafouri, M. Au, M. Kassouf, A. Mohammadi, and M. Debbabi, "Security of wide-area monitoring, protection, and control (wampac) systems of the smart grid: A survey on challenges and opportunities," *IEEE Communications Surveys and Tutorials*, vol. 25, no. 2, pp. 1294–1335, 2023.
- [23] M. W. Khan, G. Li, K. Wang, M. Numan, L. Xiong, and M. A. Khan, "Optimal control and communication strategies in multi-energy generation grid," *IEEE Communications Surveys & Tutorials*, vol. 25, no. 4, pp. 2599–2653, 2023.
- [24] A. Ghasemi and H. de Meer, "Robustness of interdependent power grid and communication networks to cascading failures," *IEEE Transactions on Network Science and Engineering*, vol. 10, no. 4, pp. 1919–1930, 2023.
- [25] X. Liu, B. Zhang, B. Chen, A. Aved, and D. Jin, "Towards optimal and executable distribution grid restoration planning with a fine-grained power-communication interdependency model," *IEEE Transactions on Smart Grid*, vol. 13, no. 3, pp. 1911–1922, 2022.
- [26] R. Chen, "Best practices to enhance cybersecurity in energy storage systems," <https://www.automation.com/en-us/articles/december-2022/>

- enhance-cybersecurity-energy-storage-systems, 2022. [Online; accessed 19-June-2024].
- [27] IEC, "Cyber security and resilience guidelines for the smart energy operational environment." [https://syc-se.iec.ch/wp-content/uploads/2021/01/IEC\\_SyC-SE\\_Cyber\\_Security\\_and\\_Resilience\\_Guidelines\\_for\\_Smart\\_Energy-1.pdf](https://syc-se.iec.ch/wp-content/uploads/2021/01/IEC_SyC-SE_Cyber_Security_and_Resilience_Guidelines_for_Smart_Energy-1.pdf), 2022. [Online; accessed 19-June-2024].
  - [28] N. I. of Standards and Technology, "Guidelines for Smart Grid Cybersecurity." <https://nvlpubs.nist.gov/nistpubs/ir/2014/NIST.IR.7628r1.pdf>, 2014. [Online; accessed 2-November-2023].
  - [29] E. U. A. for Cybersecurity (ENISA), "ENISA Smart Grid Security Recommendations." <https://www.enisa.europa.eu/publications/ENISA-smart-grid-security-recommendations/@@download/fullReport>, 2012. [Online; accessed 2-November-2023].
  - [30] J. Ostergaard, C. Ziras, H. W. Bindner, J. Kazempour, M. Marinelli, P. Markussen, S. H. Rosted, and J. S. Christensen, "Energy security through demand-side flexibility: The case of denmark," *IEEE Power and Energy Magazine*, vol. 19, no. 2, pp. 46–55, 2021.
  - [31] J. Kliem and D. McGinn, "Ingredients for a microgrid at u.s. department of defense installations," *IEEE Electrification Magazine*, vol. 8, no. 4, pp. 6–120, 2020.
  - [32] Office of Cybersecurity, "Cybersecurity Research, Development, and Demonstration (RDandD) for Energy Delivery Systems." <https://www.energy.gov/ceser/cybersecurity-research-development-and-demonstration-rdd-energy-delivery-systems>, 2022. [Online; accessed 19-June-2024].
  - [33] Office of Energy Efficiency and Renewable Energy, Department of Energy, "Solar Futures Study." <https://www.energy.gov/eere/solar/solar-futures-study>, 2023. [Online; accessed 19-June-2024].
  - [34] House of Commons Environmental Audit Committee, "Building to net zero: costing carbon in construction – report summary," *Recommendations to Government*, 2022.
  - [35] International Energy Agency, "An energy sector roadmap to carbon neutrality in china," *International Energy Agency Report*, 2021.
  - [36] India Ministry of Science and Technology, "Net zero emissions target," *Report*, 2023.
  - [37] DESERTEC, "Super grid." [https://en.wikipedia.org/wiki/Super\\_grid](https://en.wikipedia.org/wiki/Super_grid), 2010. [Online; accessed 19-June-2024].
  - [38] D. E. Olivares, A. Mehrizi-Sani, A. H. Etemadi, C. A. Cañizares, R. Iravani, M. Kazerani, A. H. Hajimiragha, O. Gomis-Bellmunt, M. Saeedifard, R. Palma-Behnke, G. A. Jiménez-Estévez, and N. D. Hatziargyriou, "Trends in microgrid control," *IEEE Transactions on Smart Grid*, vol. 5, no. 4, pp. 1905–1919, 2014.
  - [39] A. M. Jasim, B. H. Jasim, S. Mohseni, and A. C. Brent, "Consensus-based dispatch optimization of a microgrid considering meta-heuristic-based demand response scheduling and network packet loss characterization," *Energy and AI*, vol. 11, p. 100212, 2023.
  - [40] R. Machlev, L. Heistrene, M. Perl, K. Levy, J. Belikov, S. Mannor, and Y. Levron, "Explainable artificial intelligence (xai) techniques for energy and power systems: Review, challenges and opportunities," *Energy and AI*, vol. 9, p. 100169, 2022.
  - [41] Y. Zhou, "Artificial intelligence in renewable systems for transformation towards intelligent buildings," *Energy and AI*, vol. 10, p. 100182, 2022.
  - [42] M. Lehna, J. Viebahn, A. Marot, S. Tomforde, and C. Scholz, "Managing power grids through topology actions: A comparative study between advanced rule-based and reinforcement learning agents," *Energy and AI*, vol. 14, p. 100276, 2023.
  - [43] Z. Ye, Y. Gao, and N. Yu, "Learning to operate an electric vehicle charging station considering vehicle-grid integration," *IEEE Transactions on Smart Grid*, vol. 13, no. 4, pp. 3038–3048, 2022.
  - [44] N. Kharlamova, S. Hashemi, and C. Træholt, "Data-driven approaches for cyber defense of battery energy storage systems," *Energy and AI*, vol. 5, p. 100095, 2021.
  - [45] Y.-W. Chen and J. M. Chang, "Emaas: Cloud-based energy management service for distributed renewable energy integration," *IEEE Transactions on Smart Grid*, vol. 6, no. 6, pp. 2816–2824, 2015.
  - [46] Y.-W. Chen and J. M. Chang, "Fair demand response with electric vehicles for the cloud based energy management service," *IEEE Transactions on Smart Grid*, vol. 9, no. 1, pp. 458–468, 2018.
  - [47] CIGRE, "Application of 5G Technology to Smart Grids." <https://www.cigre.org/article/GB/application-of-5g-technology-to-smart-grids>, 2022. [Online; accessed 2-November-2023].
  - [48] S. Das, P. Acharjee, and A. Bhattacharya, "Charging scheduling of electric vehicle incorporating grid-to-vehicle and vehicle-to-grid technology considering in smart grid," *IEEE Transactions on Industry Applications*, vol. 57, no. 2, pp. 1688–1702, 2021.
  - [49] M. Tariq, M. Ali, F. Naeem, and H. V. Poor, "Vulnerability assessment of 6g-enabled smart grid cyber-physical systems," *IEEE Internet of Things Journal*, vol. 8, no. 7, pp. 5468–5475, 2021.
  - [50] R. Satter and C. Bing, "US pharmacy outage triggered by 'Blackcat' ransomware." <https://www.reuters.com/technology/cybersecurity/cyber-security-outage-change-healthcare-continues-sixth-straight-day-2024-02-26/>, 2023. [Online; accessed 19-June-2024].
  - [51] C. Bruno, L. Guidi, A. Lorite-Espejo, and D. Pestonesi, "Assessing a potential cyberattack on the italian electric system," *IEEE Security and Privacy*, vol. 13, no. 5, pp. 42–51, 2015.
  - [52] E-ISAC and SANS, "Analysis of the Cyber Attack on the Ukrainian Power Grid." <https://nsarchive.gwu.edu/sites/default/files/documents/3891751/SANS-and-Electricity-Information-Sharing-and.pdf>, 2016. [Online; accessed 19-June-2024].
  - [53] P. Hines, J. Apt, and S. Talukdar, "Trends in the history of large blackouts in the united states," in *2008 IEEE Power and Energy Society General Meeting - Conversion and Delivery of Electrical Energy in the 21st Century*, pp. 1–8, 2008.
  - [54] S. N. Islam, Z. Baig, and S. Zeadally, "Physical layer security for the smart grid: Vulnerabilities, threats, and countermeasures," *IEEE Transactions on Industrial Informatics*, vol. 15, no. 12, pp. 6522–6530, 2019.
  - [55] V. Havlena, P. Matoušek, O. Ryšavý, and L. Holík, "Accurate automata-based detection of cyber threats in smart grid communication," *IEEE Transactions on Smart Grid*, vol. 14, no. 3, pp. 2352–2366, 2023.
  - [56] Clayton Clem, "TVA Builds Resilience." <https://www.tdworld.com/grid-innovations/transmission/article/20970983/tva-builds-resilience>, 2018. [Online; accessed 19-June-2024].
  - [57] M. Motyka, J. Thomson, J. Nagdeo, and K. Hardin, "Electric power supply chains: Achieving security, sustainability, and resilience." <https://www2.deloitte.com/xe/en/insights/industry/power-and-utilities/supply-chain-resilience-electric-power-sector.html>, 2022. [Online; accessed 19-June-2024].
  - [58] Wikipedia, "Metcalfe sniper attack." [https://en.wikipedia.org/wiki/Metcalfe\\_sniper\\_attack](https://en.wikipedia.org/wiki/Metcalfe_sniper_attack), 2013. [Online; accessed 19-June-2024].
  - [59] I. Ilascu, "Power company Enel Group suffers Snake Ransomware attack." <https://www.bleepingcomputer.com/news/security/power-company-enel-group-suffers-snake-ransomware-attack/>, 2020. [Online; accessed 19-June-2024].
  - [60] Y. Wang, C. Deng, Y. Liu, and Z. Wei, "A cyber-resilient control approach for islanded microgrids under hybrid attacks," *International Journal of Electrical Power and Energy Systems*, vol. 147, p. 108889, 2023.
  - [61] Z. Yang, S. Zhang, C.-W. Ten, T. Liu, X. Pang, and H. Sun, "Implementation of risk-aggregated substation testbed using generative adversarial networks," *IEEE Transactions on Smart Grid*, vol. 14, no. 1, pp. 677–689, 2023.
  - [62] F. C. I. Section, "Copper Thefts Threaten U.S. Critical Infrastructure." <https://www.fbi.gov/stats-services/publications/copper-thefts>, 2022. [Online; accessed 19-June-2024].
  - [63] Department of Energy, "Electric Grid Security and Resilience." <https://www.nrel.gov/grid/security-resilience.html>, 2017. [Online; accessed 19-June-2024].
  - [64] Y. Li, J. Wang, and H. Zhang, "A survey of state-of-the-art sharding blockchains: Models, components, and attack surfaces," *Journal of Network and Computer Applications*, vol. 217, p. 103686, 2023.
  - [65] BCE, "Bell files another lawsuit against accused copper thief to help deter network vandalism." <https://www.bce.ca/news-and-media/releases/show/Bell-files-another-lawsuit-against-accused-copper-thief-to-help-deter-network-vandalism>, 2023. [Online; accessed 19-June-2024].
  - [66] CNBC, "An alarmingly simple cyberattack hit electrical systems serving LA and Salt Lake, but power never went down." <https://www.cnbc.com/2019/05/02/ddos-attack-caused-interruptions-in-power-system-operations-doe.html>, 2019. [Online; accessed 19-June-2024].
  - [67] Forbes, "Hackers hit French firm Ingerop stealing 65 GB data relating to nuclear power plants." <https://cyware.com/news/hackers-hit-french-firm-ingerop-stealing-65-gb-data-relating-to-nuclear-power-plants-f193b9ba/>, 2018. [Online; accessed 19-June-2024].
  - [68] BBC, "Ransomware hits johannesburg electricity supply." <https://www.bbc.com/news/technology-49125853>, 2019. [Online; accessed 19-June-2024].
  - [69] X. Luo, J. He, X. Wang, Y. Zhang, and X. Guan, "Resilient defense of false data injection attacks in smart grids via virtual hidden networks," *IEEE Internet of Things Journal*, vol. 10, no. 7, pp. 6474–6490, 2023.

- [70] A. A. Habib, M. K. Hasan, A. Alkhayyat, S. Islam, R. Sharma, and L. M. Alkwai, "False data injection attack in smart grid cyber physical system: Issues, challenges, and future direction," *Computers and Electrical Engineering*, vol. 107, p. 108638, 2023.
- [71] Huawei, "5G Network Slicing Enabling the Smart Grid." <https://www-file.huawei.com/-/media/corporate/pdf/news/5g-network-slicing-enabling-the-smart-grid.pdf>, 2022. [Online; accessed 2-November-2023].
- [72] 3GPP.SA3, "Study on security aspects of 5g network slicing management," *3GPP TR 33.811 V15.0.0*, 2018.
- [73] CEN-CENELEC-ETSI, "Smart Grid Information Security." [https://www.cenelec.eu/media/CEN-CENELEC/AreasOfWork/CEN-CENELEC\\_Topics/Smart%20Grids%20and%20Meters/Smart%20Grids/security\\_smartgrids.pdf](https://www.cenelec.eu/media/CEN-CENELEC/AreasOfWork/CEN-CENELEC_Topics/Smart%20Grids%20and%20Meters/Smart%20Grids/security_smartgrids.pdf), 2012. [Online; accessed 2-November-2023].
- [74] H. Zhang, Y. Liu, G. Zou, Z. Jiang, W. Huang, and K. Zhou, "Smart substation vlan configuration method based on improved search tree," in *2024 9th Asia Conference on Power and Electrical Engineering (ACPEE)*, pp. 2290–2294, 2024.
- [75] A. S. Mohamed, M. F. M. Arani, A. A. Jahromi, and D. Kundur, "False data injection attacks against synchronization systems in microgrids," *IEEE Transactions on Smart Grid*, vol. 12, no. 5, pp. 4471–4483, 2021.
- [76] 3GPP.SA3, "Technical specification group services and system aspects: security architecture and procedures for 5g system," *3GPP TS 33.501 V16.4.0*, 2020.
- [77] ETSI, "Etsi ts 133.501 v15.2.0, security architecture and procedures for 5g system," *Technical Specification Group Services and System Aspects*, 2018.
- [78] V.-L. Nguyen, P.-C. Lin, B.-C. Cheng, R.-H. Hwang, and Y.-D. Lin, "Security and privacy for 6g: A survey on prospective technologies and challenges," *IEEE Communications Surveys and Tutorials*, vol. 23, no. 4, pp. 2384–2428, 2021.
- [79] Z. Zheng, Y. Yang, X. Niu, H.-N. Dai, and Y. Zhou, "Wide and deep convolutional neural networks for electricity-theft detection to secure smart grids," *IEEE Transactions on Industrial Informatics*, vol. 14, no. 4, pp. 1606–1615, 2018.
- [80] Texas Instruments, "Smart metering tampering attacks." <https://www.ti.com/video/5396041347001>, 2017. [Online; accessed 19-June-2024].
- [81] M. Shokry, A. I. Awad, M. K. Abd-Ellah, and A. A. Khalaf, "Systematic survey of advanced metering infrastructure security: Vulnerabilities, attacks, countermeasures, and future vision," *Future Generation Computer Systems*, vol. 136, pp. 358–377, 2022.
- [82] P. Mall, R. Amin, A. K. Das, M. T. Leung, and K.-K. R. Choo, "Puf-based authentication and key agreement protocols for iot, wsns, and smart grids: A comprehensive survey," *IEEE Internet of Things Journal*, vol. 9, no. 11, pp. 8205–8228, 2022.
- [83] B. Bera, S. Saha, A. K. Das, and A. V. Vasilakos, "Designing blockchain-based access control protocol in iot-enabled smart-grid system," *IEEE Internet of Things Journal*, vol. 8, no. 7, pp. 5744–5761, 2021.
- [84] T. T. Khoei, H. O. Slimane, and N. Kaabouch, "A comprehensive survey on the cyber-security of smart grids: Cyber-attacks, detection, countermeasure techniques, and future directions," 2022.
- [85] I. Zografopoulos, J. Ospina, X. Liu, and C. Konstantinou, "Cyber-physical energy systems security: Threat modeling, risk assessment, resources, metrics, and case studies," *IEEE Access*, vol. 9, pp. 29775–29818, 2021.
- [86] E. Kovacs, "Industroyer ics malware linked to ukraine power grid attack." <https://www.securityweek.com/industroyer-ics-malware-linked-ukraine-power-grid-attack/>, 2017. [Online; accessed 19-June-2024].
- [87] J. Li, M. S. Herdem, J. Nathwani, and J. Z. Wen, "Methods and applications for artificial intelligence, big data, internet of things, and blockchain in smart energy management," *Energy and AI*, vol. 11, p. 100208, 2023.
- [88] M. Bishop, "Hypertext Transfer Protocol Version 3 (HTTP/3)," Internet-Draft draft-ietf-quic-http-34, Internet Engineering Task Force, 2021.
- [89] P. Zhuang and H. Liang, "False data injection attacks against state-of-charge estimation of battery energy storage systems in smart distribution networks," *IEEE Transactions on Smart Grid*, vol. 12, no. 3, pp. 2566–2577, 2021.
- [90] J. Hong and C.-C. Liu, "Intelligent electronic devices with collaborative intrusion detection systems," *IEEE Transactions on Smart Grid*, vol. 10, no. 1, pp. 271–281, 2019.
- [91] K. Park, J. Lee, A. K. Das, and Y. Park, "Bpps: blockchain-enabled privacy-preserving scheme for demand-response management in smart grid environments," *IEEE Transactions on Dependable and Secure Computing*, vol. 20, no. 2, pp. 1719–1729, 2023.
- [92] J. B. Leite and J. R. S. Mantovani, "Development of a self-healing strategy with multiagent systems for distribution networks," *IEEE Transactions on Smart Grid*, vol. 8, no. 5, pp. 2198–2206, 2017.
- [93] M. Nazemi, P. Dehghanian, X. Lu, and C. Chen, "Uncertainty-aware deployment of mobile energy storage systems for distribution grid resilience," *IEEE Transactions on Smart Grid*, vol. 12, no. 4, pp. 3200–3214, 2021.
- [94] M. U. Hassan, M. H. Rehmani, and J. Chen, "Optimizing blockchain based smart grid auctions: A green revolution," *IEEE Transactions on Green Communications and Networking*, vol. 6, no. 1, pp. 462–471, 2022.
- [95] M. Parandehgheibi and E. Modiano, "Robustness of interdependent networks: The case of communication networks and the power grid," in *2013 IEEE Global Communications Conference (GLOBECOM)*, pp. 2164–2169, 2013.
- [96] N. N. Tran, H. R. Pota, Q. N. Tran, and J. Hu, "Designing constraint-based false data-injection attacks against the unbalanced distribution smart grids," *IEEE Internet of Things Journal*, vol. 8, no. 11, pp. 9422–9435, 2021.
- [97] I. Zografopoulos, N. D. Hatzigiorgiou, and C. Konstantinou, "Distributed energy resources cybersecurity outlook: Vulnerabilities, attacks, impacts, and mitigations," *IEEE Systems Journal*, pp. 1–15, 2023.
- [98] M. Alshowkan, P. G. Evans, M. Starke, D. Earl, and N. A. Peters, "Authentication of smart grid communications using quantum key distribution," *Scientific Reports*, no. 12, 2022.
- [99] H. M. S. Badar, S. Qadri, S. Shamshad, M. F. Ayub, K. Mahmood, and N. Kumar, "An identity based authentication protocol for smart grid environment using physical uncloneable function," *IEEE Transactions on Smart Grid*, vol. 12, no. 5, pp. 4426–4434, 2021.
- [100] IEEE, "IEEE Standard Cybersecurity Requirements for Substation Automation, Protection, and Control Systems." <https://standards.ieee.org/ieee/C37.240/5029/>, 2022. [Online; accessed 11-November-2023].
- [101] IEC, "Standard defining communication protocols for intelligent electronic devices at electrical substations." [https://en.wikipedia.org/wiki/IEC\\_61850](https://en.wikipedia.org/wiki/IEC_61850), 2016. [Online; accessed 2-November-2023].
- [102] IEC, "Standards for for implementing and maintaining electronically secure industrial automation and control systems." <https://www.isa.org/standards-and-publications/isa-standards/isa-iec-62443-series-of-standards>, 2022. [Online; accessed 2-November-2023].
- [103] IEC, "Standard for information security management systems." <https://www.iso.org/standard/27001>, 2022. [Online; accessed 2-November-2023].
- [104] ISO/IEC, "Evaluation criteria for IT security." <https://www.iso.org/standard/72891.html>, 2022. [Online; accessed 2-November-2023].
- [105] ISO, "Vehicle to grid communication interface." <https://www.iso.org/standard/77845.html>, 2022. [Online; accessed 19-June-2024].
- [106] IEEE, "IEEE Standard for Intelligent Electronic Devices Cybersecurity Capabilities." <https://standards.ieee.org/ieee/1686/7207/>, 2022. [Online; accessed 19-June-2024].
- [107] IEEE, "Guide for Smart Grid Interoperability of Energy Technology and Information Technology Operation with the Electric Power System (EPS), End-Use Applications, and Loads." <https://standards.ieee.org/ieee/2030/11367/>, 2023. [Online; accessed 2-November-2023].
- [108] Y. Yu, J. Yang, and B. Chen, "The smart grids in china—a review," *Energies*, vol. 5, no. 5, pp. 1321–1338, 2012.
- [109] METI, "Cybersecurity Guidelines for Energy Resource." [https://www.enecho.meti.go.jp/en/category/vpp\\_dr/data/cybersecurity\\_guidelines\\_for\\_erab.pdf](https://www.enecho.meti.go.jp/en/category/vpp_dr/data/cybersecurity_guidelines_for_erab.pdf), 2019. [Online; accessed 19-June-2024].
- [110] JISEC, "IT Security Evaluation and Certification." <https://www.ipa.go.jp/en/security/jisec/index.html>, 2023. [Online; accessed 19-June-2024].
- [111] NIST, "Guide to Industrial Control Systems (ICS) Security." <https://csrc.nist.gov/pubs/sp/800/82/r2/final>, 2015. [Online; accessed 19-June-2024].
- [112] Central Electricity Regulatory Commission, "Cea (cyber security in power sector) guidelines," *Report*, 2021.
- [113] Y. Shu and Y. Tang, "Analysis and recommendations for the adaptability of china's power system security and stability relevant standards," *CSEE Journal of Power and Energy Systems*, vol. 3, no. 4, pp. 334–339, 2017.
- [114] B. Moussa, C. Robillard, A. Zugenmaier, M. Kassouf, M. Debbabi, and C. Assi, "Securing the precision time protocol (ptp) against fake

- timestamps,” *IEEE Communications Letters*, vol. 23, no. 2, pp. 278–281, 2019.
- [115] S. Vahidi, A. Amini, M. Ghafouri, M. Au, A. Mohammadi, and M. Debbabi, “Resilient event-triggered observer-based periodic wide-area control for oscillation damping in wampac systems under time synchronization attacks,” *IEEE Transactions on Industrial Informatics*, vol. 20, no. 7, pp. 9172–9185, 2024.
- [116] U.S.-Canada Power System Outage Task Force, “Final Report on the August 14, 2003 Blackout in the United States and Canada,” <https://www3.epa.gov/region1/npdes/merrimackstation/pdfs/ar/AR-1165.pdf>, 2004. [Online; accessed 19-June-2024].
- [117] R. Smith, “Assault on california power station raises alarm on potential for terrorism,” <https://www.wsj.com/articles/assault-on-california-power-station-raises-alarm-on-potential-for-terrorism-1391570879>, 2014. [Online; accessed 19-June-2024].
- [118] H. He and J. Yan, “Cyber-physical attacks and defences in the smart grid: a survey,” *IET Cyber-Physical Systems: Theory & Applications*, vol. 1, no. 1, pp. 13–27, 2016.
- [119] Wikipedia, “2015 Ukraine power grid hack,” [https://en.wikipedia.org/wiki/2015\\_Ukraine\\_power\\_grid\\_hack](https://en.wikipedia.org/wiki/2015_Ukraine_power_grid_hack), 2015. [Online; accessed 19-June-2024].
- [120] New York Times, “Attacks on Electrical Substations Raise Alarm,” <https://www.nytimes.com/2023/02/04/us/electrical-substation-attacks-nc-wa.html>, 2023. [Online; accessed 19-June-2024].
- [121] B. Miller and D. Rowe, “A survey scada of and critical infrastructure incidents,” in *Proceedings of the 1st Annual Conference on Research in Information Technology*, RIIT ’12, (New York, NY, USA), p. 51–56, Association for Computing Machinery, 2012.
- [122] M. J. M. Chowdhury, M. Usman, M. S. Ferdous, N. Chowdhury, A. I. Harun, U. S. Jannat, and K. Biswas, “A cross-layer trust-based consensus protocol for peer-to-peer energy trading using fuzzy logic,” *IEEE Internet of Things Journal*, vol. 9, no. 16, pp. 14779–14789, 2022.
- [123] K. Kaur, G. Kaddoum, and S. Zeadally, “Blockchain-based cyber-physical security for electrical vehicle aided smart grid ecosystem,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 22, no. 8, pp. 5178–5189, 2021.
- [124] International energy Agency, “Digitalisation and Energy,” <https://www.iea.org/reports/digitalisation-and-energy#downloads>, 2017. [Online; accessed 19-June-2024].
- [125] A. Mate, A. K. Barnes, R. W. Bent, and E. Cotilla-Sanchez, “Analyzing and mitigating the impacts of gmd and emp events on the electrical grid with powermodels gmd.jl,” 2021.
- [126] M. Sommerhalder, *Hardware Security Module*, pp. 83–87. Cham: Springer Nature Switzerland, 2023.
- [127] NERC, “Physical security guideline for the electricity sector,” [https://www.nerc.com/comm/RSTC\\_Reliability\\_Guidelines/Physical\\_Security\\_Guideline\\_%20Assessments\\_and\\_Resiliency\\_Measures\\_for\\_Extreme\\_Events\\_June\\_2019.pdf](https://www.nerc.com/comm/RSTC_Reliability_Guidelines/Physical_Security_Guideline_%20Assessments_and_Resiliency_Measures_for_Extreme_Events_June_2019.pdf), 2019. [Online; accessed 19-June-2024].
- [128] Paul W. Parfomak, “Physical Security of the U.S. Power Grid: High-Voltage Transformer Substations,” <https://sgp.fas.org/crs/homsec/R43604.pdf>, 2014. [Online; accessed 19-June-2024].
- [129] R. E. Institute, “Asian super grid,” [https://en.wikipedia.org/wiki/Asian\\_Super\\_Grid](https://en.wikipedia.org/wiki/Asian_Super_Grid), 2011. [Online; accessed 19-June-2024].
- [130] E. Simonov, “The risks of a global supergrid,” <https://www.eco-business.com/opinion/the-risks-of-a-global-supergrid>, 2018. [Online; accessed 19-June-2024].
- [131] U. D. of State, “Transatlantic clean network,” <https://2017-2021.state.gov/the-clean-network>, 2021. [Online; accessed 19-June-2024].
- [132] P. R. of China, “Belt and road initiative,” [https://en.wikipedia.org/wiki/Belt\\_and\\_Road\\_Initiative](https://en.wikipedia.org/wiki/Belt_and_Road_Initiative), 2023. [Online; accessed 19-June-2024].
- [133] U. D. of Commerce, “Indo-pacific economic framework,” <https://www.commerce.gov/ipef>, 2023. [Online; accessed 19-June-2024].
- [134] Wikipedia, “Stuxnet,” <https://en.wikipedia.org/wiki/Stuxnet>, 2010. [Online; accessed 19-June-2024].
- [135] A. A. Cárdenas, S. Amin, and S. Sastry, “Research challenges for the security of control systems,” in *Proceedings of the 3rd Conference on Hot Topics in Security*, HOTSEC’08, (USA), USENIX Association, 2008.
- [136] K. Jhala, P. Pradhan, and B. Natarajan, “Perturbation-based diagnosis of false data injection attack using distributed energy resources,” *IEEE Transactions on Smart Grid*, vol. 12, no. 2, pp. 1589–1601, 2021.
- [137] S. M. A. A. Abir, A. Anwar, J. Choi, and A. S. M. Kayes, “Iot-enabled smart energy grid: Applications and challenges,” *IEEE Access*, vol. 9, pp. 50961–50981, 2021.
- [138] A. Akkad, G. Wills, and A. Rezazadeh, “An information security model for an iot-enabled smart grid in the saudi energy sector,” *Computers and Electrical Engineering*, vol. 105, p. 108491, 2023.
- [139] ETSI, “Quantum safe cryptography and security,” *White Paper No. 8*, 2015.
- [140] L. D., N. Nagpal, S. Chandrasekaran, and J. H. D., “A quantum-based approach for offensive security against cyber attacks in electrical infrastructure,” *Applied Soft Computing*, vol. 136, p. 110071, 2023.
- [141] P.-Y. Kong, “A review of quantum key distribution protocols in the perspective of smart grid communication security,” *IEEE Systems Journal*, vol. 16, no. 1, pp. 41–54, 2022.
- [142] A. Aldweesh, A. Derhab, and A. Z.Emam, “Deep learning approaches for anomaly-based intrusion detection systems: A survey, taxonomy, and open issues,” *Knowledge-Based Systems*, vol. 189, p. 105124, 2020.
- [143] D. Silver, S. Singh, D. Precup, and R. S. Sutton, “Reward is enough,” *Artificial Intelligence*, vol. 299, p. 103535, 2021.
- [144] M. Zhang, X. Fan, R. Lu, C. Shen, and X. Guan, “Extended moving target defense for ac state estimation in smart grids,” *IEEE Transactions on Smart Grid*, vol. 14, no. 3, pp. 2313–2325, 2023.
- [145] A. H. M. Jakaria, M. A. Rahman, and A. Gokhale, “Resiliency-aware deployment of sdn in smart grid scada: A formal synthesis model,” *IEEE Transactions on Network and Service Management*, vol. 18, no. 2, pp. 1430–1444, 2021.
- [146] S. Al-Rubaye, E. Kadhum, Q. Ni, and A. Anpalagan, “Industrial internet of things driven by sdn platform for smart grid resiliency,” *IEEE Internet of Things Journal*, vol. 6, no. 1, pp. 267–277, 2019.
- [147] J. D. Cunningham, A. Aved, D. Ferris, P. Morrone, and C. S. Tucker, “A deep learning game theoretic model for defending against large scale smart grid attacks,” *IEEE Transactions on Smart Grid*, vol. 14, no. 2, pp. 1188–1197, 2023.
- [148] M. Wood, “How safe is defining the future of network security,” *Network Security*, vol. 2020, no. 12, pp. 6–8, 2020.
- [149] R. Wójtowicz, R. Kowalik, D. D. Rasolomampionona, and K. Kurek, “Virtualization of protection systems - tests performed on a large environment based on data center solutions,” *IEEE Transactions on Power Delivery*, vol. 37, no. 4, pp. 3401–3411, 2022.
- [150] A. Vassilev, A. Oprea, A. Fordyce, and H. Anderson, “Nist trustworthy and responsible ai,” <https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.100-2e2023.pdf>, 2024. [Online; accessed 19-September-2024].
- [151] Z. Alsulaimawi, “Enhancing security in federated learning through adaptive consensus-based model update validation,” 2024.
- [152] P. McDaniel and S. McLaughlin, “Security and privacy challenges in the smart grid,” *IEEE Security and Privacy*, vol. 7, no. 3, pp. 75–77, 2009.
- [153] P. J. Hueros-Barrios, F. J. Rodríguez Sánchez, P. Martín, C. Jiménez, and I. Fernández, “Addressing the cybersecurity vulnerabilities of advanced nanogrids: A practical framework,” *Internet of Things*, vol. 20, p. 100620, 2022.
- [154] M. B. Mollah, J. Zhao, D. Niyato, K.-Y. Lam, X. Zhang, A. M. Y. M. Ghias, L. H. Koh, and L. Yang, “Blockchain for future smart grid: A comprehensive survey,” *IEEE Internet of Things Journal*, vol. 8, no. 1, pp. 18–43, 2021.
- [155] M. Moniruzzaman, A. Yassine, and R. Benlamri, “Blockchain and cooperative game theory for peer-to-peer energy trading in smart grids,” *International Journal of Electrical Power and Energy Systems*, vol. 151, p. 109111, 2023.
- [156] Y. Jia, C. Wan, P. Yu, Y. Song, and P. Ju, “Security constrained p2p energy trading in distribution network: An integrated transaction and operation model,” *IEEE Transactions on Smart Grid*, vol. 13, no. 6, pp. 4773–4786, 2022.
- [157] Z. Li, J. Zhang, Z. Cheng, J. Si, Y. Wang, and X. Wang, “Distributed cooperative grid synchronization strategy for multiple parallel grid-supporting inverters in ac microgrid,” *International Journal of Electrical Power and Energy Systems*, vol. 155, p. 109624, 2024.
- [158] Q. Zhong, J. Yao, Y. Luo, S. Huang, and S. Chen, “Dual-sequence synchronization stability analysis and control of multi-paralleled wind farms during asymmetrical grid faults,” *IEEE Transactions on Sustainable Energy*, vol. 15, no. 1, pp. 381–397, 2024.
- [159] K. Wang, J. Wu, X. Zheng, J. Li, W. Yang, and A. V. Vasilakos, “Cloud-edge orchestrated power dispatching for smart grid with distributed energy resources,” *IEEE Transactions on Cloud Computing*, vol. 11, no. 2, pp. 1194–1203, 2023.
- [160] S. Chatzimiltis, M. Shojafar, and R. Tafazolli, “A distributed intrusion detection system for future smart grid metering network,” in *ICC 2023 - IEEE International Conference on Communications*, pp. 3339–3344, 2023.

- [161] X. Chang, Y. Xu, Q. Guo, H. Sun, and W. K. Chan, "A byzantine-resilient distributed peer-to-peer energy management approach," *IEEE Transactions on Smart Grid*, vol. 14, no. 1, pp. 623–634, 2023.
- [162] L. D. Valdez, L. Shekhtman, C. E. La Rocca, X. Zhang, S. V. Buldyrev, P. A. Trunfio, L. A. Braunstein, and S. Havlin, "Cascading failures in complex networks," *Journal of Complex Networks*, vol. 8, p. cnaa013, 05 2020.
- [163] C. Feng, B. Liang, Z. Li, W. Liu, and F. Wen, "Peer-to-peer energy trading under network constraints based on generalized fast dual ascent," *IEEE Transactions on Smart Grid*, vol. 14, no. 2, pp. 1441–1453, 2023.
- [164] T. Morstyn, A. Teytelboym, and M. D. McCulloch, "Bilateral contract networks for peer-to-peer energy trading," *IEEE Transactions on Smart Grid*, vol. 10, no. 2, pp. 2026–2035, 2019.
- [165] K. Anoh, S. Maharjan, A. Ikpehai, Y. Zhang, and B. Adebisi, "Energy peer-to-peer trading in virtual microgrids in smart grids: A game-theoretic approach," *IEEE Transactions on Smart Grid*, vol. 11, no. 2, pp. 1264–1275, 2020.
- [166] J. Kang, R. Yu, X. Huang, S. Maharjan, Y. Zhang, and E. Hossain, "Enabling localized peer-to-peer electricity trading among plug-in hybrid electric vehicles using consortium blockchains," *IEEE Transactions on Industrial Informatics*, vol. 13, no. 6, pp. 3154–3164, 2017.
- [167] T. AlSkaif, J. L. Crespo-Vazquez, M. Sekuloski, G. van Leeuwen, and J. P. S. Catalão, "Blockchain-based fully peer-to-peer energy trading strategies for residential energy systems," *IEEE Transactions on Industrial Informatics*, vol. 18, no. 1, pp. 231–241, 2022.
- [168] C. Yapa, C. de Alwis, M. Liyanage, and J. Ekanayake, "Survey on blockchain for future smart grids: Technical aspects, applications, integration challenges and future research," *Energy Reports*, vol. 7, pp. 6530–6564, 2021.
- [169] A. Paudel, K. Chaudhari, C. Long, and H. B. Gooi, "Peer-to-peer energy trading in a prosumer-based community microgrid: A game-theoretic model," *IEEE Transactions on Industrial Electronics*, vol. 66, no. 8, pp. 6087–6097, 2019.
- [170] W. Liu, D. Qi, and F. Wen, "Intraday residential demand response scheme based on peer-to-peer energy trading," *IEEE Transactions on Industrial Informatics*, vol. 16, no. 3, pp. 1823–1835, 2020.
- [171] J. Wang, D. K. Mishra, L. Li, and J. Zhang, "Demand side management and peer-to-peer energy trading for industrial users using two-level multi-agent reinforcement learning," *IEEE Transactions on Energy Markets, Policy and Regulation*, vol. 1, no. 1, pp. 23–36, 2023.
- [172] K. Gai, Y. Wu, L. Zhu, M. Qiu, and M. Shen, "Privacy-preserving energy trading using consortium blockchain in smart grid," *IEEE Transactions on Industrial Informatics*, vol. 15, no. 6, pp. 3548–3558, 2019.
- [173] E. Mengelkamp, J. Gärtner, K. Rock, S. Kessler, L. Orsini, and C. Weinhardt, "Designing microgrid energy markets: A case study: The brooklyn microgrid," *Applied Energy*, vol. 210, pp. 870–880, 2018.
- [174] N. Atzei, M. Bartoletti, and T. Cimoli, "A survey of attacks on ethereum smart contracts sok," in *Proceedings of the 6th International Conference on Principles of Security and Trust - Volume 10204*, (Berlin, Heidelberg), p. 164–186, Springer-Verlag, 2017.
- [175] Y. Huang, Y. Bian, R. Li, J. L. Zhao, and P. Shi, "Smart contract security: A software lifecycle perspective," *IEEE Access*, vol. 7, pp. 150184–150202, 2019.
- [176] R. Browne, "Accidental bug may have frozen 280 million worth of digital coin ether in a cryptocurrency wallet." <https://www.cnn.com/2017/11/08/accidental-bug-may-have-frozen-280-worth-of-ether-on-parity-wallet.html>, 2017. [Online; accessed 19-June-2024].
- [177] K. Zhou, J. Huang, H. Han, B. Gong, A. Xiong, W. Wang, and Q. Wu, "Smart contracts vulnerability detection model based on adversarial multi-task learning," *Journal of Information Security and Applications*, vol. 77, p. 103555, 2023.
- [178] A. Shabani Baghani, S. Rahimpour, and M. Khabbazi, "The dao induction attack: Analysis and countermeasure," *IEEE Internet of Things Journal*, vol. 9, no. 7, pp. 4875–4887, 2022.
- [179] A. P. Joshi, M. Han, and Y. Wang, "A survey on security and privacy issues of blockchain technology," *Mathematical Foundations of Computing*, vol. 1, no. 2, pp. 121–147, 2018.
- [180] Y.-W. Chen and L. Sutanto, "The privacy preserving framework with virtual ring and identity-based cryptography for smart grid," in *17th International Conference on Information Technology–New Generations (ITNG 2020)* (S. Latifi, ed.), (Cham), pp. 271–276, Springer International Publishing, 2020.
- [181] M. Baza, A. Sherif, M. M. E. A. Mahmoud, S. Bakiras, W. Alasmay, M. Abdallah, and X. Lin, "Privacy-preserving blockchain-based energy trading schemes for electric vehicles," *IEEE Transactions on Vehicular Technology*, vol. 70, no. 9, pp. 9369–9384, 2021.
- [182] K. Bhargavan, A. Delignat-Lavaud, C. Fournet, A. Gollamudi, G. Gonthier, N. Kobeissi, N. Kulatova, A. Rastogi, T. Sibut-Pinote, N. Swamy, and S. Zanella-Béguelin, "Formal verification of smart contracts: Short paper," in *Proceedings of the 2016 ACM Workshop on Programming Languages and Analysis for Security, PLAS '16*, (New York, NY, USA), p. 91–96, Association for Computing Machinery, 2016.
- [183] S. Jiang, X. Zhang, J. Li, H. Yue, and Y. Zhou, "Secure and privacy-preserving energy trading scheme based on blockchain," in *GLOBE-COM 2020 - 2020 IEEE Global Communications Conference*, pp. 1–6, 2020.
- [184] Y. Wang, Y. Li, W. Jiao, G. Wang, J. Zhao, Y. Qiang, and K. Li, "An efficient, secured, and infinitely scalable consensus mechanism for peer-to-peer energy trading blockchain," *IEEE Transactions on Industry Applications*, vol. 59, no. 5, pp. 5215–5229, 2023.
- [185] M. Conti, E. Sandeep Kumar, C. Lal, and S. Ruj, "A survey on security and privacy issues of bitcoin," *IEEE Communications Surveys & Tutorials*, vol. 20, no. 4, pp. 3416–3452, 2018.
- [186] M. Jafari, A. Kavousi-Fard, T. Chen, and M. Karimi, "A review on digital twin technology in smart grid, transportation system and smart city: Challenges and future," *IEEE Access*, vol. 11, pp. 17471–17484, 2023.
- [187] J. Close, J. E. Barnard, Y. John Chew, and S. Perera, "A holistic approach to improving safety for battery energy storage systems," *Journal of Energy Chemistry*, vol. 92, pp. 422–439, 2024.
- [188] O. Duman, L. Wang, M. Au, M. Kassouf, and M. Debbabi, "Hardening substations against supply chain attacks under operational constraints," in *2022 IEEE Power and Energy Society Innovative Smart Grid Technologies Conference (ISGT)*, pp. 1–5, 2022.
- [189] J. Johnson, T. Berg, B. Anderson, and B. Wright, "Review of electric vehicle charger cybersecurity vulnerabilities, potential impacts, and defenses," *Energies*, vol. 15, no. 11, 2022.
- [190] Y. Ren, H. Zhang, W. Yang, M. Li, J. Zhang, and H. Li, "Transferable adversarial attack against deep reinforcement learning-based smart grid dynamic pricing system," *IEEE Transactions on Industrial Informatics*, vol. 20, no. 6, pp. 9015–9025, 2024.
- [191] M. M. H. Sifat, S. M. Choudhury, S. K. Das, M. H. Ahamed, S. Mueen, M. M. Hasan, M. F. Ali, Z. Tasneem, M. M. Islam, M. R. Islam, M. F. R. Badal, S. H. Abhi, S. K. Sarker, and P. Das, "Towards electric digital twin grid: Technology and framework review," *Energy and AI*, vol. 11, p. 100213, 2023.
- [192] A. Takiddin, M. Ismail, and E. Serpedin, "Robust data-driven detection of electricity theft adversarial evasion attacks in smart grids," *IEEE Transactions on Smart Grid*, vol. 14, no. 1, pp. 663–676, 2023.
- [193] R. Huang and Y. Li, "Adversarial attack mitigation strategy for machine learning-based network attack detection model in power system," *IEEE Transactions on Smart Grid*, vol. 14, no. 3, pp. 2367–2376, 2023.
- [194] Y. Wang and B. C. Pal, "Destabilizing attack and robust defense for inverter-based microgrids by adversarial deep reinforcement learning," *IEEE Transactions on Smart Grid*, vol. 14, no. 6, pp. 4839–4850, 2023.
- [195] E. Catak, F. O. Catak, and A. Moldsvor, "Adversarial machine learning security problems for 6g: mmwave beam prediction use-case," 2021.
- [196] M. Rezaeimozafer, R. F. Monaghan, E. Barrett, and M. Duffy, "A review of behind-the-meter energy storage systems in smart grids," *Renewable and Sustainable Energy Reviews*, vol. 164, p. 112573, 2022.
- [197] CleanTechnica, "Gogoro batteries power 90[Online; accessed 19-June-2024].
- [198] I. E. Agency, "Global electric vehicle outlook." <https://www.iea.org/reports/global-ev-outlook-2024/executive-summary>, 2023. [Online; accessed 19-June-2024].
- [199] U. D. of Energy, "2020 smart grid system report." [https://www.energy.gov/sites/default/files/2022-05/2020%20Smart%20Grid%20System%20Report\\_0.pdf](https://www.energy.gov/sites/default/files/2022-05/2020%20Smart%20Grid%20System%20Report_0.pdf), 2020. [Online; accessed 19-June-2024].
- [200] Z. Garofalaki, D. Kosmanos, S. Moschogiannis, D. Kallergis, and C. Douligeris, "Electric vehicle charging: A survey on the security issues and challenges of the open charge point protocol (ocpp)," *IEEE Communications Surveys & Tutorials*, vol. 24, no. 3, pp. 1504–1533, 2022.
- [201] N. Bazmohammadi, A. Madary, J. C. Vasquez, H. B. Mohammadi, B. Khan, Y. Wu, and J. M. Guerrero, "Microgrid digital twins: Concepts, applications, and future trends," *IEEE Access*, vol. 10, pp. 2284–2302, 2022.
- [202] J. Hao and Y. Tao, "Adversarial attacks on deep learning models in smart grids," *Energy Reports*, vol. 8, pp. 123–129, 2022. 2021



- 6th International Conference on Clean Energy and Power Generation Technology.
- [203] Q. Song, R. Tan, C. Ren, and Y. Xu, "Understanding credibility of adversarial examples against smart grid: A case study for voltage stability assessment," in *Proceedings of the Twelfth ACM International Conference on Future Energy Systems, e-Energy '21*, (New York, NY, USA), p. 95–106, Association for Computing Machinery, 2021.
  - [204] J. Tian, B. Wang, J. Li, Z. Wang, B. Ma, and M. Ozay, "Exploring targeted and stealthy false data injection attacks via adversarial machine learning," *IEEE Internet of Things Journal*, vol. 9, no. 15, pp. 14116–14125, 2022.
  - [205] P. M. Santos, B. R. Manoj, M. Sadeghi, and E. G. Larsson, "Universal adversarial attacks on neural networks for power allocation in a massive mimo system," *IEEE Wireless Communications Letters*, vol. 11, no. 1, pp. 67–71, 2022.
  - [206] F. Wang, M. C. Gursoy, and S. Velipasalar, "Adversarial reinforcement learning in dynamic channel access and power control," 2021.
  - [207] A. Bahramali, M. Nasr, A. Houmansadr, D. Goeckel, and D. Towsley, "Robust adversarial attacks against dnn-based wireless communication systems," 2021.
  - [208] Q. Song, R. Tan, C. Ren, Y. Xu, Y. Lou, J. Wang, and H. B. Gooi, "On credibility of adversarial examples against learning-based grid voltage stability assessment," *IEEE Transactions on Dependable and Secure Computing*, vol. 21, no. 2, pp. 585–599, 2024.
  - [209] J. Wang and P. Srikantha, "Stealthy black-box attacks on deep learning non-intrusive load monitoring models," *IEEE Transactions on Smart Grid*, vol. 12, no. 4, pp. 3479–3492, 2021.
  - [210] E.-N. S. Youssef, F. Labeau, and M. Kassouf, "Adversarial dynamic load-altering cyberattacks against peak shaving using residential electric water heaters," *IEEE Transactions on Smart Grid*, pp. 1–1, 2023.
  - [211] Z. Zhang, Z. Yang, D. K. Yau, Y. Tian, and J. Ma, "Data security of machine learning applied in low-carbon smart grid: A formal model for the physics-constrained robustness," *Applied Energy*, vol. 347, p. 121405, 2023.
  - [212] M. Sadeghi and E. G. Larsson, "Adversarial attacks on deep-learning based radio signal classification," *IEEE Wireless Communications Letters*, vol. 8, no. 1, pp. 213–216, 2019.
  - [213] J. Folz, S. Palacio, J. Hees, and A. Dengel, "Adversarial defense based on structure-to-signal autoencoders," in *Proc. on Applications of Computer Vision Conf.*, p. 3568–3577, 2020.
  - [214] F. Liao, M. Liang, Y. Dong, T. Pang, X. Hu, and J. Zhu, "Defense against adversarial attacks using high-level representation guided denoiser," in *Proc. on Computer Vision and Pattern Recognition Conf.*, p. 1778–1787, 2018.
  - [215] M. Elsis, C.-L. Su, and M. N. Ali, "Design of reliable iot systems with deep learning to support resilient demand side management in smart grids against adversarial attacks," *IEEE Transactions on Industry Applications*, pp. 1–12, 2023.
  - [216] Yue Zhao, "Python library for detecting anomalies in multivariate data." <https://pyod.readthedocs.io/en/latest/>, 2017. [Online; accessed 19-June-2024].
  - [217] S. Addepalli, V. B.S., A. Baburaj, G. Sriramanan, and R. V. Babu, "Towards achieving adversarial robustness by enforcing feature consistency across bit planes," in *Proc. on Computer Vision and Pattern Recognition Conf.*, p. 1020–1029, 2020.
  - [218] H. Kannan, A. Kurakin, and I. Goodfellow, "Adversarial logit pairing," *Machine Learning*, vol. 1, 2018.
  - [219] N. Papernot, P. McDaniel, X. Wu, S. Jha, and A. Swami, "Distillation as a defense to adversarial perturbations against deep neural networks," 2016.
  - [220] M. Farajzadeh-Zanjani, E. Hallaji, R. Razavi-Far, M. Saif, and M. Parvania, "Adversarial semi-supervised learning for diagnosing faults and attacks in power grids," *IEEE Transactions on Smart Grid*, vol. 12, no. 4, pp. 3468–3478, 2021.
  - [221] A. Takiddin, M. Ismail, R. Atat, K. R. Davis, and E. Serpedin, "Robust graph autoencoder-based detection of false data injection attacks against data poisoning in smart grids," *IEEE Transactions on Artificial Intelligence*, pp. 1–15, 2023.
  - [222] E. Wong and J. Z. Kolter, "Provable defenses against adversarial examples via the convex outer adversarial polytope," 2018.
  - [223] T. Strauss, M. Hanselmann, A. Junginger, and H. Ulmer, "Ensemble methods as a defense to adversarial perturbations against deep neural networks," 2018.
  - [224] D. Gunning and D. Aha, "Darpa's explainable artificial intelligence (xai) program," *AI Magazine*, vol. 40, pp. 44–58, Jun. 2019.
  - [225] B. G. A. Tekgul and N. Asokan, "Flare: Fingerprinting deep reinforcement learning agents using universal adversarial masks," in *Proceedings of the 39th Annual Computer Security Applications Conference, ACSAC '23*, (New York, NY, USA), p. 492–505, Association for Computing Machinery, 2023.
  - [226] Z. Yang, B. Shao, B. Xuan, E.-C. Chang, and F. Zhang, "Defending model inversion and membership inference attacks via prediction purification," 2020.
  - [227] A. Athalye, N. Carlini, and D. Wagner, "Obfuscated gradients give a false sense of security: Circumventing defenses to adversarial examples," 2018.
  - [228] F. Gallego, C. Martín, M. Díaz, and D. Garrido, "Maintaining flexibility in smart grid consumption through deep learning and deep reinforcement learning," *Energy and AI*, vol. 13, p. 100241, 2023.
  - [229] D. Yao, M. Wen, X. Liang, Z. Fu, K. Zhang, and B. Yang, "Energy theft detection with energy privacy preservation in the smart grid," *IEEE Internet of Things Journal*, vol. 6, no. 5, pp. 7659–7669, 2019.
  - [230] I. U. Khan, N. Javeid, C. J. Taylor, K. A. A. Gamage, and X. Ma, "A stacked machine and deep learning-based approach for analysing electricity theft in smart grids," *IEEE Transactions on Smart Grid*, vol. 13, no. 2, pp. 1633–1644, 2022.
  - [231] L. Wu, H. Shi, S. Fu, Y. Luo, and M. Xu, "p2detect: Electricity theft detection with privacy preservation for both data and model in smart grid," *IEEE Transactions on Smart Grid*, vol. 14, no. 3, pp. 2301–2312, 2023.
  - [232] C.-C. Sun, D. J. Sebastian Cardenas, A. Hahn, and C.-C. Liu, "Intrusion detection for cybersecurity of smart meters," *IEEE Transactions on Smart Grid*, vol. 12, no. 1, pp. 612–622, 2021.
  - [233] Y. Chen, S. Lakshminarayana, and F. Teng, "Localization of coordinated cyber-physical attacks in power grids using moving target defense and deep learning," in *2022 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm)*, pp. 387–392, 2022.
  - [234] T. Cai, T. Jia, S. Adepu, Y. Li, and Z. Yang, "Adam: An adaptive ddos attack mitigation scheme in software-defined cyber-physical system," *IEEE Transactions on Industrial Informatics*, vol. 19, no. 6, pp. 7802–7813, 2023.
  - [235] Z. A. El Houda, B. Brik, and L. Khoukhi, "Ensemble learning for intrusion detection in sdn-based zero touch smart grid systems," in *2022 IEEE 47th Conference on Local Computer Networks (LCN)*, pp. 149–156, 2022.
  - [236] A. K. Al Mhdawi and H. S. Al-Raweshidy, "A smart optimization of fault diagnosis in electrical grid using distributed software-defined iot system," *IEEE Systems Journal*, vol. 14, no. 2, pp. 2780–2790, 2020.
  - [237] M. M. Badr, M. M. E. A. Mahmoud, Y. Fang, M. Abdulaal, A. J. Aljohani, W. Alasmay, and M. I. Ibrahim, "Privacy-preserving and communication-efficient energy prediction scheme based on federated learning for smart grids," *IEEE Internet of Things Journal*, vol. 10, no. 9, pp. 7719–7736, 2023.
  - [238] Y.-C. Chuang and W.-Y. Chiu, "Deep reinforcement learning based pricing strategy of aggregators considering renewable energy," *IEEE Transactions on Emerging Topics in Computational Intelligence*, vol. 6, no. 3, pp. 499–508, 2022.
  - [239] J. Qi, L. Lei, K. Zheng, S. X. Yang, and X. Shen, "Optimal scheduling in iot-driven smart isolated microgrids based on deep reinforcement learning," *IEEE Internet of Things Journal*, vol. 10, no. 18, pp. 16284–16299, 2023.
  - [240] P. M. Salmon, C. Baber, C. Burns, T. Carden, N. Cooke, M. Cummings, P. Hancock, S. McLean, G. J. M. Read, and N. A. Stanton, "Managing the risks of artificial general intelligence: A human factors and ergonomics perspective," *Human Factors and Ergonomics in Manufacturing & Service Industries*, vol. 33, no. 5, pp. 366–378, 2023.
  - [241] E. Mohammadi, M. Alizadeh, M. Asgarimoghaddam, X. Wang, and M. G. Simões, "A review on application of artificial intelligence techniques in microgrids," *IEEE Journal of Emerging and Selected Topics in Industrial Electronics*, 2022.
  - [242] A. Yayla, L. Haghnegahdar, and E. Dincelli, "Explainable artificial intelligence for smart grid intrusion detection systems," *IT Professional*, vol. 24, no. 5, pp. 18–24, 2022.
  - [243] J. Schuett, N. Dreksler, M. Anderljung, D. McCaffary, L. Heim, E. Bluemke, and B. Garfinkel, "Towards best practices in agi safety and governance: A survey of expert opinion," 2023.
  - [244] C. Strickland, C. Saha, M. Zakar, S. Nejad, N. Tasnim, D. Lizotte, and A. Haque, "Drl-gan: A hybrid approach for binary and multiclass network intrusion detection," 2023.
  - [245] F. E. Abrahamsen, Y. Ai, and M. Cheffena, "Communication technologies for smart grid: A comprehensive survey," 2021.